UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Ensuring Timely and Orderly Interconnection of Large Loads	))	Docket No. RM26-4-000

COMMENTS OF THE SOUTHEAST PUBLIC INTEREST ORGANIZATIONS

November 21, 2025

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Southern Environmental Law Center (SELC), Appalachian Voices, North Carolina Sustainable Energy Association, South Carolina Coastal Conservation League, and Southern Alliance for Clean Energy (together, Southeast Public Interest Organizations) submit these initial comments in response to the Federal Energy Regulatory Commission's (FERC or Commission) Notice Inviting Comments[1] on the Secretary of Energy's (Secretary) October 23, 2025 proposed Advance Notice of Proposed Rulemaking (ANOPR).[2]

I. INTRODUCTION

For the first time in decades, demand for electricity is growing. But this growth is not steady; it is sudden and irregular. It surges in some geographic pockets much faster than others and swells by one or more large chunks at a time. Were this growth linear or moderately paced, grid owners and operators might have the ability to expand electrical infrastructure in a rational, efficient, and proactive manner. But the lumpy velocity of large load interconnection has overburdened the capacity of existing processes. This is especially true where high-load factor demand sources require near-constant service and perform better when clustered in relatively close proximity. Taken together, the speed, size, and density of these large load interconnections overwhelm the grid's ability to serve them. And given their increasing propensity to interconnect

[1] Notice Inviting Comments, Docket No. RM26-4-000 (Oct. 27, 2025).
[2] *Ensuring Timely and Orderly Interconnection of Large Loads*, Advance Notice of Proposed Rulemaking, Secretary of Energy (2025) (ANOPR).

directly to the interstate transmission system, large loads create the need for quick and costly transmission expansion. Without changes to the traditional cost allocation and resource adequacy paradigms that may have made sense in an earlier era, increased transmission costs and heightened reliability risks will fall disproportionately on residential ratepayers, further straining their ability to afford electric service that they find to be less and less reliable.

Southeast Public Interest Organizations broadly support the nationwide concerns described by the Public Interest Organizations[3] but write separately to provide perspective from a region that has experienced these trends more acutely than others. Southeast Public Interest Organizations operate in Virginia, Georgia, North Carolina, South Carolina, Alabama, and Tennessee, among other states, all of which have seen or are expecting significant demand growth from large loads. This region, which straddles territory served by PJM Interconnection L.L.C. (PJM) and public utilities that do not belong to any Regional Transmission Organization (RTO) or Independent System Operator (ISO), is primarily served by vertically integrated utilities. All rely to some degree on Integrated Resource Planning (IRP) processes, overseen by state regulators, to determine the means by which their utilities will serve expected demand years into the future. And traditionally, these utilities have had full responsibility to coordinate the interconnection of the demand sources they serve, regardless of whether the load connects to the transmission or distribution system. To the extent this proceeding results in the Commission asserting jurisdiction over some subset of those interconnection processes, this dynamic raises unique timing, resource adequacy, and transmission planning questions for vertically integrated utilities that the Commission must consider.

[3] Public Interest Organizations, Initial Comments, Docket No. RM26-4-000 (filed Nov. 21, 2025).

Many of these states have acknowledged the scale of the issue facing them and have initiated efforts to address them through their exclusive authority over retail sales. Most of these efforts seek to commit large loads—financially and temporally—to receiving service from the utility, in order to reduce the risk that other customers bear the costs of stranded assets in the event the load does not materialize. They may also look to more equitably allocate the costs to the large loads that they cause, sparing other customer classes. But these state-level efforts vary widely and not all have made concerted efforts to address the cost and reliability risks of unrestrained large load interconnection.

Accordingly, there is ample room for the Commission to exercise jurisdiction over some aspects of large load interconnection to the transmission system. Doing so would effectively set a nationwide floor of protection against unwarranted cost shifts. And states would retain full authority to regulate their jurisdiction retail customers as they see fit. To this end, in asserting its jurisdiction over the interconnection of large loads to the transmission system, the Commission must recognize the state's necessary presence as the retail regulator in every such case. Doing so requires that the Commission not disrupt state efforts to discipline large load integration but instead complement them. At the same time, states should recognize that the transmission system is an interstate machine, and decisions affecting the grid in one state will reverberate in others, affecting ratepayers whose interests they are not bound to protect.

Truly, the interconnection of large loads to the transmission system intertwines state and federal regulation more closely than perhaps any other dynamic that FERC faces. It is therefore essential that the Commission be a good jurisdictional neighbor and take care not to intrude on any parallel state actions. To this end, Southeast Public Interest Organizations urge the Commission to give this proceeding the time and attention it deserves and not rush to meet an unreasonable deadline for final action. Asserting jurisdiction over load interconnections for the first time will require resolving many conflicting interests from all sectors of the industry and all regions of the country. These questions cannot be answered in a matter of months, at least in a manner that will protect the affordability and reliability of the transmission system. Finally, as these comments demonstrate, the issues giving rise to this rulemaking affect both RTOs/ISOs and non-RTO/ISO areas alike. The Commission must not therefore limit its reach to one or the other, as they share the right to just and reasonable transmission rates and reliable electric service, both of which must be the central pillars of this proceeding.

II. THRESHOLD ISSUES

The Commission may fashion standardized procedures for large load interconnection to the transmission system only if it can satisfy two threshold Federal Power Act (FPA) requirements. First, under section 206(a), it must affirmatively find that an existing rate, charge, classification, or practice affecting them is unjust, unreasonable, or unduly discriminatory or preferential.[4] Second, the Commission may only act to fix the identified rate, charge, classification, or practice consistent with the scope of its jurisdictional authority, which extends to "the transmission of electric energy in interstate commerce and to the sale of electric energy at wholesale in interstate commerce," including "all facilities for such transmission or sale of electric energy."[5] It does not include "facilities used for the generation of electric energy or over facilities used in local distribution or only for the transmission of electric energy in intrastate commerce, or over facilities for the transmission of electric energy consumed wholly by the

[4] 16 U.S.C. § 824e(a).
[5] 16 U.S.C. § 824(b)(1).

transmitter."[6] Below, Southeast Public Interest Organizations offer suggestions for the Commission to address both of these requirements.

A. Section 206 Step 1 Finding

Before the Commission can act in this proceeding, it must first find that an existing rate, charge, classification, or practice affecting them is unjust, unreasonable, or unduly discriminatory or preferential.[7] Making this predicate finding "is the condition precedent to FERC's exercise of its section 206 authority to change that rate."[8] In other words, "[w]ithout a showing that the existing rate is unlawful, FERC has no authority to impose a new rate."[9]

The ANOPR does not make a finding sufficient to meet this standard. It broadly describes the unprecedented growth of large loads and their increasing interconnection to the transmission system and notes the unique challenges to demand forecasting and system planning this presents.[10] The Need for Reform section adds that this growth and the importance of providing open and non-discriminatory access to the transmission system has created a need for standardized interconnection procedures and agreements.[11] But the ANOPR does not demonstrate that large loads have experienced limited or discriminatory access to the transmission system, as the Commission did in Order No. 2003.[12] Nor has it identified any existing practices that cause such discrimination. Rather, for the Commission's jurisdiction to attach, as discussed in greater detail below, the Commission must find that the procedures

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[6] *Id*.
[7] 16 U.S.C. § 824e(a).
[8] *Emera Maine v. FERC*, 854 F.3d 9, 25 (D.C. Cir. 2017) (cleaned up).
[9] *Id*.
[10] *See* ANOPR at II.A.
[11] *See id*. P 12.
[12] *Standardization of Generator Interconnection Agreements & Procs*., Order No. 2003, 68 FR 49846 (Aug. 19, 2003), 104 FERC ¶ 61,103 (2003), *order on reh'g*, Order No. 2003-A, 69 FR 15932 (Mar. 26, 2004), 106 FERC ¶ 61,220, *order on reh'g*, Order No. 2003-B, 70 FR 265 (Jan. 4, 2005), 109 FERC ¶ 61,287 (2004), *order on reh'g*, Order No. 2003-C, 70 FR 37661 (June 30, 2005), 111 FERC ¶ 61,401 (2005), *aff'd sub nom. Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 475 F.3d 1277 (D.C. Cir. 2007).

governing the physical connection of large loads to the transmission system, the agreements establishing those arrangements, and the allocation of cost responsibility for the associated upgrades are unjust unreasonable or unduly discriminatory.[13]

The Commission could do so by acknowledging: (1) the significant growth in large load interconnections to the transmission system; (2) that the pace, size, and load profiles of large load interconnections create substantial transmission upgrade costs that inequitably fall to other customer classes under current cost allocation practices; and (3) that the inherent uncertainty that large loads will materialize combined with the long lead time of transmission upgrades renders existing load forecasting processes unreliable, creating a greater risk of stranded investment, whose costs will be borne by other customers. Each of these has manifested in the Southeast.

1. Large loads are growing at an unprecedented rate across the region.

Southeast Public Interest Organizations operate in a region that has seen the greatest influx of large loads to date, a trend its public utilities only project to increase. Outside the non-jurisdictional Electric Reliability Council of Texas, PJM—which includes all of Virginia and portions of North Carolina and Tennessee—and Georgia Power Company (Georgia Power) project the greatest load increases by the end of the decade.[14] Duke Energy's affiliates in North Carolina and South Carolina are not far behind, nor is the Tennessee Valley Authority.[15]

At the region's northern tip, Virginia has long paced the world in data center integration and does not project to relinquish that title anytime soon.[16] To this end, Virginia's load serving entities (LSE) recently updated PJM on the large load additions they would like included in the

[13] *See infra* sections (II)(A)(1)-(3).
[14] *See* John D. Wilson, Zach Zimmerman & Rob Gramlich, Grid Strategies, *Strategic Industries Surging: Driving US Power and Demand*, at 24 (Dec. 2024), https://gridstrategiesllc.com/wp-content/uploads/National-Load-Growth-Report-2024.pdf.
[15] *See id*.
[16] *See* Dan Gearino and Charles Paullin, *How Did This State Become the Data Center Capital of the World?*, INSIDE CLIMATE NEWS (Oct. 26, 2025), https://insideclimatenews.org/news/26102025/virginia-data-center-capital-ai-boom/.

regional PJM load forecast going forward. For the DOM zone, which includes Dominion Energy Virginia (Dominion), Northern Virginia Electric Cooperative, Old Dominion Electric Cooperative, and Rappahannock Electric Cooperative, the LSEs requested a collective large load adjustment to 7,219.6 MW of demand for 2026, which more than doubles to 14,604.4 MW for 2030.[17] Moving out to 2035, that figure balloons to 23,986.3, and even further to 34,527.7 for 2040.[18] In pure capacity terms, those figures are significantly higher: 23,171.1 MW for 2026, 30,597.3 MW for 2030, 37,483.9 MW for 2035, and 41,780.4 MW for 2040.[19] Even for the data center capital of the world, those numbers are astronomical.

Moving south to a less traditional data center market, Duke Energy Carolinas, LLC (DEC) and Duke Energy Progress, LLC (DEP) (together, Duke Energy), which operate in North and South Carolina, also report significant large load projections. In its Fall 2025 Update to its 10-year economic development forecast, Duke Energy reported 38 projects for a "full-load" (capacity) total of 5,610 MW.[20] This figure grew by nearly 2 GW since last year's projection.[21] Likewise, in Georgia, Georgia Power recently informed investors that it has contracts with 8 GW worth of large loads out to the mid-2030s, with another 4 GW "committed."[22] In total, its large load pipeline includes over *50 GW* through the mid-2030s.[23] While all of these large loads in the

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[17] PJM Interconnection, L.L.C., Total Demand Request for Large Load Adjustment, "2026 DEMAND Requests" Tab at cells E36, I36, https://www.pjm.com/-/media/DotCom/committees-groups/subcommittees/las/2025/20250916/20250916-post-meeting---informational-only---large-load-adjustment-requests.xlsx.
[18] *See id*. at cells N36, S36.
[19] *See id*. "2026 CAPACITY Requests" Tab at cells E36, I36, N36, S36.
[20] *See* Duke Energy Carolinas, LLC and Duke Energy Progress, LLC, Semi-Annual Update Report on Large Load Customer Additions in Advanced Stages of Development – Fall 2025, N.C. Corp. Comm'n Docket No. 1-100, Sub 207, at 3 (filed Nov. 17, 2025).
[21] *See id*.
[22] *See* Southern Company, *Third Quarter 2025 Earnings Conference Call*, at 9 (Oct. 30, 2025), https://s27.q4cdn.com/273397814/files/doc_financials/2025/q3/SO-2025-Q3-Earnings-Call-FINAL.pdf.
[23] *See id*.

utilities' respective pipelines have varying degrees of commitment, the bottom-line figures are so large that even a fraction would require substantial grid buildout to accommodate.

Taken together, the scale and pace of these large load growth projections—even in a single region—are unlike anything the industry has previously seen. That alone suggests that FERC must reassess its existing processes to ensure they remain just and reasonable in a new era. Yet its ability to act requires more than the novelty of this large load growth environment; the Commission must find that specific, existing processes are unjust, unreasonable, or unduly discriminatory or preferential due to their interaction with these unprecedented conditions.

2. As a result of the scale, proximity, and pace of large load growth, large load interconnection requires substantial transmission investment, creating unique cost impacts that fall disproportionately to existing ratepayers.

The profile of large loads presents unique challenges, particularly with respect to network upgrade cost allocation. Despite the substantial transmission upgrades large load interconnections require, current cost allocation practices typically do not distinguish between network upgrades caused by these interconnections and other load. In an RTO like PJM, transmission costs are allocated at the LSE level—i.e., to Dominion as a whole—and state regulators divvy these costs broadly across retail rate classes. In non-RTO regions that utilize bundled retail rates, all transmission costs flow into the broader rate base that underlies retail rates paid by all customer classes. In both cases, the costs of network upgrades caused by large load interconnections are generally socialized across all customers.

In the past, this practice may not have had an appreciable effect on customer bills, but large loads present a different case. For one, as the label suggests, they are large. Data centers, the most prevalent type of large load in development, can range from 5-10 MW to 200 MW or more,[24] with some significantly larger projects proposed. xAI has billed its Colossus 2, currently in development in South Memphis, Tennessee, as the "first gigawatt AI training cluster."[25] Second, they have high load factors, as they require near-constant service to power various computing processes.[26] Finally, they tend to cluster in locations that share desirable attributes, such as affordable power supply, high network connectivity via access to fiber optic infrastructure, abundant water resources for cooling, and governmental considerations like tax incentives and zoning restrictions.[27] The confluence of all of these factors led to Northern Virginia becoming the preeminent data center host,[28] but data center clusters have also emerged in metro areas like Atlanta and in the Carolinas.[29] Taken together, the size, load profile, and geographic proximity of large loads can place acute strain on electric infrastructure in relatively localized areas. Combine those factors with the pace and volume of requests and the associated impacts must necessarily rise beyond the distribution system to the interstate transmission system.

As one would expect given its prominence in the data center boom, Virginia is currently experiencing a surge in data center-related transmission upgrades. In its recent 2024 IRP,

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[24] *See* Attachment A, London Economics International, LLC, *Uncertainty and Upward Bias are Inherent in Data Center Electricity Demand Projections*, at 7 (July 7, 2025) (LEI Report).

[25] *See* Neil Strebig, *Elon Musk Says xAI's Colossus 2 Supercomputer in Memphis Close to Coming Online*, Commercial Appeal (July 22, 2025), https://www.commercialappeal.com/story/money/business/development/2025/07/22/xai-colossus-2-memphis-supercomputer/85325723007/.

[26] *See* John D. Wilson & Zach Zimmerman, GRID STRATEGIES, *The Era of Flat Power Demand is Over*, at 18 (Dec. 2023).

[27] *See* Tim Tuberville, *Data Center Site Selection: Key Considerations*, PROCORE (Sept. 23, 2025), https://www.procore.com/library/data-center-site-selection?msockid=34571de7a8da60f710ff0c8da96a6196#water.

[28] *See* Dan Gearino and Charles Paullin, *How Did This State Become the Data Center Capital of the World?*, INSIDE CLIMATE NEWS (Oct. 26, 2025), https://insideclimatenews.org/news/26102025/virginia-data-center-capital-ai-boom/.

[29] *See* DorMiya Vance, *Tax incentives are drawing data centers to Atlanta's south suburbs, worrying residents*, NPR (Oct. 12, 2025), https://www.npr.org/2025/10/12/nx-s1-5537109/tax-incentives-are-drawing-data-centers-to-atlantas-south-suburbs-worrying-residents; Brian Gordon, *Data centers in NC: Secret locations, hyperscale giants and Raleigh's next 'edge' site*, THE NEWS & OBSERVER (Mar. 12, 2025), https://www.newsobserver.com/news/business/article299051160.html.

Dominion included a list of planned transmission projects slated to go into service during the planning period.[30] Pursuant to a Virginia State Corporation Commission (SCC) directive, Dominion specified "whether the need for the transmission project is primarily being driven by data center load growth."[31] Of the 203 total projects listed, 89 were designated as directly caused by data centers.[32] More specifically, they were either "(1) . . . initiated by a Delivery Point ('DP') request specifically indicating that the interconnection was for a new data center load or, (2) resolve 'harm' associated with the interconnection of new data center DPs."[33] The total estimated cost of these projects came to $2.435 billion.[34]

Dominion also listed 37 multi-driver projects that "may be driven by increased loads within a region (including those from data centers) but cannot be directly attributed to a specific customer type."[35] There is significant overlap between these projects and those selected as part of the PJM Regional Transmission Expansion Plan's (RTEP) 2022 Window 3 solicitation. Per PJM, the purpose of many of these 2022 Window 3 lines was (1) "Local Constraints: Resulting from directly serving the data center loads in APS and Dominion zones through the respective 230 kV networks and into the points of delivery" and (2) "Regional Constraints: Resulting from imports into load center areas (500 kV primarily)."[36] Thus, while a direct nexus between the need for these multi-driver upgrades and individual data center customers cannot be made, facilitating service to large loads was a primary consideration. Dominion estimated the total cost

[30] *See* Virginia Electric & Power Company, 2024 IRP Supplement, Va. State Corp. Comm'n Case No. PUR-2024-00184, at Supplemental App. 2C-2 (Dominion 2024 IRP Supplement).
[31] *Id*. at 6.
[32] *See id*. at 6, Supplemental App. 2C-2.
[33] *Id*. at 6.
[34] *See id.* at Supplemental App. 2C-2.
[35] *Id*. at 7.
[36] PJM Interconnection, L.L.C., Reliability Analysis Report: 2022 RTEP Window 3, at 6 (Dec. 8, 2023), https://www.pjm.com/-/media/DotCom/committees-groups/committees/teac/2023/20231205/20231205-2022-rtep-window-3-reliability-analysis-report.pdf.

of these upgrades at $3.329 billion.[37] That brings the total cost of upgrades that can be attributed to data centers in some fashion to $5.764 billion. That figure represents 75.9 percent of the $7.595 billion total cost of the entire Dominion transmission plan.

Looking at PJM more broadly, the Union of Concerned Scientists (UCS) recently issued a report that sought to identify the Supplemental Projects planned to serve data centers directly. It found that "costs for the electric connections that extend from the existing transmission lines to the data center customers," which consist of the "costs for directly connecting the data centers, often including new transformers and gear on the property of the data center," amounted to $4.3 billion in 2024 alone.[38] In its recent update to the report, UCS found the following costs associated with Virginia: 60 projects totaling $2.0215 billion in 2024, 26 projects totaling $1.4225 billion in 2025, and at least 44 projects planned for beyond 2025.[39]

As Supplemental Projects in PJM, the costs for these facilities are fully allocated to the applicable zone, in this case DOM.[40] The larger, regional projects are allocated 50 percent to the zone and 50 percent to the entire PJM region, based on load ratio share.[41] All are included in the applicable Transmission Owner's rate base and recovered through network transmission rates, which are broadly paid by LSEs who pass the costs to retail customers. The responsibility to determine the share of transmission costs among retail customer classes then falls to state utility

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[37] Dominion 2024 IRP Supplement at Supplemental App. 2C-2.

[38] Mike Jacobs, UNION OF CONCERNED SCIENTISTS, *Data Centers Are Already Increasing Your Energy Bills. We Have the Receipts.* (Sept. 29, 2025), https://blog.ucs.org/mike-jacobs/data-centers-are-already-increasing-your-energy-bills/ (*Data Centers Are Already Increasing Your Energy Bills. We Have the Receipts*).

[39] Mike Jacobs, UNION OF CONCERNED SCIENTISTS, *Finally, Something Everyone Agrees On: Data Centers Should Cover Their Own Costs* (Nov. 6, 2025), https://blog.ucs.org/mike-jacobs/finally-something-everyone-agrees-on-data-centers-should-cover-their-own-costs/.

[40] PJM Interconnection, L.L.C., Intra-PJM Tariffs, OATT, Schedule 12(b)(ii), (iii), (v) (15.0.0).

[41] PJM Interconnection, L.L.C., Intra-PJM Tariffs, OATT, Schedule 12(b)(i) (15.0.0).

commissions. In Virginia, residential ratepayers pay 48 percent of transmission capital costs under Rider T1.[42]

Part of the reason that data center-specific upgrade costs fall on all customer classes in Virginia is that its line extension policies—whereby individual loads pay the costs of facilities needed to serve them—were designed for distribution-level facilities and do not apply to transmission facilities:

> Dominion's current line extension tariff assesses only the costs of additional investment in local facilities, such as approach and branch lines, to serve the new customer load. The line extension tariff does not currently assign the upstream costs, such as transmission system network upgrades or substation costs, directly to the connecting customer. . . . The [SCC] has concluded that Section XXII "applies only to electric distribution, not transmission, facilities." Thus, customers would not be directly responsible for the costs of normal transmission facilities required to serve new load.[43]

Combine the traditional state practice of broadly allocating transmission costs with PJM's broad allocation of large load-specific network upgrade costs to all transmission customers, and the result is that residential ratepayers are covering costs directly caused by large loads.

This issue is not limited to RTOs. In Georgia, the Public Service Commission (Georgia PSC) recently held a proceeding to assess Georgia Power's 2025 IRP. As part of that proceeding, Georgia Power submitted the ten-year expansion plan for the Integrated Transmission System (ITS), the collective proposed facilities of all transmission-owning utilities in the state. The 2025 ten-year plan included several large transmission lines that Georgia Power explicitly identified as

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[42] Virginia Electric & Power Company, Direct Testimony of William J. Caffall, Va. State Corp. Comm'n Case No. PUR-2025-00076, at 4 (filed May 1, 2025).

[43] Sierra Club, Direct Testimony of Ron Nelson, Va. State Corp. Comm'n Case No. PUR-2025-00058, at 51:16-52:11 (filed Sept. 4, 2025) (citing Virginia Electric & Power Company, Application for Approval of Electric Transmission Facilities: Haymarket 230 kV Double Circuit Transmission Line & 230-345 kV Haymarket Substation, Case No. PUE-2015-00107, Interim Order at 19 n.63 (Apr. 6, 2017)).

planned to meet surging load growth, including the following that Georgia Power itself intends to build:

- Ashley Park – Wansley 500kV line – 35 miles[44]
- Farley – Tazewell 500 kV line – 120 miles[45]
- Hatch – Wadley 500 kV line – ~65 miles[46]
- McGrau Ford – Middle Fork 500 kV line – 65 miles[47]

The ten-year plan also included additional large strategic projects that other ITS transmission owners would build but whose cost responsibility Georgia Power would share. In addition to these facilities explicitly driven by large loads, an expert witness's distribution factor analysis found additional major lines that also "show[ed] strong operational connections to large loads."[48] While the costs of all of these lines are protected trade secrets, it follows that nearly 300 miles of 500 kV lines will cost multiple billions of dollars, and this is only a subset of the full ten-year plan driven by large loads. Indeed, the number and miles of transmission lines that Georgia Power has included in its ten-year plan has ballooned in recent years, in large part due to the proliferation of large loads. To illustrate, over the ten-year period in the 2018 plan, before the data center boom, the ITS owners planned 24.2 miles of new transmission lines requiring new rights of way.[49] The 2024 plan, in contrast, called for 1,142 miles.[50]

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[44] *See* Georgia Power Company, Integrated Resource Plan Vol. III, Ga. Pub. Serv. Comm'n Docket No. 56002, at internal p. 272, PDF p. 442 (filed Jan. 31, 2025).
[45] *See id*. at internal p. 277, PDF p. 447.
[46] *See id.* at internal p. 281, PDF p. 451.
[47] *See id.* at internal p. 285, PDF p. 455.
[48] Natural Resources Defense Council, Sierra Club, and the Southern Alliance for Clean Energy, Direct Testimony of Matthew Richwine, Ga. Pub. Serv. Comm'n Docket No. 56002, at 24:4-5 (filed May 2, 2025).
[49] Georgia Power Company, Integrated Resource Plan Vol. III, Ga. Pub. Serv. Comm'n Docket No. 42310, at internal p. 5, PDF p. 110 (filed Jan. 31, 2019).
[50] Georgia Power Company, Integrated Resource Plan Vol. III, Ga. Pub. Serv. Comm'n Docket No. 56002, at internal p. 5, PDF p. 175 (filed Jan. 31, 2025).

Likewise, in DEP's South Carolina territory, an expert witness identified over $10 million in transmission investment added to rate base that could be specifically attributed to serving new customers, including new delivery points.[51] Under DEP's 12 CP methodology, these costs are spread across all customers, which results in residential customers covering roughly half the costs of transmission upgrades.[52] And like Dominion, DEP's line extension policies also fail to cover transmission facilities.[53] South Carolina and Georgia both utilize bundled retail rates, in which transmission costs are included in the broader rate base alongside generation and distribution costs. They are thus spread across all retail customer classes. In these jurisdictions, the Commission has even less visibility into the transmission upgrade costs created by new large load customers and the manner in which those costs are allocated to existing customers.

Originating in an earlier era, these transmission upgrade cost allocation practices were not developed to accommodate the unique transmission impacts that large, high-load factor, clustered loads cause, especially when they interconnect at volume in a short period of time. Where these large loads create the need for substantial transmission investment, the Commission must assess whether status quo practices, which broadly allocate these costs to all load, violate the cost causation principle. As the Commission recently recognized, this fundamental tenet of just and reasonable transmission rates requires "costs to be allocated to those who cause the costs to be incurred and reap the resulting benefits."[54] It is beyond dispute that large loads are creating

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[51] *See* Sierra Club, Direct Testimony of Ron Nelson, S.C. Pub. Serve. Comm' Docket No. 2025-154-E, at 55:7-9 (filed Sept. 16, 2025).
[52] *See id*. at 35:9.
[53] *See id*. at 65:1-5.
[54] *Building for the Future Through Electric Regional Transmission Planning and Cost Allocation*, Order No. 1920, 187 FERC ¶ 61,068, at n.2786 (2025), *order on reh'g*, Order No. 1920-A, 189 FERC ¶ 61,126 (2024), *order on clarification*, Order No. 1920-B, 191 FERC ¶ 61,026 (2025) (citing *S.C. Pub. Serv. Auth. v. FERC*, 762 F.3d 41, 87 (D.C. Cir. 2014); *Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 475 F.3d 1277, 1285 (D.C. Cir. 2007); *Ill. Com. Comm'n v. FERC*, 576 F.3d 470, 476 (7th Cir. 2009) (parentheticals omitted)); *K N Energy, Inc. v. FERC*, 968 F.2d 1295, at 1300 (D.C. Cir. 1992) ("FERC and the courts have added flesh to these bare statutory bones, establishing what has become known in Commission parlance as the 'cost-causation' principle. Simply put, it has been

enormous transmission costs. This prompts the necessary question: who benefits from them and how much?

Historically, FERC has allocated the costs of network upgrades associated with transmission service requests and generator interconnection to all customers due to the broad benefits they provide.[55] But the Commission's transmission pricing policy has long sought to protect load from the cost impacts of excessive network upgrades:

> When rolling in the cost of Network Upgrades would cause the embedded cost rate paid by existing transmission customers to increase, we permit the non-independent Transmission Provider to charge an incremental rate (i.e., the rate associated with the costs of the Network Upgrades divided by the Interconnection Customer's units of service) to the Interconnection Customer. This will fully insulate existing customers from the cost of the Network Upgrades.[56]

While the Commission has credited the costs of these network upgrades back to the cost causer in the generator interconnection context (at least outside RTOs/ISOs), it did so because it determined that all load would still derive some benefit, as "a competitive transmission system, with barriers to entry removed or reduced, is in the public interest."[57] Yet those same competitive benefits do not exist when expanding the transmission system to accommodate retail load, as opposed to opening the wholesale market to a more diverse pool of sellers.

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traditionally required that all approved rates reflect to some degree the costs actually caused by the customer who must pay them.").

[55] *See W. Mass. Elec. Co. v. FERC*, 165 F.3d 922, 927 (D.C. Cir. 1999) ("The Commission's position with regard to assignment of costs is, so far as we can tell, part of a consistent policy to assign the costs of system-wide benefits to all customers on an integrated transmission grid. We have approved the underlying rationale of this policy. When a system is integrated, any system enhancements are presumed to benefit the entire system.").

[56] *Standardization of Generator Interconnection Agreements and Procedures*, Order No. 2003, 104 FERC ¶ 61,103 (2003), *order on reh'g* Order No. 2003-A, 106 FERC ¶ 61,220, at P 586 (2004).

[57] *Entergy Servs., Inc. v. FERC*, 319 F.3d 536, 543-44 (D.C. Cir. 2003).

To warrant a continuation of existing cost allocation practices that spread the costs of large load network upgrades to all load equally, the Commission must find that existing customers derive benefits commensurate to the enormous costs these upgrades entail.[58] In doing so, it must avoid any circular reasoning that would assign benefits to customers for upgrades that fix issues caused by the large load interconnections themselves. For example, while expanding the transmission system often produces broad reliability benefits,[59] existing customers derive no net benefit when the large loads caused the reliability issues that the upgrades ultimately address. Likewise, transmission upgrades can reduce economic congestion.[60] But if that congestion is caused by the interconnection of multiple data centers in a load pocket, existing ratepayers do not meaningfully benefit if upgrades ease congestion that would not exist but for the large loads' interconnections.

That is not to say that transmission upgrades do not create broader reliability and economic benefits. Order No. 1920 recently found that transmission providers must consider such benefits when conducting "Long-Term Regional Transmission Planning, which accounts for multiple drivers of Long-Term Transmission Needs and results in Long-Term Regional Transmission Facilities that produce a broader set of benefits."[61] There, the Commission appropriately required a more holistic approach to cost allocation upon recognizing that all users

[58] *See Ill. Com. Comm'n v. FERC*, 576 F.3d at 477.
[59] *See* Order No. 1920, 187 FERC ¶ 61,068 at P 90, n. 191 (stating that "a robust, well-planned transmission system is foundational to ensuring an affordable, reliable supply of electricity") (citing US DOE, Initial Comments on Advanced Notice of Proposed Rulemaking, Docket No. RM21-17-000 at 1 (filed Oct. 12, 2021) (stating that "strengthening and expanding existing transmission infrastructure, particularly the development of regional and inter-regional transmission projects, is key to continued access to reliable, resilient, lower-cost, and clean electricity for all")).
[60] *See id*. P 90 ("Proactive, forward-looking transmission planning that considers both evolving reliability needs and other drivers of transmission needs more comprehensively can enable transmission providers to identify potential reliability problems and economic constraints[.]"); *id.* P 301 ("[W]hen transmission providers are working to identify Long-Term Transmission Needs, areas of significant congestion on the transmission system–where Long-Term Regional Transmission Facilities could reduce congestion and in turn facilitate production cost savings–may indicate a Long-Term Transmission Need.").
[61] *Id.* P 126.

of the electric system will benefit from a proactive, comprehensive planning process that (1) selects regional upgrades only *after* comparing costs to benefits and (2) allocates costs commensurate with these demonstrable benefits. By contrast, the size and speed of large load interconnections cause the immediate need for substantial network upgrades, which only perpetuates dependence on the piecemeal grid expansion processes that Order No. 1920 sought to correct. Indeed, the Commission found that overreliance on these processes "can result in relatively inefficient or less cost-effective transmission development for customers, which contributes to rates for transmission that are unjust and unreasonable."[62] Where large load interconnection deepens reliance on these processes—perhaps to the exclusion of long-term, right-sized grid expansion—existing ratepayers should not bear the unjust and unreasonable cost delta.

As such, the Commission may find that existing practices that broadly allocate the costs of network upgrades caused by large load interconnections are unjust, unreasonable, and unduly discriminatory because they neither account for the unique transmission needs created by large load interconnections nor allocate costs roughly commensurate with benefits.

3. The uncertain nature of large loads and the long lead time of transmission development makes load forecasting unreliable, increasing the likelihood of stranded transmission investment whose costs must be borne by existing customers.

In addition to their size, load factor, and penchant for clustering, large loads—particularly data centers—are beset by uncertainty. While these large loads dominate utilities' load interconnection pipelines, it is widely accepted that a large percentage of them will never come

[62] *Id*. P 103.

to fruition.[63] But to the extent that utilities are duty-bound to serve them, those utilities must prepare to do so well before they can be sure the loads will materialize. That is because arranging the necessary generation capacity and expanding the transmission system—particularly for large sources of demand—take years to accomplish. If the utility makes these capital expenditures and the large loads never show up, the costs of these otherwise stranded assets will fall upon all other ratepayers. As such, load forecasting processes have taken on heightened importance in the era of large load growth. But unless incentives for both large load customers and public utility transmission owners meaningfully change, current load forecasting practices will remain an imperfect tool for consumer protection.

Earlier this year, SELC commissioned a report from London Economics International LLC (LEI) to explore the uncertainty associated with large loads' demand projections, particularly with respect to data centers.[64] LEI first found that large loads have structural incentives to duplicate requests for electric service to understand their least-cost siting options. As noted above, many considerations go into data center siting, including land costs, access to fiber capacity, latency, and reliable, cost-effective power service.[65] An increasing number of markets tick these boxes, giving data center developers options.[66] Unlike other types of large loads, data centers can be constructed relatively quickly—as short as six to nine months in some cases—so developers like to preserve locational flexibility.[67] The costs to maintain this flexibility are low, as developers can submit electric service requests to multiple utilities to gauge the associated electricity costs. Even where data center developers must sign a binding Energy

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[63] *See* Brian Martucci, *A fraction of proposed data centers will get built. Utilities are wishing up.*, UTILITY DIVE (May 15, 2025), https://www.utilitydive.com/news/a-fraction-of-proposed-data-centers-will-get-built-utilities-are-wising-up/748214/.
[64] *See generally* LEI Report.
[65] *See id*. at 10.
[66] *See id*.
[67] *See id*. at 11.

Services Agreement (ESA) and subject themselves to financial penalties if they do not choose the contracted supplier, this typically does not occur until after the developer has a sense for the required facility costs and timing.[68] Given the minimal cost of doing so, it is therefore in developers' best economic interests to engage in this form of price discovery. Yet while many of these requests for service will never turn into actual large load customers, the time lag involved in developing the necessary infrastructure often dictates that these requests be factored into load projections before any financial commitments attach.[69] On the other side of the equation, investor owned utilities that operate under traditional cost-of-service ratemaking have economic incentives not to question the firmness of large load service requests, as they "expect[] to benefit financially from making investments that expand [their] assets and contribute to increases in rate base."[70]

Beyond these structural incentives, LEI identified numerous factors suggesting that current large load projections are overstated. These include: (1) equipment bottlenecks, particularly for transformers and gas turbines, which could delay service to the large load;[71] (2) the reexamination or possible repeal of tax benefits and other incentives that have drawn data centers to some jurisdictions;[72] (3) increasing local opposition that has delayed or entirely derailed some data center projects;[73] and (4) increasing options for data center flexibility that may limit the ultimate capacity utilities need to serve.[74] Perhaps most significantly, LEI found that the global capacity to manufacture semiconductor chips does not support current data center load projections:

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[68] *Id*. at 13.
[69] *See id*. at 14.
[70] *Id*. at 16.
[71] *See id*. at 18-19.
[72] *See id*. at 19-20.
[73] *See id*. at 20.
[74] *See id*. at 20-21.

> [E]ven if global AI semiconductor chip manufacturing were to grow at an average rate of 10.7% annually (significantly faster rate than the 6.1% growth rate over the past decade) it could only satisfy an incremental 63 GW of data center-related demand globally over the six years from 2025 to 2030. The implied RTO/ISO/BA demand projection . . . is for US-only data center demand growth of 57 GW over the six years from 2025 to 2030. This would amount to more than 90% of the new total global manufacturing capability being earmarked for US data centers. In other words, for the forecasts to hold water, this would mean the US would require more than 90% of the world's new supply of semiconductor chips from 2025 through 2030.[75]

LEI found this scenario to be unrealistic, as the U.S. accounts for "less than 50% of global semiconductor chip demand, and other regions in the world are also projecting strong demand for semiconductor chips to support their region's growth in data centers."[76] This led LEI to conclude that "over-forecasting data center growth and therefore the associated electricity demand is significant at the national level in the United States."[77]

Whereas the potential for gain accrues to the large load customer and utility in this scenario, other customers bear the risk. First, as explained above, the lack of spare transmission headroom and idle generation capacity dictates that new investment in both will be required to serve large loads.[78] To the extent utilities make such capital expenditures in anticipation of large loads coming online and those loads fail to materialize, existing customers will bear the expense. Second, data centers have less certain permanence than other large load types, such as more traditional industrial customers. Certain data centers do not require much fixed investment, allowing them to relocate quickly if they find more favorable terms elsewhere.[79] And third, the life span of a data center is significantly shorter than the life of the capital assets built to serve

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[75] *Id*. at 22.
[76] *Id*. at 22-23.
[77] *Id*. 23.
[78] *See id*. at 17.
[79] *See id*. at 17-18.

them.[80] Even if these large loads pay the associated expense for half or more of these assets' useful lives, other customers will have to pick up the back end. As such, the uncertainty surrounding large loads creates a uniquely high potential for stranded transmission assets, whose cost burden will fall to existing customers.

Given this risk and the proliferation of large load interconnections, load forecasting has assumed an immense importance. Commissioner Rosner recognized this significance in requesting information from the RTOs and ISOs regarding their load forecasting practices.[81] As the response from PJM suggests, RTO/ISO efforts to standardize and improve load forecasting remain a work in progress.

In an attempt to strengthen its practices, PJM published a new Load Adjustment Request Implementation Document earlier this year. However, as it acknowledges up front, "PJM does not specify a particular methodology for forecasting large loads" and instead allows Electric Distribution Companies (EDC) and LSEs to "utilize any methodology that is reasonable and appropriate."[82] While it can set expectations for the types of information it must receive, such as "capacity and demand values, ramp rates . . ., levels and details relating to electric service obligations to loads, construction commitments, and other potentially relevant agreements,"[83] PJM ultimately depends on its EDCs and LSEs to provide accurate information on large load requests.[84] This leads to "a wide range of approaches to load adjustment processes" from entities

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[80] *See id*. at 18.
[81] FERC, *Chairman Rosner's Letter to the RTOs/ISOs on Large Load Forecasting* (Sept. 18, 2025), https://www.ferc.gov/news-events/news/chairman-rosners-letter-rtosisos-large-load-forecasting .
[82] PJM Interconnection, L.L.C., *Load Adjustment Request Implementation*, at 3 (July 1, 2025), https://www.pjm.com/-/media/DotCom/committees-groups/subcommittees/las/postings/load-adjustment-request-implementation.pdf.
[83] PJM Interconnection, L.L.C., Pre-Technical Conference Remarks of Jason Connell, Docket No. AD25-8-000, at 2 (filed Oct. 20, 2025) (PJM Load Forecasting Comments).
[84] *Id*. at 4.

with varying levels of experience with large load additions.[85] Across all LSEs and EDCs, however, the common denominator is that "[d]uplicative requests are generally not being explicitly accounted for[.]"[86] Blame does not lie entirely with PJM or the EDCs/LSEs, as large load customers often refuse to volunteer this information for competitive reasons.[87] And PJM has attempted to nevertheless improve this process by assigning different probability factors to loads for inclusion in the regional load forecast based on when they project to come online and the firmness of any financial commitments that bind them.[88]

PJM recognizes that its hand are largely tied, because even though large loads are now "interconnected in many cases at the transmission level," "data centers and other large loads are retail loads" and "the load interconnection process is generally administered through the application of retail tariffs."[89] PJM thus finds itself at the mercy of a "diversity of state engagement on retail load interconnections and forecasting,"[90] which it demonstrates in a table showing the different (and sometimes nonexistent) state efforts on this front.[91] Nevertheless, these disparate practices surrounding data collection and load forecasting, which originate outside PJM's control, directly lead to (1) M-3 Supplemental Projects, whose costs are allocated across entire zones and (2) the wholesale load forecast that underlies the RTEP process each year.

Again, these issues are not limited to RTO/ISO areas. Georgia Power's load forecast represented the primary flashpoint during its IRP proceeding earlier this year before the Georgia PSC. There, intervenors raised numerous concerns with Georgia Power's novel load forecasting approach, which it developed out of recognition that traditional load forecasting processes based

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[85] *Id*.
[86] *Id*.
[87] *Id.* at 5.
[88] *See id*. at 2.
[89] *Id*. at 4.
[90] *Id*. at 5.
[91] *See id* at 8-11.

on historic trends were unsuited for the unique type and scale of load growth it was experiencing.[92] Since developing its load forecasting method, Georgia Power had begun publishing granular reports on load requests outcomes.[93] But despite having information in hand that "suggest[ed] some of Georgia Power's assumptions are too aggressive, which would cause load projections to be too high," it nevertheless stuck to its model.[94] In doing so, it underestimated project delays, mistakenly assumed that ramp rates were certain, and failed to reflect the high dropout rates shown by the actual data.[95] These issues persist in Georgia Power's pending request for 10 GW of new capacity.[96]

These flaws have significant ramifications on transmission expansion, as Georgia Power's load forecast "serves as the basis" for its ten-year transmission plan, "which incorporates updates to generation and load growth."[97] As discussed above, the ten-year transmission plan included nearly 300 miles of 500 kV lines explicitly planned to serve load growth, as well as additional large upgrades that would have the same effect.[98] Utilizing load forecasting processes divorced from observed large load trends, Georgia Power's approach created the "risk that one or more of the new Strategic Transmission Projects identified . . . may be significantly under-utilized in the 10-year planning horizon if the new large load projects fail to move forward or are delayed, which is a significant possibility . . . because . . . much of this

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[92] *See* Georgia Interfaith Power & Light, Direct Testimony of Chelsea Hotaling, Ga. Pub. Serv. Comm'n Docket No. 56002, at 3:22-4:6 (filed May 2, 2025).
[93] *See id*. at 6:21-7:2.
[94] *See id*. at 9:14-16.
[95] *Id*.
[96] *See generally*, Direct Testimony of Trokey, Drugan, and Pol on behalf of Ga. PSC Public Interest Advocacy Staff, Ga. Pub. Serve. Comm'n Docket Nos. 56298 and 56310 (filed Nov. 12, 2025).
[97] Georgia Power Company, Integrated Resource Plan, Ga. Pub. Serv. Comm'n Docket No. 56002, at 112 (filed Jan. 31, 2025).
[98] *See supra* section (II)(A)(2).

projected load growth is highly speculative at this stage."[99] Most troublingly, Georgia Power did "not perform[] its transmission planning analysis considering a scenario with a lower level of load growth."[100] And, since the costs of these large transmission projects are added to the rate base that determines bundled retail rates for all existing customers, the heightened risk of overbuilding falls entirely upon them.

To be sure, the act of load forecasting is inextricably linked to retail electric service and typically the duty of retail providers. But given the unique uncertainty and potential for stranded investment that large loads carry, the Commission may find that transmission providers need increased certainty as to the magnitude of large load growth—a significant driver of transmission costs—in order to effectively plan system expansion and ensure just and reasonable transmission rates. Specifically, the Commission can find that the lack of uniform conditions placed on large load interconnections to the transmission system fails to limit speculative interconnection requests, which could lead to overbuilding, a heightened risk of stranded investment, and unjust and unreasonable rates borne by transmission customers.

As the Commission recently found in Order No. 2023: "[T]ransmission providers may face uncertainty regarding the size and makeup of the interconnection queue and the commercial viability of the projects in the interconnection queue, creating inefficiencies in the study process, increasing interconnection study costs, and delayed study results. Such uncertainty . . . [is] ultimately passed through to consumers through higher transmission or energy rates."[101] There, the Commission similarly sought to address incentives to submit speculative interconnection

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[99] Sierra Club, Direct Testimony of Matthew, Ga. Pub. Serv. Comm'n Docket No. 56002, at 28:5-9 filed May 2, 2025) (quotations omitted).
[100] *Id*. at 28:9-11.
[101] *Improvements to Generator Interconnection Procedures and Agreements*, Order No. 2023, 184 FERC ¶ 61,054, at P 43 (2023), *order on reh'g*, 185 FERC ¶ 61,063 (2023), *order on reh'g*, Order No. 2023-A, 186 FERC ¶ 61,199, *errata notice*, 188 FERC ¶ 61,134 (2024).

requests, finding that the lack of (1) access to information by interconnection customers about points of interconnection prior to submitting interconnection requests and (2) "meaningful financial commitments . . . for interconnection customers to enter and stay in the interconnection queue . . . incentivize interconnection customers to submit speculative interconnection requests that contribute to interconnection study backlogs, delays, and uncertainty, and, in turn, unjust and unreasonable Commission-jurisdictional rates."[102] Large load interconnection to the transmission system presents similar uncertainty, similar incentives, and a similar potential for unjust and unreasonable transmission rates.

B. Jurisdiction

Once it has found that current rates or practices are unjust, unreasonable, unduly discriminatory or preferential, "the Commission shall determine the just and reasonable rate, charge, classification, rule, regulation, practice, or contract to be thereafter observed and in force."[103] However, the Commission may only impose a replacement rate that falls within its defined purview, which extends to "the transmission of electric energy in interstate commerce and to the sale of electric energy at wholesale in interstate commerce," including "all facilities for such transmission or sale of electric energy." [104] Excluded from Commission authority are "facilities used for the generation of electric energy or over facilities used in local distribution or only for the transmission of electric energy in intrastate commerce, or over facilities for the transmission of electric energy consumed wholly by the transmitter."[105]

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[102] *Id*. P 48.
[103] 16 U.S.C. §824e(a).
[104] 16 U.S.C. § 824(b)(1).
[105] *Id*.

1. The Commission can credibly assert jurisdiction over the interconnection of large loads to the interstate transmission system.

The ANOPR sets forth four bases for the Commission to assert jurisdiction over large load interconnections to the transmission system. First, analogizing generator interconnection, the ANOPR characterizes large load interconnections as a "critical component to open access transmission service that require minimum terms and conditions to ensure non-discriminatory transmission service."[106] But as retail customers, large loads typically do not purchase transmission service directly, at least in RTOs/ISOs. Rather, they pay for transmission service alongside all other retail customers to an LSE that arranged the service on their behalf. As such, this provides a tenuous basis for Commission jurisdiction.

Second, the ANOPR finds that the interconnection of large loads to the transmission system is a practice that directly affects wholesale electricity rates.[107] It is undoubtedly true that large loads affect wholesale electricity rates by creating demand that an LSE must serve by procuring power supply in the wholesale market. This would be the case, however, whether the large load interconnects to the transmission system or the distribution system. Moreover, small residential loads affect wholesale rates in the same way, albeit on a much smaller scale. If the Commission asserts jurisdiction on this basis, it must justify its decision to limit these reforms only to large loads interconnecting to the transmission system when its jurisdiction would theoretically extend much further.

The third and fourth bases asserted in the ANOPR provide much more clearly defensible premises on which the Commission can exert its jurisdiction. Third, the ANOPR states that asserting jurisdiction over large loads to the transmission system "does not impinge on States'

[106] ANOPR at P 13 (citations omitted).
[107] *See id*. P 14.

authority over retail electricity sales."[108] Fourth, the ANOPR concludes that any "contrary view of the proposed reforms conflicts with the FPA's core purposes," which includes the Commission's "exclusive jurisdiction over the transmission of electric energy in interstate commerce, including the rates, terms, and conditions of transmission service, and all facilities for such transmission or sale of electric energy at wholesale in interstate commerce."[109] Merged, these bases assert that the Commission has exclusive jurisdiction over transmission rates and facilities used for transmission in interstate commerce and that physical large load interconnections to those facilities is separate and apart from state authority over retail sales.

The plain language of the FPA supports the Commission's jurisdiction over "all facilities for such transmission [in interstate commerce]."[110] Section 205 also directs the Commission to ensure the justness and reasonable of "[a]ll rates and charges made, demanded, or received by any public utility for or in connection with the transmission . . . of electric energy subject to the jurisdiction of the Commission, and all rules and regulations affecting or pertaining to such rates or charges[.]"[111] As documented throughout these comments, the physical connection of large loads to the transmission system requires enlargement of those facilities, which directly affects the rates applicable to transmission service.

As the ANOPR suggests, the Commission's development of standard procedures and agreements governing generator interconnection to the transmission system provides a useful analogue. There, the Commission addressed "interconnections to the facilities of a public utility's Transmission System that, at the time the interconnection is requested, may be used either to transmit electric energy in interstate commerce or to sell electric energy at wholesale in

[108] *Id*. P 15.
[109] *Id*. P 16.
[110] 16 U.S.C. § 824(b)(1).
[111] 16 U.S.C. § 824d(a).

interstate commerce."[112] The United States Court of Appeals for the D.C. Circuit blessed this construction, characterizing interconnections as "relationships between parties with respect to electricity flowing over facilities," and finding that Order No. 2003 and its progeny "by their terms control the agreements governing those relationships."[113] The court added that, "[b]y establishing standard agreements FERC has exercised its jurisdiction over the *terms* of those relationships."[114] Likewise, by establishing standardized procedures and agreements for the interconnection of large loads to the transmission system, the Commission would be exercising its jurisdiction over the relationship between parties with respect to energy flowing across jurisdictional transmission facilities as well as the enlargement facilities needed to enable that flow.

Crucially, this authority does not interfere with the sale of power between the LSE and the large load retail customer, even if it directly enables that sale. It is critical to note here that the FPA governs two distinct aspects of electricity service. First, it outlines the jurisdictional demarcation between different types of *facilities*: FERC has authority over interstate transmission facilities, while states have authority over facilities used for local distribution and generation. Second, the FPA sets another bright line dividing authority over *sales*: FERC has authority over wholesale sales in interstate commerce while states have authority over retail sales. These two spheres of authority are distinct even though they may overlap in many instances, and the Commission's jurisdiction over one must be considered independently of the other.

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[112] Order No. 2003, 104 FERC ¶ 61,103 at P 804.
[113] *Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 475 F.3d 1277, 1280 (D.C. Cir. 2007).
[114] *Id*. (emphasis in original).

For the purposes of this proceeding, the Commission may regulate transmission facilities, even where they facilitate a state-jurisdictional retail sale. The United States Supreme Court has previously held as much:

> This statutory text thus unambiguously authorizes FERC to assert jurisdiction over two separate activities—transmitting and selling. It is true that FERC's jurisdiction over the *sale* of power has been specifically confined to the wholesale market. However, FERC's jurisdiction over electricity *transmissions* contains no such limitation. Because the FPA authorizes FERC's jurisdiction over interstate transmissions, without regard to whether the transmissions are sold to a reseller or directly to a consumer, FERC's exercise of this power is valid.[115]

It has also determined that "a FERC regulation does not run afoul of § 824(b)'s proscription just because it affects—even substantially—the quantity or terms of retail sales."[116] The Commission may thus separately exert its authority over transmission facilities, rates, and practices directly affecting them, even where they involve an end use retail customer.

2. The Commission must ensure that it does not conflict with state regulation over retail service.

Although FERC authority over the interconnection of large loads to the transmission system does not inherently encroach on states' authority over retail sales, it creates the clear potential for jurisdictional conflict. That is because every single large load interconnection to the transmission system involves a retail sale, which will cause FERC and state commissions to regulate in close proximity. While establishing procedures and agreements governing large load interconnection would create a helpful nationwide baseline to protect existing customers from unjust and unreasonable cost shifts, the Commission must not restrict the states' ability to regulate large loads in a stricter manner if they so choose. Likewise, the Commission must not

[115] *New York v. FERC*, 535 U.S. 1, 19-20 (2002) (emphasis in original).
[116] *FERC v. Elec. Power Supply Ass'n*, 577 U.S. 260, 281 (2016).

stray into the terms and conditions of the retail sale, a distinction which may not always appear obvious.

The Commission recently encountered a similar dynamic when it rejected Tri-State Generation and Transmission Association's (Tri-State) proposed large load tariff. In doing so, the Commission found that aspects of Tri-State's proposal "appear to present an impermissible intrusion on retail rate regulation and thus are not subject to the Commission's authority pursuant to section 205 of the FPA."[117] These included provisions "set[ting] requirements for the terms of energy sales" between a member cooperative and its end-use customer, "minimum load and energy requirements for retail purchases," and "monthly caps for retail energy or demand."[118] The Commission rightly found that "these requirements appear to dictate certain terms and conditions for retail sales from a Utility Member to its end-use customer."[119]

In fashioning standard procedures and agreements governing large load interconnections, the Commission must ensure that all terms and conditions apply strictly to the transmission-related aspects of the interconnection. These should include procedures for studying and addressing impacts to the transmission system, as well as provisions establishing cost responsibility for associated transmission upgrades. To the extent the Commission offers expedited interconnection in exchange for the large load agreeing to implement flexibility measures, the Commission must limit such curtailment to wholesale power supply, transmission service, or interconnection service, as opposed to distribution or retail power supply. Likewise, the Commission should refrain from establishing minimum service terms or demand charges, all

[117] *Tri-State Generation & Transmission Ass'n, Inc*., 193 FERC ¶ 61,070, at P 45 (2025).
[118] *Id*. P 49.
[119] *Id*.

of which could have some transmission aspects, but are too intertwined with retail service to cleanly isolate.

As it develops a replacement rate in this proceeding, the Commission should recognize that states can be willing partners in this effort. The interconnection of large loads has caused significant upheaval to state-level authority over generation facilities and retail electric service. As a result, many states have taken affirmative steps to mitigate any negative impacts these large loads may have on other customers. For example, the SCC is currently considering creating a separate rate class for Dominion's large load customers, which would entail a minimum contract term, deposits, and a demand charge that would require the large load to pay minimum portion of the generation, transmission, and demand costs associated with its requested level of service.[120] Rather than disrupt these and similar efforts, the Commission must strive to complement them.

More broadly, the Commission must ensure that any reforms it adopts in this proceeding do not impair reliability by creating a timing mismatch between when large loads interconnect to the transmission system and when LSEs can provide them retail service. This potential presents particular concerns for territories served by vertically integrated utilities, like those in which the Southeast Public Interest Organizations operate. Charged with providing every aspect of electricity service, vertically integrated utilities are accustomed to procuring the necessary generation capacity and expanding their transmission and distribution grids to meet the demand of their present and future retail customers. This has traditionally entailed complete control over the interconnection of all load, which allows the utilities to sync the commencement of service with the necessary resource procurement and grid expansion. To the extent the Commission asserts jurisdiction over a subset of these interconnections, it could create a disconnect between

[120] *See* Virginia Electric and Power Company, Consolidated Post-Hearing Brief and Issues Matrix, Va. State Corp. Comm'n Docket No. PUR-2025-00058, at 39-46 (filed Oct. 24, 2025).

these processes if not handled carefully. Accordingly, the Commission must ensure that interconnecting large loads to the transmission system does not outpace the retail utility's ability to reliably serve its new large load customers or its existing customers.

Otherwise, if the Commission does not respect state authority over retail electric service, this proceeding could exacerbate the reliability and cost shift concerns that disorderly large load interconnection creates.

3. The Commission must extend the applicability of its reforms to RTO/ISO areas and non-RTO/ISO areas alike.

As these comments have documented, the unique effects that large loads have on the transmission system and rates due to their size, load factors, physical proximity, and uncertainty affect RTO/ISO areas as well as non-RTO/ISO areas. Virginia and Georgia have both experienced an influx of large load interconnection requests that may accelerate over the coming decade. Given the widespread effects these large loads have on the transmission system regardless of regional location, the Commission must impose uniform requirements for large load interconnection on their respective transmission systems.

Potentially complicating this action is the use of bundled retail rates in non-RTO/ISO areas, particularly the Southeast. The Commission in Order No. 888 declined to assert jurisdiction over the transmission portion of bundled retail rates, finding that doing so was neither necessary to its purpose in that proceeding nor worth untangling the thorny jurisdictional questions it raised.[121] In finding this approach acceptable, the Supreme Court nevertheless surmised that the FPA erected no barrier to FERC exercising jurisdiction over bundled retail transmission if it made the requisite finding: "Were FERC to investigate this alleged discrimination and make findings concerning undue discrimination in the retail electricity

[121] *New York v. FERC*, 535 U.S. at 26.

market, § 206 of the FPA would require FERC to provide a remedy for that discrimination. . .. And such a remedy could very well involve FERC's decision to regulate bundled retail transmissions."[122]

To the extent the Commission must exercise jurisdiction over bundled retail transmission in order to equally apply large load interconnection procedures, it must not hesitate. First, it could do so only as narrowly as the reforms adopted require. For instance, the Commission could assign full cost responsibility for network upgrades to the large load that causes them by coordinating cost responsibility for these upgrades contractually, outside broadly applicable rates. This would not otherwise detract from the state commission's responsibility to assess the prudence of all other transmission cost investment and facilitate cost recovery through rates. It would also have the effect of equitably quarantining these costs from rate base, protecting customers while assigning the costs to the large load.

Second, because non-RTO/ISO areas and RTO/ISO areas are experiencing similar effects of large load interconnection, applying different standards would unduly discriminate between customers in each, in contravention of the FPA. Unlike generator interconnection, which in non-RTO/ISO contexts presents the opportunity for incumbent transmission owners to engage in anti-competitive conduct, large load interconnection does not implicate similar competitive concerns that would warrant differential treatment.

Finally, were the Commission to establish different standards for large load interconnection in non-RTO/ISO areas and RTO/ISO areas, it would funnel large loads to whichever jurisdiction they perceive to have more favorable processes. This could overwhelm

[122] *Id*. at 27.

the capacity of the receiving jurisdictions to serve them and deprive the departing jurisdictions of valuable economic development opportunities.

Accordingly, the Commission should apply its adopted reforms uniformly to RTO/ISO areas and non-RTO/ISO areas alike.

III. COMMENTS ON ANOPR PRINCIPLES FOR REFORM

The Commission must ensure that each reform it adopts in this proceeding furthers its mission to ensure just and reasonable and not unduly discriminatory or preferential transmission rates,[123] a duty that the Supreme Court has reiterated is meant to further the public interest.[124] In tackling the challenge of rationalizing large load interconnection to the transmission system, the Commission must therefore adhere strictly to this statutory obligation to ensure just and reasonable rates in the public interest. This demands that existing customers do not unduly bear costs for which they derive no meaningful benefit. It also requires ensuring that large load interconnection does not come at the expense of bulk power system reliability.

The ANOPR sets forth a list of principles meant to guide the ultimate reforms the Commission adopts in this proceeding, but these principles raise more questions than answers. The answers the Commission devises to each will impact reliability and cost, two issues that directly implicate the public interest. Therefore, the Commission must prioritize the public interest in converting these broad principles into concrete reforms.

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[123] *See* 16 U.S.C. §824(a) ("It is declared that the business of transmitting and selling electric energy for ultimate distribution to the public is *affected with a public interest*[.]" (emphasis added)).
[124] *Nat'l Ass'n for Advancement of Colored People v. FPC*, 425 U.S. 662, 670-71 (1976) (discussing what it means for the Commission to ensure "just and reasonable rates in the public interest"); *see also Pa. Water & Power Co. v. FPC*, 343 U.S. 414, 418 ("A major purpose of the whole [Federal Power] Act is to protect power consumers against excessive prices.").

A. Principle 3: Establishing a MW Threshold

At this time, Southeast Public Interest Organizations do not support any particular size threshold to govern which large loads will be subject to the reforms the Commission adopts. This will be a live issue in multiple state-level regulatory dockets in the coming months, each of which has its own unique concerns. Instead, Southeast Public Interest Organizations provide considerations that the Commission may take into account as it develops a threshold requirement.

1. Although Order No. 2003's size threshold is informative, the Commission should not rely solely thereon.

Order No. 2003 provides guidance as to where the Commission should draw the jurisdictional line for transmission-connected large loads. In drawing the line between small and large generators at 20 MW, the Commission there explained that it expected "that the majority of interconnections to [distribution-level facilities] will be made by generators no larger than 20 MW."[125] In other words, the Commission expected that few generators smaller than 20 MW would connect to the Commission-jurisdictional transmission system. So, the Commission drew the 20 MW line with jurisdictional principles in mind. Other sources also suggest that 20 MW is a reasonable threshold that finds a basis in current industry practice. For example, Dominion's Facility Interconnection Requirements document states that "[t]he use of transmission Facilities should be considered for . . . [a]ll loads over 20 MW."[126] Accordingly, since loads 20 MW and above may consider interconnection to the transmission system, it may be reasonable for the

[125] Order No. 2003, 104 FERC ¶ 61,103 at P 806.
[126] Dominion Energy, *Dominion Energy Virginia–Electric Transmission Facility Interconnection* Requirements, at 35 (last accessed Nov. 12, 2025), https://www.pjm.com/-/media/DotCom/planning/planning-criteria/dominion-planning-criteria.pdf.

Commission to establish a 20 MW threshold below which any reforms resulting from this proceeding do not apply.

Yet Order No. 2003 may not provide perfect guidance for addressing the challenges presented in this docket. The Commission did not draw the line between small and large generators based purely on jurisdictional reasoning; the Commission adopted separate rules for generators under 20 MW in part to avoid "unduly hinder[ing] the development of Small Generators" with complicated study processes better suited for the interconnection of larger, more complicated generators.[127] In other words, the Commission also based the 20 MW threshold on practical reasons that may or may not apply in the context of interconnecting large loads. Indeed, the Commission issued Order No. 2003 in 2003. At the time, cloud computing was still in its infancy, hyperscale data centers did not exist, and the interconnection of large loads was not the burgeoning issue it is today. Although the interconnection of generating facilities and large loads may be analogous in some ways such that Order No. 2003 provides a helpful benchmark, the two scenarios raise different challenges. Therefore, the Commission should consider other approaches in evaluating the applicable threshold in this proceeding.

2. The Commission should consider thresholds to which states have applied large load regulations.

Although the Commission has never before addressed large load interconnection, utilities in several states have developed rules for large loads that may help inform the Commission's decision making. Nationwide, utilities have adopted a wide range of thresholds for large load rules,[128] which may support a flexible approach at the federal level. The Commission should

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[127] Order No. 2003, 104 FERC ¶ 61,103 at PP 16-17.
[128] *See, e.g.*, SMART ELECTRIC POWER ALLIANCE, *Database of Emerging Large-Load Tariffs (DELTa)* (last accessed Nov. 13, 2025), https://sepapower.org/large-load-tariffs-database/ (providing a map of states with accepted and proposed large load tariffs and their various size thresholds).

consider the rationales underpinning these decisions as it develops its own threshold applicable to standardized large load interconnections.

Utility large load reforms in the Southeast offer a range of examples. Appalachian Power Company has proposed reforms for individual loads over 100 MW and aggregated loads over 150 MW.[129] Similarly, the Georgia PSC has established rules for Georgia Power's handling of large loads that apply to customers over 100 MW.[130] Towards the lower end, Duke Energy proposed to the South Carolina Public Service Commission a 50 MW threshold for the application of large load interconnection reforms.[131] In first proposing this threshold, DEP reasoned that loads over 50 MW would require "significant transmission or distribution investment[.]"[132]

Offering ample support, Dominion proposed a still lower threshold of 25 MW for a new large load rate class in its 2025 rate case.[133] In explaining its choice to establish a lower threshold, Dominion asserted that any such threshold should reflect the unique circumstances faced by each utility in a given state. Dominion explained that "given the fact that Virginia's data center proliferation is matched by nowhere else in the world, the relative risks of that should

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129 Appalachian Power Co., Motion to Accept Stipulation, Va. Corp. Comm'n Docket No. PUR-2025-00057, at 2 (filed Nov. 10, 2025).
130 Ga. Pub. Serv. Comm'n, Order Approving Revisions to Georgia Power Company's Rules and Regulations, Ga. Pub. Serv. Comm'n Docket No. 44280, at 1 (Jan. 28, 2025). Georgia Power's parent company, Southern Company, however, also requires heightened scrutiny of transmission-connected large loads at or above 50 MW. Southern Company, Integration Process for Transmission Connected Large Loads (Jan. 14, 2025).
131 Duke Energy Progress LLC, Agreement and Stipulation of Partial Settlement Regarding Large Load, S.C. Pub. Serv. Comm'n Docket No. 2025-154-E, at 6 (filed Oct. 27, 2025); Duke Energy Carolinas, LLC, Agreement and Stipulation of Partial Settlement Regarding Large Load, S.C. Pub. Serv. Docket No. 2025-172-E, at 6 (filed Nov. 11, 2025).
132 Duke Energy Progress, LLC, Rebuttal Testimony of Jonathan Byrd, S.C. Pub. Serv. Comm'n Docket No. 2025-154-E, at 16:17-17:1 (filed Oct. 7, 2025).
133 Dominion Energy, Rebuttal Testimony of Stan Blackwell, Va. Pub. Serv. Comm'n Docket No. PUR-2025-00058, at 1:10-18 (filed Sept. 11, 2025) (Dominion Blackwell Rebuttal Testimony).

be considered by the Commission in evaluating the High Load Customer Proposal."[134]

Dominion's expert added that

> Northern Virginia was among the earliest U.S. data-center hubs, which has led to a proliferation of smaller facilities across [Dominion's] service territory. A lower 25 MW threshold ensures that those established, yet still large customers are captured and subject to the proposed High Load Customer tailored contract terms. It is entirely appropriate for individual utilities to set thresholds that reflect their unique customer base, grid investments, and policy objectives.[135]

A lower threshold may then be appropriate to capture regions of the country experiencing greater proliferation of large load growth.

3. In addition to establishing a MW threshold, the Commission should consider other factors.

Applying large load regulations to loads based solely on size may risk sweeping in loads that do not tax the transmission system to a degree that raises reliability and cross-subsidization concerns. This is because certain large loads, like some large industrial facilities, may have lower load factors, meaning they only place high demands on the transmission system during a small percentage of hours. Accordingly, the Commission could aim to regulate the interconnection of loads based on the ways in which they interact with the grid, rather than focusing solely on size. To do so, the Commission can follow the lead of utilities that have adopted thresholds for load factor as well as size. This would help the Commission address the impacts of loads with consistently high demands that are less flexible or responsive to grid conditions. It could also avoid unnecessarily subjecting less demanding large loads to overly rigorous interconnection requirements.

[134] Dominion Energy, Rebuttal Testimony of Steven Wishart, Va. Pub. Serv. Comm'n Docket No. PUR-2025-00058, at 9:12-15 (filed Sept. 11, 2025).
[135] *Id.* at 9:15-21.

For example, Dominion established a 75 percent load factor threshold in addition to the 25 MW threshold for its proposed large load rate class.[136] Dominion explained that, in setting these thresholds, it sought "to isolate a subset of customers with distinct usage and cost causation characteristics so that their cost allocation and contribution can be transparently developed and monitored[.]"[137] Applying large load interconnection rules to "higher demand customers that require higher levels of investment" would also help "[p]rotect against the risks of substantial stranded cost" for which existing ratepayers would otherwise pay.[138]

Finally, the Commission should consider whether it should dispense with a size-related requirement and instead assert jurisdiction over all interconnections to the transmission system. If it takes this course, it should consider whether the Seven Factor Test provides a sufficient basis to do so or whether a higher voltage threshold might provide a better and less jurisdictionally contentious alternative.

B. Principle 4: Standardized Interconnection Procedures

Like Order No. 2003, the more recent Order No. 2023 can help inform the Commission's approach to standardized interconnection procedures and requirements. In Order No. 2023, the Commission adopted several reforms to address the growing number of exploratory, speculative interconnection requests generators made "seekin[ing] to secure valuable queue positions as early as possible, even if they are not prepared to move forward with the proposed generating facility."[139] As described above, utilities today are similarly struggling to contend with a glut of speculative large load interconnection requests. In the case of large load interconnection, the lack of standardized processes contributes to applicants' inability "to understand interconnection

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[136] Dominion Blackwell Rebuttal Testimony at 1:10-18.
[137] *Id.* at 8:7-9.
[138] *Id.* at 8:11-13; *see also id.* at 9:4-12.
[139] Order No. 2023, 184 FERC ¶ 61,054 at P 47.

timelines, costs, and technical requirements–another reason many applications may be speculative in nature."[140] This leads to duplicative load interconnection requests, inflated load projections, and the possibility of overbuilt transmission facilities whose costs fall to existing customers. The unpredictability that results "discourages collaboration, erodes stakeholder trust, and may lead to disputes or regulatory intervention."[141]

Although the motivations of large loads and generators submitting speculative interconnection requests may differ, adopting standardized interconnection processes can help address the inefficiencies that result from speculative interconnection requests in both cases.[142]

1. The Commission should adopt Order No. 2023's requirements for financial commitments and readiness requirements.

In Order No. 2023, the Commission found that, under previous interconnection requirements, customers could "proceed through the generator interconnection process without having shown evidence to the transmission provider of meaningful progress toward achieving commercial viability."[143] As a result, transmission providers could not assess whether any of the interconnection requests in the queue would actually come to fruition. To address this, the Commission adopted "more stringent financial commitments and readiness requirements for interconnection customers to remain in the interconnection queue to discourage speculative interconnection requests and allow transmission providers to focus on processing viable interconnection requests[.]"[144]

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[140] Ryan Quint, et al., ELEVATE ENERGY CONSULTING AND GRIDLAB, *Practical Guidance and Considerations for Large Load Interconnections*, at 27 (May 2025) (*Practical Guidance and Considerations for Large Load Interconnections*).
[141] *Id.*
[142] *See, e.g.*, *Practical Guidance and Considerations for Large Load Interconnections* at 27-28.
[143] Order No. 2023, 184 FERC ¶ 61,054 at P 490.
[144] *Id.*

These measures included requirements for study deposits, readiness deposits, site control, and withdrawal penalties.[145] To start, the Commission required transmission providers to collect a single study deposit upon a generator's entry into the cluster.[146] In setting deposit levels, the Commission adopted a tiered approach with larger generators paying more "because larger proposed generating facilities within a cluster generally cost more to study than smaller proposed generating facilities within a cluster."[147] Additionally, the Commission required interconnection customers to submit a commercial readiness deposit at the beginning of each study in the cluster study process.[148] In adopting this reform, the Commission reasoned that readiness deposits would help reduce the number of speculative requests by "encourag[ing] interconnection customers not ready to proceed through the interconnection process . . . to withdraw earlier in the process[.]"[149]

The Commission also required withdrawal penalties "that increase in amount as interconnection customers proceed through the interconnection process in order to ensure that interconnection customers continue to evaluate whether their proposed generating facilities are commercially viable, thereby reducing the number of late-stage withdrawals[.]"[150] These provisions would likewise help limit speculative large load interconnection requests, increasing both the reliability of load forecasts and efficiencies for large load interconnection. The Commission should therefore implement study deposits, readiness deposits, and withdrawal penalties in any standardized large load interconnection procedures.

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[145] *Id.* P 5.
[146] *Id.* P 503.
[147] *Id.* P 504.
[148] *Id.* P 690.
[149] *Id.* P 691.
[150] *Id.* P 781.

In addition to adopting these reforms, the Commission should set appropriately robust financial commitments for large load customers.[151] Large loads tend to be highly sophisticated, well-resourced, and thus less sensitive to price signals. To achieve the desired result of reducing speculative requests, the Commission should require large loads to pay higher study deposits and withdrawal penalties than those applicable to generator interconnections. In their report on best practices for large load interconnection, Elevate and GridLab suggest "[s]ignificantly higher non-refundable application fees and deposits, sometimes based on $/MW demand capacity[,]" which could "allow utilities to hire adequate staff to perform increasingly complex studies" required to interconnect large loads.[152]

In addition to these financial commitments and penalties, the Commission should also require demonstration of site control in order to enter the load interconnection queue.[153] In Order No. 2023, the Commission required interconnection customers to "provide evidence of 90% site control for the generating facility at the time of submission of the interconnection request and . . . evidence of 100% site control for the generating facility at the time of execution of the facilities study agreement[.]"[154]

To help limit the number of speculative large load interconnection requests, the Commission should similarly adopt site control requirements for large loads looking to interconnect to the transmission system.

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[151] *Practical Guidance and Considerations for Large Load Interconnections* at 32.
[152] *Id.*
[153] Order No. 2023, 184 FERC ¶ 61,054 at P 583.
[154] *Id.* P 594; *see also Practical Guidance and Considerations for Large Load Interconnections* at 32 (advocating for the same).

2. The Commission should apply Order No. 2023's cluster study requirement to large load interconnections.

In Order No. 2023, the Commission provided several reasons that cluster studies would increase efficiencies for generators and transmission providers. In explaining the superiority of cluster studies over serial studies, the Commission stated that "transmission providers can perform larger interconnection studies encompassing many proposed generating facilities, rather than separate studies for each individual interconnection customer."[155] The Commission further explained that "[t]he cluster study process will provide greater certainty to interconnection customers, regarding both the timing of studies and the magnitude of network upgrade costs," and limit withdrawals.[156]

Cluster studies can similarly benefit efforts to interconnect large loads to the transmission system. As with generator interconnection queues, barriers to submitting interconnection requests are low, which has led to a high volume of speculative large load interconnection requests. Accordingly, "utilities may be too overwhelmed to conduct detailed studies for all applications. Moving toward a cluster study-based approach, similar to the generator interconnection queue reforms, could alleviate the serial challenges in the interconnection queue."[157] This is because, as with generator interconnection, "[b]y evaluating multiple large load requests together, utilities can assess system impacts more holistically, streamline study efforts, and fairly allocate necessary network upgrades."[158]

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[155] Order No. 2023, 184 FERC ¶ 61,054 at P 177.
[156] *Id.*
[157] *Practical Guidance and Considerations for Large Load Interconnections* at 28.
[158] *Id.* at 35.

3. The Commission should require transmission providers interconnecting large loads to consider use of Advanced Transmission Technologies (ATTs) before large scale network upgrades.

ATTs can have an outsized impact on integrating new large loads, especially where speed to power is at a premium. In 2024, the Department of Energy (DOE) studied the ability of ATTs to enhance the grid's capacity to support increasing demand more efficiently and cost-effectively than large scale network upgrades. DOE found that "[d]eploying the advanced grid solutions available today could cost effectively increase the capacity of the existing grid to support 20-100 GW of incremental peak demand when installed individually, while improving grid reliability, resilience, and affordability."[159] To illustrate the speed at which ATTs could come online, DOE highlighted Dynamic Line Ratings, which "can be scaled in fewer than three months after initial implementation to increase effective transmission capacity by an average of 10-30% at less than five percent of the cost of rebuilding the line to expand capacity."[160] Other ATTs can be deployed in under three to five years, which is still significantly shorter than the amount of time it takes to build a new transmission line.[161] Due to their ability to increase transmission capacity efficiently and cost-effectively, ATTs should be the first solutions considered to increasing large load demand.[162]

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[159] U.S. Dep't of Energy, *Pathways to Commercial Liftoff: Innovative Grid Deployment*, at 3 (Apr. 2024).
[160] *Id.* at 5.
[161] *Id.*; *see also* Emilia Chojkiewicz, et al., GRIDLAB, *2035 and Beyond: The Report – Reconductoring with Advanced Conductors Can Accelerate the Rapid Transmission Expansion Required for a Clean Grid*, at 35 (Apr. 2024) ("[R]econductoring with advanced conductors can typically increase transmission capacity of existing lines by up to 110% while still being implemented in a relatively short time-frame (typically 18-36 months)"); *id.* 17 for examples of other ATTs, the typical amount they increase transmission capacity, and the amount of time they take to complete.
[162] *Practical Guidance and Considerations for Large Load Interconnections* at 65 ("Deploying these advanced technologies can improve system utilization and flexibility, reduce network constraints and congestion in real-time, minimize the need for curtailment, and enable faster interconnection of large loads. A combination of these solutions can improve hosting capacity for large loads without requiring extensive new infrastructure.").

Requiring transmission providers to consider ATTs in this proceeding would parallel requirements that the Commission imposed in Order Nos. 1920 and 2023. In Order No. 2023, the Commission required transmission providers to consider ATTs during the study process because they "may reduce interconnection costs by providing lower cost transmission solutions . . . and may allow for a faster interconnection by providing solutions that can be implemented more quickly."[163] The Commission also required transmission providers to consider ATTs as part of Long-Term Regional Transmission Planning in Order No. 1920.[164] Building off of these recent precedents, the Commission should go a step further and require transmission providers to prioritize ATTs as solutions when processing large load interconnection requests and only select larger scale solutions when ATTs are not feasible.

C. Principles 8 and 9: Responsibility for Network Upgrade Costs

1. The Commission should allocate the cost of network upgrades required for interconnection to the responsible large loads.

As discussed at length above, current cost allocation processes cause existing ratepayers to cover the costs of transmission facilities that enable the interconnection of large loads to the transmission system.[165] Upon finding that this cost allocation is unjust and unreasonable, the Commission should directly assign the full cost of network upgrades caused by large load interconnections to the large load(s) that caused them.

Commission precedent requires "generation developers [] to be allocated the costs for transmission system upgrades that would not have been made but for the interconnection of the developers, minus the cost of any facilities that the [transmission operator]'s regional plan

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[163] Order No. 2023, 184 FERC ¶ 61,054 at P 1583.
[164] Order No. 1920, 187 FERC ¶ 61,068 at P 1239.
[165] *See supra* section (II)(A).

dictates would have been necessary anyway for load growth and reliability purposes."[166] Consistent with this standard, the Commission should not permit large loads to share the costs of network upgrades that enable their interconnection with other customers unless it can be shown that the other customers clearly benefit from them. Typically, transmission providers build local transmission upgrades solely for the purpose of integrating new loads rather than attempting to achieve multiple purposes that would create benefits for the region at-large.[167] Because these load-driven network upgrades tend not to produce broader regional benefits, the Commission should adopt a default policy whereby large loads pay 100 percent of the costs for transmission upgrades required for their interconnections. From there, if the large load can demonstrate that the transmission upgrades resolve a need that preexisted its large interconnection request, the costs can be shared amongst a broader customer base, with existing customers paying costs roughly commensurate to the benefits they receive.

Even where the Commission finds that large load interconnection facilities provide regional benefits, the Commission should carefully scrutinize the degree to which existing ratepayers actually benefit from them. Although many network upgrades may provide some systemwide benefits, existing customers should pay nothing when "benefits [] are trivial in

[166] *Midwest Indep. Transmission Sys. Operator Inc.*, 129 FERC ¶ 61,019, at P 23 (2009) (citation omitted); *see also Ill. Commerce Comm'n v. FERC*, 576 F.3d 470, 476 (7th Cir. 2009) ("FERC is not authorized to approve a pricing scheme that requires a group of utilities to pay for facilities from which its members derive no benefits, or benefits that are trivial in relation to the costs sought to be shifted to its members.").
[167] *See, e.g.*, Order No. 1920, 187 FERC ¶ 61,068 at P 54 ("[T]he expansion of the high-voltage transmission system is increasingly occurring outside of the regional transmission planning process through other mechanisms such as the generator interconnection process, which results in piecemeal transmission development."); *see id.* P 103 ("Because these other processes–specifically, generator interconnection processes and local transmission planning processes–are generally designed to address discrete, shorter-term needs, and do not comprehensively asses either broader transmission needs or solutions to those needs, overreliance on those processes can result in relatively inefficient or less cost-effective transmission development for customers, which contributes to rates for transmission that are unjust and unreasonable."); Public Interest Organizations, Comments, Docket No. EL25-49-000, at 13 (filed Apr. 23, 2025) (PIOs Co-Location Comments) (providing a chart with examples of transmission upgrades built to accommodate large loads and explaining how those upgrades did not provide broader, regional benefits).

relation to the costs sought to be shifted" to them.[168] In an era when individual large loads may have the electricity demand the size of cities or even small states,[169] the costs to interconnect a single one of these facilities can quickly exceed tens of millions of dollars.[170] Should the Commission continue to allow existing ratepayers to subsidize the interconnection costs of new large loads, the shift in costs to these ratepayers would still likely dwarf any reliability benefits they would enjoy.[171]

The Commission should therefore establish a default policy of requiring large loads to pay for 100 percent of the costs the network upgrades required to facilitate their own interconnection. Large loads could escape full cost responsibility only if they can clearly demonstrate, on a case-by-case basis: (1) that the facilities would resolve a need that preexisted their interconnection, and (2) that the benefit the interconnection facilities would provide existing customers is not trivial. Only if both conditions are satisfied would the large load be eligible to recoup its investment through credits.

Longstanding precedent exists for this cost responsibility paradigm. For example, the Midcontinent Independent System Operator's (MISO) tariff directly assigns to the generator interconnection customer 90 percent of the cost of network upgrades with a voltage of 345 kV and above and 100 percent of the cost of network upgrades under 345 kV.[172] The Commission originally accepted MISO's cost allocation method in 2009, when MISO explained that:

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[168] *Ill. Commerce Comm'n v. FERC*, 576 F.3d at 476.
[169] *Data Centers Are Already Increasing Your Energy Bills. We Have the Receipts.*; *see also* Blackridge Research & Consulting, *Top 10 Largest Data Centers in the US (2025)* (Sept. 15, 2025), https://www.blackridgeresearch.com/blog/largest-top-biggest-data-centers-in-united-states-list (notably showing that two data centers, in Oregon and Iowa, exceed 1000 MW of electricity demand).
[170] *Data Centers Are Already Increasing Your Energy Bills. We Have the Receipts.* showing interconnection costs that exceed $35 million for a single, 300 MW load).
[171] PIOs Co-Location Comments at 12-13 ("[M]any early examples of transmission upgrades driven by data center loads have not led to system benefits, but rather, only provide the benefit of allowing the large load to operate–disproportionately benefiting the co-location arrangement.").
[172] *Midcontinent Indep. Sys. Operator, Inc.*, 187 FERC ¶ 61,170, at P 7 (2024).

> The current rule, which tends to allocate a large share of Network Upgrade costs to the pricing zone where a new generator interconnects, has proved to be unjust and unreasonable, as developers have sought to add to northwestern Midwest ISO zones thousands of megawatts of new generation that vastly exceed the loads in these zones. As a result, the rural communities, farmers, and small businesses in this area face increases of thousands of percent in the new transmission costs assigned to their zone, just from projects expected to execute generation interconnection agreements this summer.[173]

MISO adopted this reform, and the Commission approved it, because the volume of interconnection customers looking to connect to MISO's transmission system was so large and required such significant upgrades that broadly spreading their costs would have substantially increased existing customers' rates. A similar dynamic is occurring in the context of large load interconnections.

MISO is not the only region that directly assigns the costs of network upgrades to interconnection customers: the Southwest Power Pool, the Independent System Operator of New England, and PJM's tariffs all utilize this cost allocation approach for network upgrades.[174] Due to the high and increasing cost of interconnecting large loads that would otherwise fall to existing customers under existing cost allocation practices, the Commission should require large loads to pay for 100 percent of the costs associated with their interconnections.

2. The Commission should allow large load customers to exercise the Option to Build network upgrades.

The Commission should allow interconnection customers to exercise the Option to Build and initially fund network upgrades. In the context of generator interconnection, the Commission's *pro forma* Large Generator Interconnection Agreement provides interconnection customers an option to build as an alternative to the standard option. Under this option to build,

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[173] Midcontinent Independent System Operator, Inc., Transmittal, Docket No. ER09-1431-000, at 2 (filed July 9, 2009).
[174] *Midcontinent Indep. Sys. Operator, Inc.*, 187 FERC ¶ 61,170 at P 33.

interconnection customers may choose "the option to assume responsibility for the design, procurement and construction of Transmission Provider's Interconnection Facilities and Stand Alone Network Upgrades[.]"[175]

Extended to large load interconnections, this option would allow large loads to set the pace of network upgrade planning and construction on their own schedule. It could also reduce costs because the interconnection customer would be incentivized to choose more efficient or cost-effective construction methods and materials, especially where they have full cost responsibility for the upgrades. The Commission has found that the reverse scenario, where the transmission owner initially funds network upgrades, "incentivizes a transmission owner to find higher-cost vendors in order to inflate the rate base on which it earns a rate of return, i.e., 'gold-plating,' which would further increase costs for the interconnection customer."[176] As such, allowing the interconnection customer to exercise the Option to Build the network upgrades for which it has cost responsibility should limit overall spending on those network upgrades.

D. Principle 7: Large Load Demand Flexibility

To the extent it avoids the need to build additional transmission or generation capacity, large load flexibility can reduce costs and enhance reliability. This flexibility can come in many forms. Large loads can elect interruptible service, which "[a]llows for defined curtailment rights in exchange for lower cost of service" and possibly lower network upgrade costs.[177] Large loads can also enter into peak load curtailment agreements, which "[i]ncentivize[] demand reduction

[175] *Pro forma* LGIA § 5.1.3.
[176] *Midcontinent Indep. Sys. Operator*, 187 FERC ¶ 61,170 at P 50. The Commission has held that "the Transmission Provider cannot include the cost of the Network Upgrades in its transmission rates until it has provided credits to the Interconnection Customer, and as long as any part of the cost of the Network Upgrades remains the responsibility of the Interconnection Customer, that part of the cost cannot be recovered in transmission rates." Order No. 2003-A, 106 FERC ¶ 61,220 at P 657.
[177] *Practical Guidance and Considerations for Large Load Interconnections* at 66.

during grid stress events through financial or reliability-based incentives."[178] Additionally, transmission providers can implement flexible interconnection frameworks, which "[m]ay include phased load deployments as generation or transmission expansion occurs, mobile generation or other interim solutions, expedited permitting, etc."[179] Since utilities can curtail load in different ways, the Commission should clarify what it means for a large load to be "curtailable" in a final rule.

Given the potential reliability and cost-reduction benefits load flexibility can provide, the Commission should consider encouraging large loads to be flexible or requiring them to do so before any needed network upgrades are in place. By agreeing to be flexible during a relatively limited number of hours, large loads can reduce pressure on reserve margins and the transmission grid and thus contribute to reducing the need to procure generation or expand facilities to meet peak demand. Duke University's Nicholas Institute for Energy, Environment, and Sustainability (Nicholas Institute) found that "76 GW of new load . . . could be integrated with an average annual load curtailment rate of 0.25% (i.e., if new loads can be curtailed for 0.25% of their maximum uptime)[.]"[180] In other words, integrating 76 GW of new load could be achieved with minimal curtailment time, comparable to that of existing U.S. demand response programs. The Nicholas Institute researchers further found that "[t]he average duration of load curtailment (i.e., the length of time the new load is curtailed during curtailment events) would be relatively short, at 1.7 hours when average annual load curtailment is limited to 0.25%, 2.1 hours at a 0.5% limit,

[178] *Id.*
[179] *Id.*
[180] Tyler Norris, et al., DUKE NICHOLAS INSTITUTE FOR ENERGY, ENVIRONMENT, & SUSTAINABILITY, *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems* at 2 (Feb. 2025).

and 2.5 hours at a 1.0% limit."[181] Accordingly, encouraging large load flexibility could yield significant benefits with minimal burden on the large loads themselves.

If the Commission offers large load flexibility in exchange for expedited interconnection, however, the Commission must be realistic about interconnection timelines. Currently, large load interconnection can take up to several years in some regions.[182] Although not perfectly comparable, generation interconnection wait times can also take several years.[183] Even if flexible interconnection limits the necessary network upgrades, it is simply unrealistic to expect transmission providers to complete interconnection studies for large loads in 60 days. Moreover, attempting to meet demanding timelines for large load interconnection would force transmission providers to divert significant resources—which are already limited—away from studying generation interconnection. Not only would this unduly discriminate against generation that has waited in the queue for years, but it would only further clog interconnection queues across the country that already face years-long backlogs. Finally, should the Commission create an expedited interconnection process for large loads, it should require large loads to pay study deposits significant enough to enable transmission providers to hire adequate staff to conduct interconnection studies.[184]

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[181] *Id.*; *see also* GRIDLAB, *Bringing Data Center Flexibility into Resource Adequacy Planning*, at 35 (Sept. 2025), https://gridlab.org/portfolio-item/data-center-flexibility-nv-energy-case-study-report/ ("Assuming a fleet of data centers, the number of times load response is required is small, and the durations are short.").
[182] *See, e.g.*, Josh Saul, *Data Centers Face Seven-Year Wait for Dominion Power Hookups*, BLOOMBERG (Aug. 29, 2024), https://www.bloomberg.com/news/articles/2024-08-29/data-centers-face-seven-year-wait-for-power-hookups-in-virginia.
[183] *See, e.g.*, Dana Ammann, NATURAL RESOURCES DEFENSE COUNCIL, *Breaking Through the PJM Interconnection Queue Crisis* (May 18, 2023), https://www.nrdc.org/bio/dana-ammann/breaking-through-pjm-interconnection-queue-crisis ("Projects [in PJM] now face an average delay of nearly four years just to get connected.").
[184] *Practical Guidance and Considerations for Large Load Interconnections* at 32.

E. Principles 3, 5, 6, and 10: Regulating Co-Location[185]

If the Commission allows transmission-connected load to co-locate with generation, it must protect existing ratepayers and other interconnection customers from cost, reliability, and competitive impacts. In February, the Commission initiated a section 206 proceeding to address various aspects of co-location in PJM.[186] Since the Commission received substantial comments in that proceeding, these comments do not repeat the substance of those comments in depth but raise some high-level issues for the Commission's consideration in this docket.

First, the Commission should establish requirements for interconnection studies that enable the transmission provider to accurately identify the costs for which the co-locating load is responsible. PJM, for example, uses studies that ultimately allow co-locating load to share transmission upgrade costs with existing load. Among the series of processes that PJM conducts to analyze the interconnection of large load are (1) the Transmission Owner Load Integration Study, which is used to identify upgrades to resolve any local transmission reliability needs, and (2) the RTEP to identify needs arising from the aggregation of collocation arrangements.[187] Although co-located load typically causes the need for interconnection-related upgrades, PJM still rolls those costs into its regional plan and spreads those costs among other ratepayers even where the associated upgrades do not provide broader system benefits.[188] Accordingly, to the extent transmission providers lack study processes that enable them to isolate facilities that will

[185] Southeast Public Interest Organizations note that the ANOPR uses the term "hybrid facilities" to refer to large loads that "share a point of interconnection with new or existing generating facilities[.]" ANOPR at P 12. For the purpose of these comments, the Southeast Public Interest Organizations will refer to the same as "co-located facilities" to conform to the terminology that the Commission used during its technical conference considering large loads that share points of interconnection with generating facilities. *See, e.g.*, *PJM Interconnection, L.L.C. v. PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,115, at P 3 (2025) (defining "co-located load arrangements.").
[186] *PJM Interconnection, L.L.C. v. PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,115 (2025).
[187] PJM Interconnection, L.L.C., Answer, Docket Nos. AD24-11-000 and EL25-20-000, at 6-8 (filed Mar. 24, 2025) (PJM Co-Location Comments).
[188] *See, e.g.*, PIOs Co-Location Comments at 13 (providing a table of transmission upgrades driven by large loads that have not led to broader system benefits).

serve large loads alone, there is a substantial risk that co-locating loads underpay for transmission and existing ratepayers overpay for benefits they do not receive. Therefore, the Commission should require transmission providers to develop processes that enable them to accurately identify and isolate upgrades driven by large loads.

In addition to underpaying for transmission upgrades, co-located loads may underpay for transmission services. Even where the generation at a co-located facility satisfies the large load's full demand, the large load still benefits from transmission services that the grid offers.[189] Large loads, regardless of co-location status, use grid support and consumer ancillary services like frequency regulation, black start service, and reactive service.[190] The Commission should require large loads to pay the applicable rates in exchange for these benefits.

The Commission must also ensure that bilateral contracts between large loads and existing generation resources do not unjustly shift costs onto existing ratepayers or jeopardize resource adequacy. Even in restructured states where ratepayers do not pay for generation through utility cost of service, removing generation from capacity markets through bilateral arrangements between large loads and existing generation will likely raise capacity prices and ultimately increase costs for existing ratepayers.[191] Moreover, as existing resources pull out of capacity markets to serve large loads, this will reduce the capacity available to serve all other load and could jeopardize reliability. This unjust and unreasonable result would exacerbate already excessive capacity costs in regions like PJM.[192] Accordingly, in order to adhere to its statutory mandate, the Commission must require transmission providers to scrutinize bilateral

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[189] PJM Co-Location Comments at 42.
[190] *Id.* at 4.
[191] *See, e.g.*, PIOs Co-Location Comments at 27.
[192] Eliza Martin and Ari Peskoe, Harvard Law School Environmental & Energy Law Program, *Extracting Profits from the Public: How Utility Ratepayers Are Paying for Big Tech's Power*, at 21 (Mar. 2025), https://eelp.law.harvard.edu/wp-content/uploads/2025/03/Harvard-ELI-Extracting-Profits-from-the-Public.pdf.

contracts between existing generation and large loads to protect existing ratepayers from unjust and unreasonable rates.

IV. OTHER ISSUES

A. The Commission must allow time for fulsome consideration of the issues underlying this rulemaking.

The issues raised in this proceeding are complex and novel. Whereas the Commission has previously addressed generation interconnection through industry-wide rules,[193] it has never considered load interconnection. The integration of large loads raises similarly difficult questions around cost-causation and reliability. Yet large loads interact with the transmission system in a distinct manner such that the Commission cannot simply repurpose generator interconnection rules to apply to large loads. Moreover, large load interconnection will have wide-ranging impacts on system reliability and electricity rates, especially if allowed to proceed without guardrails. Rushing this rulemaking without fulsome consideration of the many issues involved could have devastating impacts on costs to customers and grid reliability.

Nor is the Commission obligated to act with undue haste. Although DOE directed the Commission to issue a final rulemaking by April 30, 2026, section 403(b) of the DOE Organization Act states that "[t]he Commission . . . shall consider and take final action on any proposal made by the Secretary . . . in accordance with such *reasonable* time limits as may be set by the Secretary."[194] The timeline the Secretary provided for the Commission to act in this proceeding is extremely limited relative to the scale and complexity of the issues it raises. The Commission must consider comments on this ANOPR, issue a Notice of Proposed Rulemaking and invite further comment, consider those comments, and then develop a final rule. The

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[193] *See, e.g.*, Order No. 2023, 184 FERC ¶ 61,045; Order No. 2003, 104 FERC ¶ 61,103.
[194] 42 U.S.C. § 7173 (emphasis added).

Secretary provided the Commission six months to complete these tasks, a process that usually takes multiple years.

The limited time offered for public comment on the ANOPR illustrates the constrained nature of this timeline. The Commission allowed the public 29 days to submit initial comments and an additional 14 days to submit reply comments on the ANOPR. In comparison, the Commission allowed the public 75 days for initial comments and an additional 30 days for reply comments on the ANOPR that preceded Order No. 1920.[195] That rule, while complicated, iterated on previous transmission planning rulemakings. By contrast, this rulemaking treads fresh ground and raises equally, if not more, complicated issues. Accordingly, the Commission should allow appropriate time for public comment in this proceeding, commensurate with the importance of the issues involved. This will more enable fulsome consideration of the novel and complex matters raised.

B. In conjunction with issuing a final rule in this docket, the Commission should ensure rigorous compliance with Order No. 1920.

Establishing robust interconnection procedures is one important aspect of ensuring the orderly connection of large loads; implementing and following robust transmission planning practices to ensure that there are adequate transmission facilities to which large loads can interconnect is another. In establishing its section 206 authority to issue Order No. 1920, the Commission acknowledged that one factor necessitating transmission planning reform was the changing demand on the transmission system.[196] Specifically, the Commission stated that

> [a]fter many years of flat or minimal load growth in regions across the country, demand, on both a national and a regional basis, is projected to significantly increase in the coming decades, and it will require an increasingly robust transmission system to reliably serve this load growth . . . changes in electric

[195] *Building for the Future Through Electric Regional Transmission Planning and Cost Allocation and Generator Interconnection*, 176 FERC ¶ 61,024, at P 183 (2021).
[196] Order No. 1920, 187 FERC ¶ 61,068 at P 95.

> demand and associated load profiles are occurring as load-serving entities work to meet increasing needs due to . . . new large loads associated with evolving industrial and commercial needs, such as growth in data centers.[197]

The Commission concluded that "[t]o serve more load, the capacity of the already-oversubscribed transmission system will need to increase."[198] In other words, the Commission clearly contemplated that the reforms it adopted in Order No. 1920 would contribute to the portfolio of solutions needed to tackle the burgeoning challenge of large loads on the transmission system. As part of ensuring the orderly interconnection of large loads, in addition to enabling fulsome consideration of issues raised in this docket, the Commission should require rigorous compliance with Order No. 1920.

C. The Commission should consider providing guidance on forecasting large load demand.

As discussed above, forecasting large load demand is inherently uncertain.[199] The mismatched timelines for developing data centers and electrical infrastructure exacerbates this difficulty and creates the potential for severe reliability and rate impacts. Given the direct and substantial effect that demand forecasting can have on transmission rates and reliable transmission service, the Commission would be well within its authority to provide standards for utilities to follow to help manage uncertainty.[200] Large load forecasts dictate utility planning decisions, so inflated large load forecasts can result in overbuilding and thus unjust and unreasonable Commission-jurisdictional rates.[201] Moreover, uncertainty regarding the size and makeup of the interconnection queue and the commercial viability of projects in it create

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[197] *Id.*
[198] *Id.*
[199] *See supra* section (II)(A)(3).
[200] In Order No. 1920, the Commission similarly acknowledged the inherent uncertainty involved in long-term regional transmission planning. Yet, the Commission concluded that rigorous transmission planning processes can help mitigate uncertainty. Order No. 1920, 187 FERC ¶ 61,068 at P 229. These principles can be applied in the context of demand forecasting for large loads.
[201] *See supra* section (II)(A)(3).

inefficiencies in the study process, which increase interconnection study costs that also result in unjust and unreasonable Commission-jurisdictional rates.[202]

While a standardized large load interconnection process that involves strict financial commitments may help to discipline large load speculation, the Commission should also establish load forecasting requirements that allow transmission providers to mitigate the uncertainty in large load demand forecasting for large loads to ensure just and reasonable rates.

V. CONCLUSION

For the foregoing reasons, Southeast Public Interest Organizations urge the Commission to prioritize the public interest as it develops reforms in this proceeding. To this end, the primary considerations must be the costs borne by existing customers and the reliability of the transmission system they rely on. While the Commission should strive to create an efficient large load interconnection process, it must not compromise its duty to ensure just and reasonable rates to all users of the electric system.

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[202] Order No. 2023, 184 FERC ¶ 61,054 at P 43.

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Attachment A

Uncertainty and Upward Bias are Inherent in Data Center Electricity Demand Projections

by London Economics International LLC

(July 7, 2025)

UNCERTAINTY AND UPWARD BIAS ARE INHERENT IN DATA CENTER ELECTRICITY DEMAND PROJECTIONS

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July 7, 2025

Uncertainty and upward bias are inherent in data center electricity demand projections

prepared for Southern Environmental Law Center by London Economics International LLC

July 7, 2025

London Economics International LLC ("LEI") was engaged by the Southern Environmental Law Center ("SELC") to evaluate current projections of growth in power demand from a particular type of large load electricity customer: data centers. Data centers are large users of electricity, and projections of their growth are driving utility plans for system expansions. If data center growth does not materialize at the expected levels, the cost of system expansions by vertically integrated utilities would likely fall on other ratepayers.

Utilities, like any other business sector, must make investment decisions under uncertainties about the future. In the case of uncertainty over future load, a particular problem is that data centers are large consumers, so the same general forecast error applied to a large load will be more impactful on the success or failure of investment plans. Based on the analysis documented in this report, uncertainties inherent in outlooks for data center electricity demand reflect a bias to overstating future demand. For example, data center developers have incentives to duplicate requests in different jurisdictions for electric interconnection for the same facility, and industry reports indicate this has been ongoing. In addition, because of the US-wide and even global nature of the options for data center developers to site their facilities, it is difficult to determine in which jurisdictions the projected growth of data center electricity demand would meet, exceed, or fall short of such projections.

LEI's analysis also shows that projections of electric power demand by data centers in the United States exceed the capability of global chip manufacturers to supply the semiconductor chips that data centers need, given the demand from other locations globally. This is further evidence that data center developers are submitting more requests for service for new facilities in the United States than they plan to build.

The uncertainty and upward bias in data center electricity demand projections create the risk that new energy infrastructure (including electricity generation and transmission, and gas pipeline capacity) proposed to meet the needs of these large load customers could become underutilized and lead to higher costs for other customers than what they would have paid if the large load materialized.

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Acronyms

AI	Artificial intelligence
AMD	Advanced Micro Devices, Inc.
BA	Balancing authority
CAGR	Compounded annual growth rate
CEC	California Energy Commission
CSA	Customer service agreement
EIA	Energy Information Administration
ERCOT	Electric Reliability Corporation of Texas
ESA	Electricity service agreement
FERC	Federal Energy Regulatory Commission
FLOPS	Floating point operations per second
FLOPS/W	Floating point operations per second per watt
GPC	Georgia Power Company
GPSC	Georgia Public Service Commission
GPU	Graphic processing unit
GW	Gigawatt
GWh	Gigawatt hour
IBM	International Business Machines Corporation
IEA	International Energy Agency
Intel	Intel Corporation
IRP	Integrated Resource Plan
ISO	Independent system operator
ISO-NE	ISO New England
IT	Information technology
LBNL	Lawrence Berkeley National Laboratory
LEI	London Economics International, LLC
MISO	Midcontinent Independent System Operator
mm2	Square millimeter
MW	Megawatt
MWh	Megawatt hour
NPCC	Northwest Power and Conservation Council
NTP	Notice to proceed
NYISO	New York Independent System Operator
PJM	PJM Interconnection, LLC
PUE	Power usage effectiveness
ROE	Return on equity
RTO	Regional transmission organization
SELC	Southern Environmental Law Center
SPP	Southwest Power Pool

SXM	Server PCI Express Module
TDP	Total design power
TOU	Time-of-use
US	United States
W	Watt

1 Executive summary

Recent growth in electricity demand from data centers has been rapid in the United States and other regions throughout the world. A data center can require electric service on a much larger scale compared to other commercial or industrial customers. Though in the past a typical average data center had a load of 5-10 megawatts ("MW"), hyperscale data centers (defined in more detail later in this report) have loads of 100 MW or more, and even a 200 MW facility is now considered typical.[1] As a point of comparison, 200 MW is enough capacity to power over 109,000 households on a peak summer day in the state of Georgia.[2] The large scale of demand represented by even one new data center customer in the footprint of a vertically integrated utility makes it important for the utility to carefully evaluate the drivers and assumptions of such projected demand growth and the risks around such growth forecasts. Federal and state regulators will similarly want to ensure that planning processes and authorizations recognize inherent risks and uncertainties in electric demand growth projections.

As demonstrated in this report, forecasts of electricity demand stemming from data center growth are beset with uncertainty—the extent and pace of data center demand growth remains highly uncertain, and the specific locations even more difficult to predict. For the reasons outlined in this report, LEI believes that not all the electricity demand growth associated with new data centers projected for the United States or for any given individual jurisdiction in the United States will necessarily materialize.

The potential for overestimating electricity demand from new data centers has important implications for evaluating utility plans for generation and transmission capacity expansion, and in some cases related proposals for new interstate gas pipeline infrastructure to serve new gas-fired electricity generation. Currently, many utilities are projecting substantial growth in electricity demand from potential data centers within their territories. These demand projections in turn can drive a significant portion of the utilities' generation capacity expansion plans, and, in some cases, prompt a projected need for additional firm transportation capacity on interstate gas pipelines. However, if the projected data center load fails to fully materialize, then the cost of these assets — whether power plants or pipeline infrastructure—would be borne by other utility customers.

[1] International Energy Agency ("IEA"). "What the data centre and AI boom could mean for the energy sector." October 18, 2024. <https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector>; and McKinsey & Company. "AI power: Expanding data center capacity to meet growing demand." October 29, 2024. <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand>

[2] LEI calculated the number of households based on the total number of residential customers for Georgia Power Company ("GPC") in 2023 and the total GWh consumption by GPC residential customers at a recent peak month, July 2023. This results in an average hourly peak for the month, not the peak hour for the month. (Source: Georgia Public Service Commission ("GPSC") Docket No. 56002. GPC. 2025 IRP Technical Appendix Volume 1: B2025 Load and Energy Forecast. January 2025. <https://psc.ga.gov/search/facts-document/?documentId=221233)>

1.1 Organization of the report

With this report, LEI examines the drivers of data center electricity demand in the United States using a two-part approach. First, in Section 2, LEI examines the uncertainties and incentives that impact forecasts of electricity demand growth from data centers. Second, in Section 3, LEI applies a critical lens to the projected growth of data center load in the United States by comparing aggregated US electricity demand projections from select jurisdictions against global semiconductor chip supply projections to discover whether US data center electricity demand growth could credibly be served by the projected global supply of semiconductor chips, a key component to data center operations. Third, Section 4 discusses implications for utility ratepayers; Section 5 concludes the report. Appendix A provides the reader with a basic understanding of data centers. Appendix B provides an index of works relied upon and cited in this report.

1.2 LEI finds uncertainty and inherent bias in data center load outlooks

Based on its analysis, LEI concludes that data center electricity demand projections remain highly uncertain and currently reflect a bias to overestimating growth in the number of data centers that will be built, and therefore also overestimating future electricity demand. In addition, because of the US-wide and even global nature of the options for data center developers to site plants, it is challenging to determine in which jurisdictions the data center-induced electricity demand growth would meet, exceed, or fall short of projections. This uncertainty creates substantial risks that new energy infrastructure proposed to serve large load customers may never become fully utilized as intended and could lead to higher costs for existing ratepayers.

LEI's review of the drivers of data center electric demand projections (detailed in Section 2) shows that:

1) **Data center developers have incentives to duplicate requests for electric service, and evidence shows they are doing so.** This reflects the variety of locations where a data center can choose to locate, in the United States and across the world; the incentive to duplicate requests and evidence of duplication of requests; and recent attrition in data center load based on announcements from such customers;
2) **Vertically integrated electric utilities do not have an incentive to be very skeptical of requests.** A vertically integrated utility under traditional cost-of-service regulation expects to benefit financially from making investments that expand its assets and contribute to increases in rate base; and
3) **Most data center interconnection requests do not reflect large financial commitments from potential customers.** Utilities perform some level of probabilistic analysis of demand requests, and some have adopted contracting terms to reduce risk of shifting of costs to existing customers, which can also help to reduce the incentive of data centers to submit multiple requests for interconnection, so that even if the utility adds new system assets which are underutilized for a time, the contract with the data center may prevent cost shifting to other ratepayers. However, many data center interconnection requests driving current utility load forecasts are at the stage of the interconnection process at which no contracts have yet been signed, so financial commitments are minimal;

4) **Other drivers result in over-forecasting of data center electricity demand.** Bottlenecks in electric generation, transmission, and distribution equipment are slowing the pace of electric sector infrastructure growth that would serve new data centers. If the generation cannot be built in time, the new load must be deferred or move to another location and will grow more slowly than current data center electricity demand projections would imply. Local opposition to data centers is growing and has resulted in delays and cancellations. At the state level, the cost of tax incentives for data centers is increasing as data center development booms. Finally, demand flexibility from future large load customers may reduce the need for new generation capacity.[3] Data centers often incorporate on-site generation and, where appropriate, the capability to shift operations across multiple facilities in far-flung regions. This supports flexibility in hourly consumption of electricity, moderating electricity demand from new data center customers, especially if the customer is on a tariff which incentivizes reduction in electricity demanded from the utility during periods of scarcity and/or high peak demand.

LEI's quantitative analysis (detailed in Section 3) compares the data center load projections of US jurisdictions with the projected physical supply of global chip manufacturers. If total projections for data center electricity demand exceed the number of semiconductor chips that global manufacturing can supply over the same period, then it is likely that not all the announced or planned data centers incorporated into the US electricity demand forecasts could be built. LEI finds that this is, in fact, the case. For all the data centers announced in the United States for 2025 through 2030 to go forward, it would require 90% of incremental global chip supply for that period be directed to the United States market. This is unrealistic, as the United States currently accounts for less than 50% of global chip demand, and other regions in the world are expanding demand for chips.

[3] Norris, T. H., T. Profeta, D. Patino-Echeverri, and A. Cowie-Haskell. 2025. *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems.* NI R 25-01. Durham, NC: Nicholas Institute for Energy, Environment & Sustainability, Duke University. <https://nicholasinstitute.duke.edu/publications/rethinking-load-growth>

2 Uncertainties beset data center electric load forecasts

LEI finds that there are many uncertainties based on demand-side factors (on the part of data center developers) and on supply-side factors (on the part of generation-owning electric utilities) that impact the outlook for data center demand growth. These are discussed below, and each uncertainty currently points to the likelihood that industry observers, utilities, and system operators will over-forecast rather than under-forecast growth of data center electricity demand because the potential errors all point in the same direction.

2.1 The data center development business model leads to over-forecasting of electricity demand growth potential

As detailed below, data center developers have a variety of choices as to where to site their facilities and they have incentives to submit multiple requests for service to different utilities. Therefore, one would expect attrition relative to initial requests, and this has recently become evident.

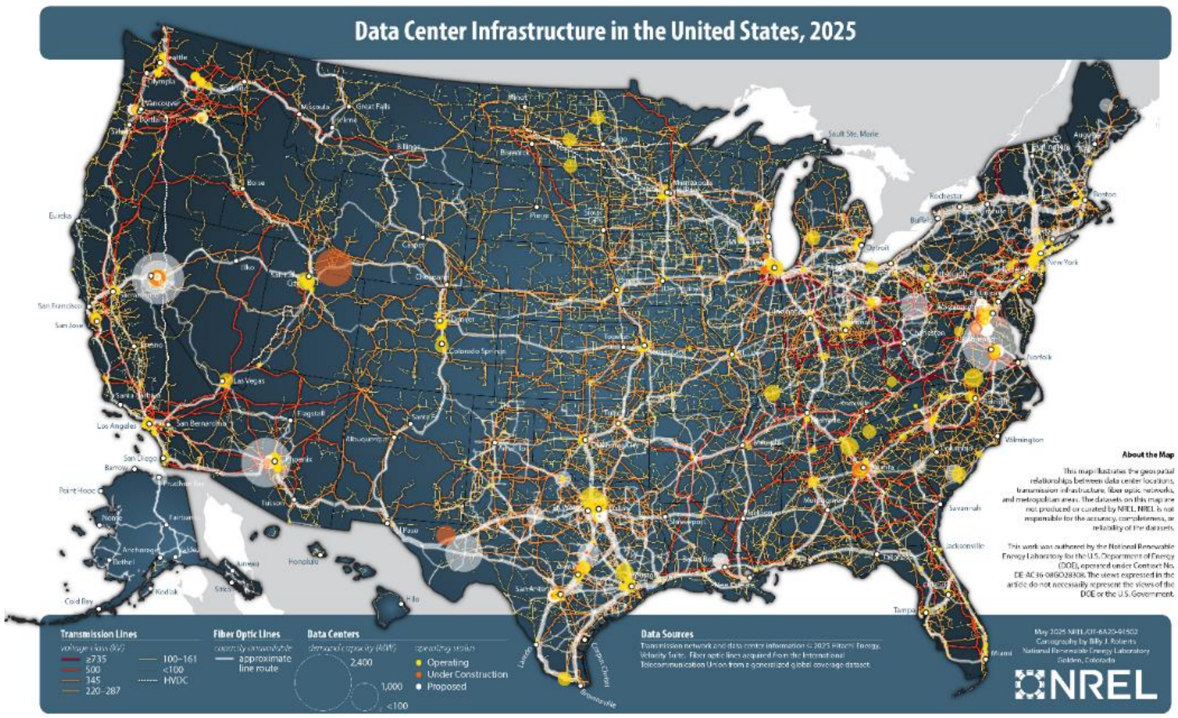
2.1.1 Data center developers have many choices as to where to locate

The design of a data center may be for a specific activity and therefore varies depending on the specific needs of the tenants. But, at its most basic, data center developers must consider certain essential requirements: availability of affordable land, accessibility to fiber capacity, adequate latency (the time delay in data transmission and processing within a network or system), water for cooling (if the design uses water cooling), and high quality and cost-effective electricity service. There are many locations in the United States that meet these criteria and are home to a large and growing data center sector (see Figure 1).

Data center developers are not only interested in the United States. The International Energy Agency ("IEA") estimated that in 2024, US data center total installed capacity amounted to 42 gigawatts ("GW"), while other countries' data center total installed capacity amounted to 55 GW.[4] Edge data centers (data centers which serve low-latency applications like streaming services, and real-time content delivery) need to be close to their customers, so there will be more data centers built around urban centers and areas of high commercial activity around the world.

[4] IEA. *Energy and AI*. April 2025. P. 259. Table A.2. <https://www.iea.org/reports/energy-and-ai> Note that IEA's "installed capacity" refers to the maximum theoretical capacity the facility can support when fully populated with information technology ("IT") equipment and operating at its design limits. This is higher than its "installed IT capacity" which reflects redundancy requirements, operational safety margins, or partial buildouts.

Figure 1. Data centers in the United States as of May 2025



Source: NREL. "Data Center Infrastructure in the United States, 2025." May 2025. <https://docs.nrel.gov/docs/gen/fy25/94502.jpg>

Developers want flexibility to choose between different sites because the facilities can be developed quickly compared to the time it typically takes to develop and build energy generation, transmission, and distribution infrastructure. Data centers are reported to take two to three years to design, permit, and build, though some potential customers are looking to build large data centers in as little as six to nine months.[5] In contrast, the planning, approval, and construction of electric infrastructure such as generation, transmission, and distribution typically involves a rigorous and interdependent process (to ensure reliability, resilience, and system integrity) which can take longer. For example, the median time for clean energy resources in the

5 Levitt, Ben. "AI and Energy, the Big Picture." S&P Global. December 2024. <https://www.spglobal.com/en/research-insights/special-reports/look-forward/ai-and-energy>; and Bain and Company. "Utilities Must Reinvent Themselves to Harness the AI-Driven Data Center Boom." <https://www.bain.com/insights/utilities-must-reinvent-themselves-to-harness-the-ai-driven-data-center-boom>

United States to move from transmission interconnection request to commercial operation was five years for projects which entered commercial operations in 2023.[6]

2.1.2 Submitting duplicate requests is a low-barrier, low-cost, and low-risk strategy despite efforts to secure commitments

To serve a large load customer, a utility's large load tariff is often paired with a bilateral contract referred to as a customer service agreement ("CSA") or electricity service agreement ("ESA"). These contractual agreements specify a variety of terms and conditions, in addition to the rates that will be applicable under the terms and conditions of the tariffs. A customer who has signed such a contract has "skin in the game" (i.e., some financial exposure) if the contract includes terms that commit the customer to pay for costs at the contracted level of demand requested and/or penalize the customer for not using the level of demand that it initially requested. This creates an incentive for the customer not to overstate the service it will need and supports a more accurate load forecast by the utility.

However, these agreements are only as strong or as weak as the terms of the CSA or ESA, which can vary widely:

- **Minimum demand requirements**: A number of utilities establish demand charges based on a minimum percentage of the customer's contracted capacity. This is essentially a take-or-pay arrangement. The intent is to prevent a customer from requesting more service than it plans to use. This minimum billing demand ranges widely, from 50% to 85% of contracted capacity across several jurisdictions examined by LEI.[7]
- **Contract period:** The length of time the new large customer must commit to its minimum billed demand varies also across jurisdictions, from as little as one year to proposals ranging from 8 to 30 years. A longer contract term increases the cost of walking away from an ESA (and discourages the customer from requesting more service than it will use) because the customer commits to paying its minimum demand charges for longer.
- **Capacity exceedance penalties**: Several jurisdictions include penalties for exceeding the contracted capacity. Such penalties provide a disincentive for a large customer to use more

[6] Silverman, A., Wendling, Dr. Z.A., Rizal, K., Samant, D. *Outlook for Pending Generation in the PJM Interconnection Queue.* Columbia University CGEP. May 2024. P. 9. <https://www.energypolicy.columbia.edu/wp-content/uploads/2024/05/PJM-Interconnection-CGEP_Report_042924-2.pdf>

[7] Sources: Dominion Energy: "Schedule 10 Large General Service." Virginia State Corporation Commission. December 20, 2022; "Terms and Conditions Section XXII. Electric Line Extensions and Installations." Virginia State Corporation Commission. March 8, 2024; and "Terms and Conditions Section IX. Deposits." Virginia State Corporation Commission. December 12, 2013; Georgia Power Company: "Customer Renewable Supply Procurement Schedule CRSP-1." Georgia Public Service Commission. October 2023; "Rules Regulations and Rate Schedules for Electric Service," Georgia Public Service Commission. January 2023; "Power and Light Large Schedule PLL-18." Georgia Public Service Commission. April 2025; and GPSC Docket No. 44280. GPC. *Rules & Regulations Update 12-11-2024*. December 2024. <https://psc.ga.gov/search/facts-document/?documentId=220667>'; Indiana Michigan Power Company: "Submission of Unopposed Settlement Agreement and Unopposed Motion for Acceptance of Out of Time Filing." Indiana Utility Regulatory Commission Cause No. 46097. November 22, 2024.

capacity than it has contracted for; or, alternatively, to contract for less capacity than it expects to use.

- **System allocation fee:** A system allocation fee is an upfront deposit required to secure a given level of system (generation and transmission) capacity. This is currently not widely used, though Portland General Electric proposed such a fee in Oregon Public Utility Commission Docket UE-430.

The effectiveness of the terms described above will depend on the specific levels of demand requirements, contract periods, and other details. Nevertheless, even with contract terms that are meant to winnow potential data center requests, such contracts are typically signed **after** a request to interconnect or an announcement of a new data center facility is released. Of course, the developer is aware of the terms of the ESA it will have to sign, but the utility does not require a customer to sign the ESA at the front end. The developer will wait to hear from the utility on what the costs of any necessary new substation expansions would be and the expected in-service date, and only then does the developer decide if it will sign an ESA.

Therefore, in the pre-ESA early stages of the interconnection process, the cost is low to submit multiple requests. A former commissioner at the Texas Public Utility Commission noted that the cost of putting forward an initial request for service is typically minimal: "[t]he phantom load problem arises because the cost of getting in a queue is lower than the weighted likelihood that they'll want to use their position. When it's cheaper to buy a queue position than not to use your queue position, you'll buy queue positions all day long."[8] In addition, potential customers who have requested service or simply announced tentative plans to build a data center (but do not have a significant financial commitment—i.e., have not signed a CSA or ESA), do not have the same level of financial obligation as a customer who has executed a CSA or ESA. Nevertheless, because of the longer timeframe required to plan for and build electricity infrastructure, such early-stage customers (which are often more numerous than the data center customers who have made financial commitments through an ESA) impact utility projections of future electricity load. Such early-stage requests are considered in utility load forecasts, which are then used for generation, transmission and distribution system planning. LEI has observed that utilities increasingly apply weighting factors to early-stage projects, to reflect a lower probability of an early-stage request materializing compared to the probability of a project with an executed CSA or ESA in place. However, with a very short history of such potential customers and their ultimate attrition, utilities must rely on limited information to create such weighting factors.

Therefore, even assuming many data centers are built in the United States (rather than elsewhere in the world), unless a new customer signs a contract, and the contract commits the customer to a specific level of demand and a material financial commitment, there is no guarantee that the data center would be built in any given utility's territory versus somewhere else.

[8] Martucci, Brian. "A fraction of proposed data centers will get built. Utilities are wising up." Utility Dive. May 15, 2025. <https://www.utilitydive.com/news/a-fraction-of-proposed-data-centers-will-get-built-utilities-are-wising-up/748214>

2.1.3 Data center developers submit redundant requests for electric service across multiple jurisdictions

The timing mismatch between data center development and the comparatively slower pace of energy infrastructure buildout, coupled with the typically low cost of submitting a request for service to a utility, creates a strong incentive for potential data center customers to submit multiple, redundant requests for service across several utility jurisdictions.[9] This approach gives developers optionality for negotiating and selecting a site that meets their needs, including timetable for achieving commercial operations. Industry experts have observed that "[D]ata center developers consider multiple states as possible locations for data centers, and they query multiple utilities simultaneously for electricity rates and incentives prior to making a final selection."[10] Meta's former energy strategy director remarked that technology companies themselves are "getting the same project bid into them multiple times," making it difficult to distinguish viable projects from speculative ones.[11]

This dynamic is increasingly evident, including in regional planning processes. As parties to a recent Federal Energy Regulatory Commission ("FERC") docket noted "… PJM has no way to cross-check whether a data center in, for example, Exelon's service territory has also made the same proposal in Dominion's territory, and both proposals end up in PJM's forecast even though only one will be built. … [I]n a recent presentation at the Pennsylvania Environmental Law Forum, PJM's own Senior Manager of Government Services, Stephen Bennett, stated that data center companies 'are pitching the same data centers in different locations.'"[12]

A former Google senior director of software engineering said there are "five to 10 times more interconnection requests than data centers actually being built."[13] Microsoft expressed concerns that "over-forecasting demand from data centers could lead to procuring excessive carbon-intensive generation," and recommended that the Georgia Public Service Commission ("GPSC") only approve near-term resource planning decisions in Georgia Power Company's 2023

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[9] Giacobone, Bianca March 2025. "Phantom data centers are flooding the load queue." March 26, 2025. <https://www.latitudemedia.com/news/phantom-data-centers-are-flooding-the-load-queue/>

[10] Koomey, Jonathan (Koomey Analytics), Schmidt, Zachary (Koomey Analytics), and Das, Tania (Bipartisan Policy Center). *Electricity Demand Growth and Data Centers: A Guide for the Perplexed*. February 2025. P. 10. </https://bipartisanpolicy.org/download/?file=/wp-content/uploads/2025/02/BPC-Report-Electricity-Demand-Growth-and-Data-Centers-A-Guide-for-the-Perplexed.pdf>

[11] Giacobone, Bianca March 2025. "Phantom data centers are flooding the load queue." March 26, 2025. https://www.latitudemedia.com/news/phantom-data-centers-are-flooding-the-load-queue/.

[12] FERC Docket No. EL25-49-000. Public Interest Organizations. *Comments of Public Interest Organizations in response to PJM Interconnection, L.L.C.'s 03/24/2025 Answer to FERC's 02/20/2025 Order under EL25-49.* April 23, 2025. P. 16-17. <https://elibrary.ferc.gov/eLibrary/filedownload?fileid=9D23F7BC-5484-CAA9-9030-96645FF00000>

[13] Martucci, Brian. "A fraction of proposed data centers will get built. Utilities are wising up." *Utility Dive*. May 15, 2025. <https://www.utilitydive.com/news/a-fraction-of-proposed-data-centers-will-get-built-utilities-are-wising-up/748214/>

Integrated Resource Plan ("IRP") Update based primarily on "known, mature projects that have made firm commitments to Georgia Power."[14]

The problem reverberates beyond electric utilities. Because of concerns over potential duplication of requests, executives in the US natural gas industry have publicly voiced skepticism for growth in gas demand from data center electric power customers. The Vice President of New Ventures for pipeline company The Williams Companies ("Williams") noted at an industry event: "… if you look at how these [data center] projects are coming into different organizations, there is double and triple [counting] … it is the same project because you have different players that are developing pieces."[15] "It's creating a lot of problems for these regulators and utilities because how do you differentiate between a real project and a fake project?" the president of a shale gas producer remarked at the same event, adding that he expects only 10% of data center projects that have been announced to be built.[16]

2.1.4 Expected attrition is in evidence

With multiple requests for service and the implied intent of developers to ultimately build only a subset of the proposed data centers, some attrition among large load projects should be expected (and therefore warrant downward adjustment to current electricity demand growth prospects). There is already evidence of this attrition. A recent industry report noted that Microsoft put on hold three data centers in Ohio (a total of $1 billion in spending), as well as portions of a data center in Wisconsin.[17] Microsoft is reported to be slowing down or pausing projects, including those for which it has "secured energy and all necessary approvals"[18] though this distinction does not necessarily apply to the Ohio or Wisconsin projects (in other words, Microsoft did not report whether or not the Ohio and Wisconsin projects had already secured energy and necessary approvals). The President of Microsoft Cloud Operations noted in April 2025, "[b]y nature, any significant new endeavor at this size and scale requires agility and refinement as we learn and grow with our customers. What this means is that we are slowing or pausing some early-stage projects. While we may strategically pace our plans, we will continue to grow strongly and

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[14] GPSC Docket No. 55378. *Comments on Georgia Power's 2023 Integrated Resource Plan Update*. Microsoft. Docket No. 55378 at 1, 4. Apr. 1, 2024. <https://psc.ga.gov/search/facts-document/?documentId=218199>

[15] Energy Intelligence. "US Gas Companies Temper Data Center Demand Expectations." *Natural Gas Week*, Vol. 41, No. 11. March 14, 2025.

[16] Ibid.

[17] Mannion, Annemarie. "Microsoft Hits Pause Button on $1B in Data Centers in Ohio." *Engineering News-Record*. April 10, 2025. <https://www.enr.com/articles/60572-microsoft-hits-pause-button-on-1b-in-data-centers-in-ohio#:~:text=By%20Annemarie%20Mannion,contractors%20had%20not%20been%20announced>

[18] Patel, Dylan; Jeremie Eliahou Ontiveros; Maya Barkin. "Microsoft's Datacenter Freeze - 1.5GW Self-Build Slowdown & Lease Cancellation Misconceptions." *SemiAnalysis*. April 28, 2025. <https://semianalysis.com/2025/04/28/microsofts-datacenter-freeze/>

allocate investments that stay aligned with business priorities and customer demand."[19] This statement seems intended to provide Microsoft with the flexibility to pause investment as and where needed, without alarming its shareholders.

Data center developers face their own uncertainties and may be evaluating changes in their market environment, some of which may be contributing to a slowdown in commitments. In addition to long waits for interconnection, these include, but are not limited to, uncertainty about the impact of future artificial intelligence ("AI") models that are more energy-efficient and cheaper to train and run than existing AI models; the US administration's proposed $500-billion AI-powered Stargate project; and concerns over tariffs and the cost of inputs.[20]

2.2 Utilities and shareholders benefit from strong load growth

Although utilities have some processes in place to reduce the number of "phantom load" requests by data centers—by assigning probabilities to requests for service and requiring varying levels of monetary commitments in CSAs and ESAs— a vertically integrated utility under traditional cost-of-service regulation expects to benefit financially from making investments that expand its assets and contribute to increases in rate base. As detailed below, investments in assets to support service to new customers will increase rate base, which in turn increases allowed returns under a cost-of-service regulatory model. The utility and its shareholders therefore benefit from an expanded rate base, with only some of the risk of under-utilization of assets borne by shareholders (if the investment is deemed imprudent), and the rest of the risk borne by the utility's customers.

2.2.1 Utility investments are ultimately paid for by customers

A public utility has an obligation to serve customers in its territory but cannot jeopardize the reliability of the service it provides to existing customers to take on new customers. Therefore, the utility periodically performs forward-looking load studies, which must incorporate assumptions about the interest of new potential customers, and the load growth which will ultimately materialize. A vertically integrated electric utility (which owns generation, transmission, and distribution assets) will plan capacity expansions of its system to meet this expected load. If the utility is regulated under a cost-of-service model, and the investment is deemed prudent based on available information, then the utility will be allowed to recover the costs of such new infrastructure in its rates. The costs will include a regulator-approved return on equity ("ROE") on the rate base, which contributes to the profits that the utility and its

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[19] Noelle Walsh, President Microsoft Cloud Operations and Innovation | Nouryon, Non-Executive Director. <https://www.linkedin.com/posts/noelle-walsh-b29356108_microsoftcloud-datacenters-activity-7315439628562423808-W67e/>

[20] Williams, Kevin. "AI data center boom isn't going bust but the 'pause' is trending at big tech companies." CNBC. April 27, 2025. <https://www.cnbc.com/2025/04/27/ai-data-center-boom-isnt-going-bust-but-the-pause-is-trending.html>

shareholders earn. The cost of the investments as well as the ROE are paid for by the utility's customers.

While the utility expects to benefit financially from the investments it must make to serve new customers, it still faces some investment risk. This risk to the utility may come in the form of prudency reviews and/or performance incentives or multi-year plans whereby a utility will face regulatory lag and may not be able to adjust its rate base immediately. These practical implications temper the zeal to build assets too quickly, though the utility still has a fundamental motivation to expand investment in its system.

On balance, then, the utility does not have a strong incentive to be skeptical regarding forecasted electricity load growth (whether from data centers or other types of customers) given the opportunity to expand its rate base. Nevertheless, utilities often perform analyses in which they adjust the total requests for service of prospective large load customers such as data centers for the possibility that some load will not materialize. However, experience with attrition of large load customers is still limited, so difficult for a utility to predict.

2.2.2 Risk of over-forecasting may impact ratepayers more than the utility

Large incremental capital expenditures for large load customers can impact existing ratepayers as a result of several sources of risk:

- **The first risk stems from the lack of spare generation capacity and/or transmission system headroom in some utility systems in the United States.** Such utility systems would need new investment to serve new large loads such as data centers. New investment in system generation and/or other network resources (transmission) would cause rates to rise for all customers (but for the new large load, an investment in transmission or generation could be deferred for some time and therefore current customers would not have seen their rates increase). If a system has spare generation and transmission capacity, it can add new customers without a large increase in rates (all else equal), but as noted above, some utility systems do not have spare capacity. Under some circumstances a new large load customer could benefit other ratepayers, if it materializes and is sustained. Large loads increase energy consumption and demand over which to recover certain fixed costs. In addition, large load customers can create economies of scale for the electric system, because the utility may be able to use larger transformers with a lower cost per unit of capacity. As the new infrastructure investment made to increase capacity to serve a large load customer comes online, the average cost to serve customers can decline both because the larger investments cost less per unit, and fixed operating costs can be spread among a larger base of customers. So, it is possible that large loads can, under certain circumstances, lead to a lower dollar rate per unit of energy consumed or per unit of peak demand. However, these benefits are contingent on the load materializing and remaining on the system long enough to offset the initial investment cost, and on the relative cost of the new investment(s) versus the embedded cost of existing infrastructure.
- **The second risk stems from the uncertain permanence of new large loads.** Some data centers do not require as much complex fixed investment as, for example, an auto

manufacturer, so not only can a data center be built quickly, it can also relocate quickly. If the data center leaves the system, and the system investments related to new generation plants and transmission lines were triggered by that data center's load, the recovery of the fixed costs of those new assets will be shifted to remaining customers. This also happens if, for example, population declines and/or households migrate—but because data centers are such large users of electricity, the loss of single data center customer will have a disproportionate impact.

- **A third risk is the shorter expected life span of a data center as compared to the operational life and typical cost recovery period for electricity infrastructure.** Even if the data center does materialize as proposed and does not re-locate, data centers have shorter operational lifespans—approximately 15 years[21]—compared to the energy facilities built to serve them. The economic life of power assets is about 30 to 40 years for gas generation, 30 years for transmission and distribution facilities, and 35 years for natural gas pipelines.[22] Assuming data center facilities are built and the electricity demand materializes as planned, the data center customer may not be part of the system for the entire lifetime of the transmission and generation assets that were built by the utility to provide service. This mismatch in infrastructure lifetimes versus economic life of customer facilities means that some burden may be shifted to other customers.

2.3 Other factors indicate that growth in data centers may be weaker than implied by current announcements

Apart from evidence of multiple requests for service, there are other factors that suggest that electricity demand from data centers will fall well below the levels implied based on review of requests for service or announced plans of data centers.

2.3.1 Equipment bottlenecks limit the pace of development of gas-fired generation resources

Tight supplies of equipment for the next few years could become a bottleneck to serving potential new load from data centers, particularly with natural gas-fired generation. Siemens and GE Vernova, two industry leaders in gas turbine manufacturing, are seeing lead times increase for

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[21] Data Center Knowledge. "Hyperscalers in 2024: Where Next for the World's Biggest Data Center Operators?" February 28, 2024. <https://www.datacenterknowledge.com/hyperscalers/hyperscalers-in-2024-where-next-for-the-world-s-biggest-data-center-operators->

[22] Gas generation: Florida Power & Light, Power Generation Division. "Depreciation Analysis for Power Generation." February 2016. <https://www.floridapsc.com/library/filings/2016/07554-2016/Support/OPCs%201st-38-Attachment%203.pdf>; Transmission and Distribution: Thomson Reuters Onvio. MACRS asset life table. Item 49.14, Electric Utility Transmission & Distribution. <https://onvio.us/ua/help/us-en/staff/fixed-assets/depreciation/macrs-asset-life-table.htm>; Pipelines: Thomson Reuters Onvio. MACRS asset life table. Item, 49.21, Gas Utility Distribution Facilities <https://onvio.us/ua/help/us-en/staff/fixed-assets/depreciation/macrs-asset-life-table.htm>

turbine delivery; the lead time for a 200 MW Siemens turbine has increased to three years,[23] and GE Vernova has a company backlog through 2028 for delivery times on orders.[24] Unless they have already put orders in, utilities and independent power producers may face not only delays but also higher prices in bringing new gas-fired power plants online in the next few years. The CEO of NextEra Energy (one of the largest power generation companies in the United States) recently noted that "We built our last gas-fired facility in 2022, at $785 [per kilowatt ("kW")]. If we wanted to build that same gas-fired combined cycle unit today [the cost would be] $2,400/kW," continuing, "[t]he cost of gas-fired generation has gone up three-fold."[25] Timing is also an issue, as he added, "[w]hen you look at gas as a solution…you're really looking at 2030 or later." Industry observers also report an average lead time of three years for delivery of large transformers (a key part of many transmission and distribution projects); and in some cases, lead times as long as five years.[26]

Until the utility places orders for necessary equipment it needs to procure and a notice to proceed ("NTP") on construction of substation or other required facilities is authorized, there is no clear timeline of when the utility can serve the new load. If the receipt of necessary transmission, distribution, or generation equipment is delayed, then bringing on new customers will, naturally, have to be deferred.

2.3.2 Some state authorities are re-examining tax incentives

Most states offer tax incentives to attract data centers. This is intended to boost economic development driven by the employment opportunities at data centers, and the indirect impacts on the local economy for services that support data centers and their workers. However, a recent study found that in some cases the cost to the state treasury exceeded the benefits to the state's economy, and some states have begun to re-examine the value of such tax incentives.[27] In Georgia, the state legislature passed a bill in 2024 halting tax breaks to data centers for two years, though

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[23] Malik, Naureen S. "Gas Power Won't Provide an Easy Fix for AI Boom." *Bloomberg*. January 8, 2025. <https://www.bloomberg.com/news/newsletters/2025-01-08/gas-power-won-t-provide-an-easy-fix-for-ai-boom>

[24] Casey, Simon. "GE Vernova CEO Sees Order Backlog Stretching Into 2028." *Bloomberg*. March 11, 2025. <https://www.bloomberg.com/news/articles/2025-03-11/ge-vernova-ceo-sees-order-backlog-stretching-into-2028?utm_source=chatgpt.com&embedded-checkout=true>

[25] Cunningham, Nicholas. "Costs to build gas plants triple, says CEO of NextEra Energy." Gas Outlook. March 25, 2025. <https://gasoutlook.com/analysis/costs-to-build-gas-plants-triple-says-ceo-of-nextera-energy/>

[26] Seiple, Chris. "Gridlock: The demand dilemma facing the US power industry." *Wood Mackenzie*. October 2024. <https://www.woodmac.com/horizons/gridlock-demand-dilemma-facing-us-power-industry/>

[27] LeRoy, Greg and Tarczynska, Kasia. *Cloudy with a loss of Spending Control*: *How Data Centers Are Endangering State Budgets*. goodjobsfirst.org. April 2025. </https://goodjobsfirst.org/wp-content/uploads/2025/04/Cloudy-with-a-Loss-of-Spending-Control-How-Data-Centers-Are-Endangering-State-Budgets.pdf>

the bill was later vetoed by the Governor. The Governor's Office of Planning and Budget expects the tax breaks to cost the state $327 million in 2025.[28]

As the data center industry grows, so does the cost of providing tax incentives. In Texas, the sales tax exemption program for data centers cost the state $157 million in 2023; the state estimates the cost will increase by an order of magnitude, to $1 billion in 2025.[29]

2.3.3 Local opposition has derailed and/or delayed data center projects

Industry observers report that a large number of proposed data centers have been cancelled or delayed, owing to local opposition.[30] Residents object to local impacts such as noise, and fears over water resource depletion and pressure on electricity rates. Local authorities may deny permits because of such concerns.

2.3.4 Data center demand flexibility may reduce peak demand

Data centers serve a variety of users and applications. Certain types of data centers can temporarily reduce their energy consumption and may do so to avoid high prices during periods of surging demand, similar to how some other industrial customers of electricity act under demand response programs. Some data centers have flexibility in their operations and can provide demand response. They can lower energy consumption from the grid by reverting to on-site generation, and/or shifting computational work to other locations, or even reducing operations.[31] These strategies may be limited, however. On-site generation is subject to local environmental laws, where diesel (which is used in back-up generation) may only run a few hundred or less hours per year because of emissions limitations. In co-tenant sites, leases often have pass-through clauses where utility costs are passed on to tenant, so that the landlord has limited incentive to be part of a utility demand response program.

To incentivize a data center customer to provide a demand response when needed, a utility can offer it a tariff tailored to reflect the value of such flexibility, and/or it can offer a time-of-use

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[28] Chow, Andrew R. "Why Tax Breaks for Data Centers Could Backfire on States." April 25, 2025. <https://time.com/7280058/data-centers-tax-breaks-ai/>

[29] LeRoy, Greg and Tarczynska, Kasia. *Cloudy with a loss of Spending Control*: *How Data Centers Are Endangering State Budgets*. goodjobsfirst.org. April 2025. </https://goodjobsfirst.org/wp-content/uploads/2025/04/Cloudy-with-a-Loss-of-Spending-Control-How-Data-Centers-Are-Endangering-State-Budgets.pdf>

[30] Eddy, Nathan. "Local Opposition Hinders More Data Center Construction Projects." Data Center Knowledge. May 15, 2025. <https://www.datacenterknowledge.com/regulations/local-opposition-hinders-more-data-center-construction-projects>

[31] See for example: Bloomberg News. "How 'Load Shifting' May Help Improve Data Center Sustainability." *Data Center Knowledge*. February 26, 2024. <https://www.datacenterknowledge.com/sustainability/how-load-shifting-may-help-improve-data-center-sustainability>; and Judge, Peter. "Google tests system to cut data center power use during grid problems." *Data Center Dynamics*. October 5, 2023. <https://www.datacenterdynamics.com/en/news/google-tests-system-to-cut-data-center-power-use-during-grid-problems/>

(“TOU”) tariff which automatically incentivizes shifting load away from peak hours. This flexibility could reduce the level of expansion of generation resources a utility would need to meet load growth.

3 US data center load growth projections cannot be met by global chip supply

The data center boom in the United States is part of a larger global phenomenon, but that global phenomenon will ultimately be constrained by the supply of intrinsic components that data centers require, most critically semiconductor chips. By comparing the load projections from a set of US jurisdictions with available data (covering 77% of US electricity generation and demand) with the physical supply constraints of global chip manufacturers, LEI sought to assess the credibility of the total US-wide data center demand outlook. LEI hypothesized that if the sum of data center demand projections for these jurisdictions of the United States electricity market exceeds the global capacity to manufacture semiconductor chips over the same period, then it is likely that not all the announced or planned data centers incorporated into those US electricity demand forecasts could be built. If not all announced or planned data centers in the US jurisdictions surveyed could be built, then some of the implied attrition would impact specific states.

LEI developed an approach to examining the reasonableness of data center forecasts by working backward from projections for data center electricity demand to the implied need for semiconductor chips, the primary electricity-using component in a data center. The purpose of this analysis is not to forecast the number of AI-related chips ordered or chip production capacity in the future. Instead, it is meant to be a sanity check on the scale of projected data center growth and by extension the forecast of the aggregated electricity demand growth reported by the US regional transmission organizations ("RTOs"), independent system operators ("ISOs"), and balancing authorities ("BAs"). In other words, if the aggregated US RTOs', ISOs', and BAs' data center demand forecasts were to materialize, how many semiconductor chips would this require, and is it reasonable to expect the needed chip-making capacity would materialize and be available to meet that forecasted demand?

As explained below, LEI found evidence that the high-level forecast is not credible. LEI found that, even if global AI semiconductor chip manufacturing were to grow at an average rate of 10.7% annually (significantly faster rate than the 6.1% growth rate over the past decade)[32] it could only satisfy an incremental 63 GW of data center-related demand ***globally*** over the six years from 2025 to 2030. The implied RTO/ISO/BA demand projection, detailed below, is for ***US-only*** data center demand growth of 57 GW over the six years from 2025 to 2030. This would amount to more than 90% of the new total global manufacturing capability being earmarked for US data centers. In other words, for the forecasts to hold water, this would mean the US would require more than 90% of the world's new supply of semiconductor chips from 2025 through 2030.

Such a scenario is unrealistic. The United States currently accounts for slightly less than 50% of global semiconductor chip demand, and other regions in the world are also projecting strong

[32] Semiconductor Industry Association. "Strengthening the U.S. Semiconductor Supply Chain." May 2024. P. 2. Figure 1. <https://www.semiconductors.org/wp-content/uploads/2024/05/Emerging-Resilience-in-the-Semiconductor-Supply-Chain_SIA-Summary.pdf>

demand for semiconductor chips to support their region's growth in data centers.[33] LEI concludes that over-forecasting data center growth and therefore the associated electricity demand is significant at the national level in the United States.

As described in this section, LEI focused its analysis on AI semiconductor chips. While most current data centers in the US are not used for AI applications, most future data centers, especially those with high electricity demand, will be used for AI.[34]

3.1 LEI's approach compares demand outlooks to the global supply of chips

LEI's approach worked backward from projections for data center electricity demand to the implied need for semiconductor chips—the fundamental elements in the data center supply chain. LEI's approach followed a four-step process (see Figure 2):

- First, LEI summed the data center demand forecasts published by the RTOs, ISOs, and BAs in the United States that could be obtained. LEI took a conservative approach to this, meaning that LEI referred to electricity demand growth based on the slower rather than faster end of the range offered by each jurisdiction, if the jurisdiction provided a range of electricity demand growth forecasts rather than a single forecast. Slower growth is more conservative in that it is more likely to be met by the global supply of semiconductor chips and less likely to be constricted by semiconductor chip shortages.
- Second, LEI estimated the electricity needed to power an AI-related chip. LEI also accounted for energy used for non-computing activities, like cooling (especially important in warm climates such as in the southeastern United States). LEI made assumptions as to the use of energy for non-computing activities, and the rate of efficiency improvement of semiconductor chips. Efficiency and computing capabilities are discussed in more detail in Appendix A.
- Third, LEI calculated the implied number of semiconductor chips needed annually based on the data center demand outlook from Step 1.
- Fourth, LEI compared this annual demand for semiconductor chips to global chip manufacturing capacity (including expansion of that capacity). Where LEI had to make assumptions, for example, as to the rate of annual expansion of chip manufacturing capacity, LEI made conservative assumptions—in this case, assuming that chip manufacturing growth would be rapid (to allow fast growth in data centers).

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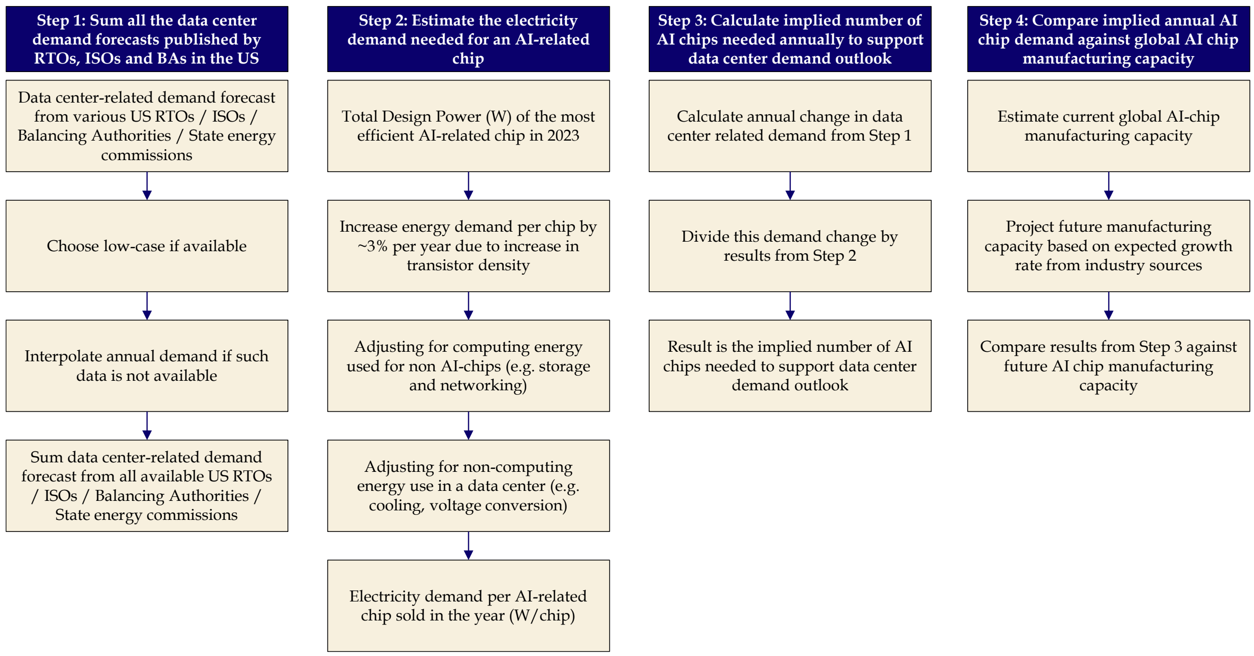
[33] IEA. *Energy and AI*. April 2025. P. 259. Table A.2. <https://www.iea.org/reports/energy-and-ai>

[34] See, for example, Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M.A.B., Holecek, B., Koomey, J., Masanet, E., Sartor, D. 2024. 2024 *United States Data Center Energy Usage Report.* Lawrence Berkeley National Laboratory, Berkeley, California. LBNL-2001637. P. 31.

Figure 2. LEI's approach to examining reasonableness of data center load forecast by selected US RTOs, ISOs, and utilities



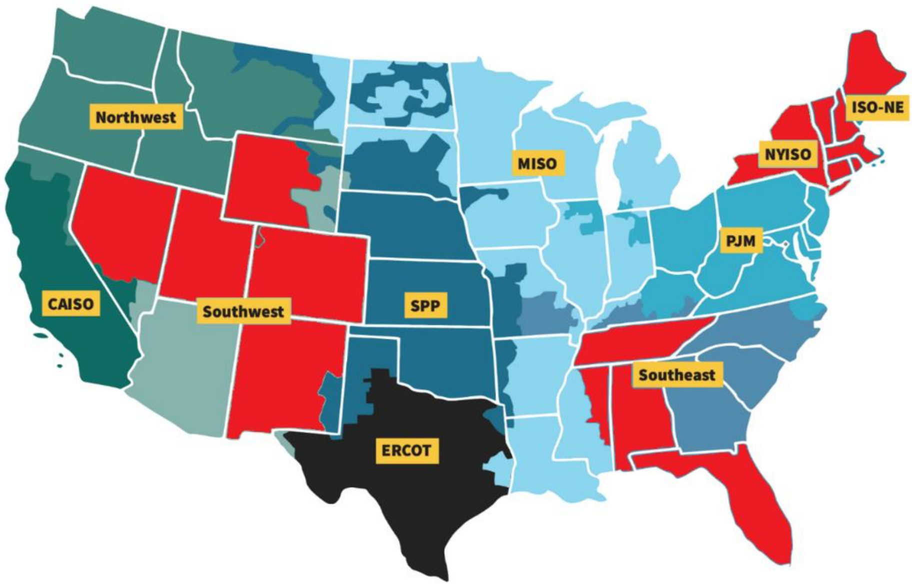
The following sections detail each of the steps of LEI's methodology.

3.1.1 Step 1: Aggregate available load forecasts across US RTOs, ISO, and BAs

First, LEI summed all the available data center demand forecasts published by the RTOs, ISOs, and BAs in the United States that LEI was able to obtain from public sources. Data was available for Midcontinent Independent System Operator ("MISO"), Electric Reliability Corporation of Texas ("ERCOT"), Southwest Power Pool ("SPP"), California Energy Commission ("CEC"), the Northwest Power and Conservation Council ("NPCC"), Arizona Public Service, PJM (AEP, APS, PSEG, and Dominion), North Carolina and South Caroline (Duke Energy Carolinas and Duke Energy Progress), and Georgia Power Company ("GPC"). LEI was unable to obtain publicly available forecasts for several jurisdictions (New England, New York, parts of the Southeast United States, and parts of the Western United States) (see Figure 3). LEI's aggregation accounts for 77% of net summer electric capacity in the United States.[35]

[35] Based on data from the Energy Information Administration, LEI's jurisdictions omit 23.49% of US net summer capacity (MW) for 2023; they omit 23.55% of US consumption (MWh) for 2023.

Figure 3. US RTOs and ISOs—jurisdictions in red are not covered in LEI's analysis



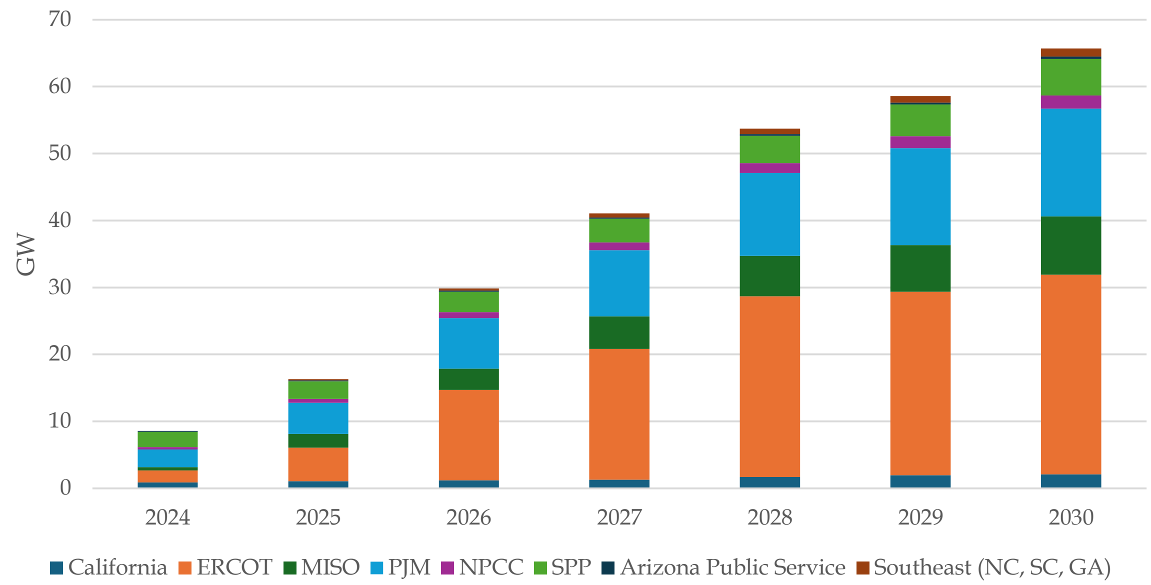
Note: LEI was not able to obtain specific data center-related demand forecast in the latest published integrated resource plans or load forecasts in the jurisdictions colored in red. Therefore, the red colored areas, which represent approximately 23% of US electricity capacity, are not included in LEI's data center-related demand calculations. The other colors indicate RTO or BA boundaries.

The forecasts from the jurisdictions differed in issue dates and in periodicity. If a forecast was not available for every year, LEI interpolated using the cumulative average growth rate ("CAGR") between two years. If the forecasts had an outlook timeframe shorter than required, LEI assumed data center-related load growth would be 50% of the change in demand (in GW terms) between the last two years of the available forecasts. For example, if an electricity demand forecast ended in 2029, with data center-related demand being 7 GW in 2028 and 8 GW in 2029, then LEI extended the forecast to 2030 by adding 50% of the growth between 2028 and 2029 (i.e., 50% of 1 GW which is equal to 0.5 GW) to the projected data center-related electricity demand in 2029 (8 GW), resulting in a 2030 projection of 8.5 GW. This approach is conservative because it uses a slower growth rate than in the prior years forecasted by the jurisdiction.

Applying this approach to US data center-related demand to the 77% of the US electricity sector with available data results in an electricity demand of somewhat less than 9 GW in 2024. This is the total aggregation based on LEI's methodology described in Step 1, and because it includes only 77% of the US sector, it is a conservative (low) estimate for the US overall.

Using the same methodology for the ensuing years results in electricity demand of 65 GW by 2030 (see Figure 4).

Figure 4. RTO/ISO/BA data center-driven electricity demand outlook*



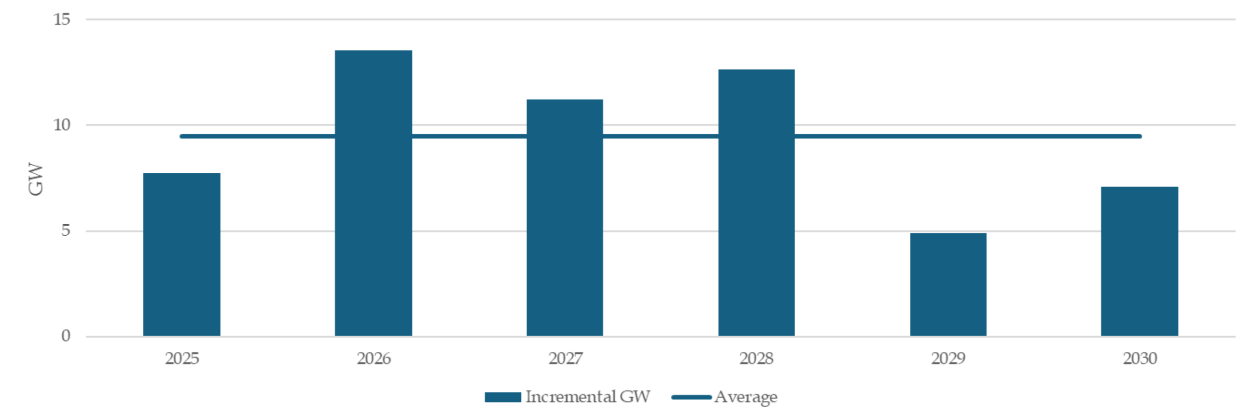
*US RTOs and utilities with data available as noted in the text—this covers 77% of the US power sector by capacity and consumption.

Note: Forecasts for California, MISO and NPCC are low cases. NPCC's forecast is only available until 2029, therefore LEI assumed 2030 demand would be 2029 demand plus 50% of the increment of data center-related demand growth from 2028 to 2029.

For some utilities, although the demand outlook may be small compared to other regions, the share of forecasted data center demand could be large compared with electricity demanded by existing customers. This adds risk to the smaller utilities' customers, because a potential buildout of new infrastructure to meet demand (some of which might not materialize) has fewer customer billable units over which to spread the cost. All else equal, risk to customers will be higher in jurisdictions with vertically integrated utilities where investments in generation assets are also paid for by customers. By contrast, in some deregulated wholesale electricity markets, some of the risk of not recovering capital expenditures stays with the generation owners (rather than being backstopped by the utility's customers).

In terms of annual increments to projected load growth (the conservative outlook shown in Figure 4 above), the growth from 2025 to 2030 is uneven but averages 9.5 GW per year (see Figure 5). Again, this is a conservative (low) estimate for the US overall, because it only includes 77% of the electricity sector, and because it is based on the low-end forecasts provided by the RTOs, ISOs, and BAs, as noted previously.

Figure 5. Total annual incremental electricity demand implied by RTO/ISO/BA US data center outlook



To reiterate, LEI ensured its aggregation of US demand forecasts is conservative because:

- LEI used the lowest data center demand estimate from the forecasting entity if multiple outlooks were available (this applied to California, MISO, and NPCC);
- US coverage is not complete. The available data covers 77% of the market as noted above, because not all utilities or BAs report electricity demand growth associated with data centers. Data center-related demand forecasts for Colorado, Florida, Utah, New Mexico, the area served by Tennessee Valley Authority (including Tennessee and parts of Alabama, Georgia, Kentucky, and Mississippi), New York state (for New York Independent System Operator ("NYISO")), and the six states covered by ISO-NE, were not publicly available at the time LEI performed this analysis; and
- Not all utilities within regions that LEI included provided data center-related load growth forecasts. For example, the parts of Arizona that are not covered by Arizona Public Service are not included. Data centers may be developed in these regions in the future,[36] which will create demand for AI-related chips that are not included in LEI's total. Therefore, LEI's analysis is conservative in relation to the hypothesis being tested.

3.1.2 Step 2: Estimate the electricity demand per AI chip and other uses

The data center energy demand forecasted by the US utilities, RTOs, and ISOs comes from expectations that new data centers will be built, which implies that new AI-related semiconductor

[36] For example, under the provisions of Tennessee Code Ann. Section 67-6-206(c), qualified data centers may purchase electricity at a reduced rate of sales tax, and under Section 67-6-102, qualified data centers are also eligible for sales tax exemption on certain computers and devices. This is intended to incentivize data centers to locate in Tennessee and would create data center-related demand for Tennessee Valley Authority that is not included in LEI's analysis.

chips must be purchased and installed. Fundamentally, the world must therefore be able to produce sufficient AI semiconductor chips for this forecasted demand to materialize.

To calculate the number of AI semiconductor chips needed to meet the energy demand forecast, LEI first quantified the energy need of AI semiconductor chips. To estimate energy use per AI semiconductor chip, LEI made the conservative assumption that a falling number of AI semiconductor chips will be needed over time to drive the same MWs of data center-related demand. While the energy efficiency of AI-related semiconductor chips measured in floating point operations per second per watt ("FLOPS/W") has improved over the past decade, each AI-related semiconductor chip consumes more energy because the density of transistors has increased. From the first generation of AI-related semiconductor chips in 2016[37] to the most recent generation in 2024,[38] the total design power ("TDP") (in watts ("W")) per square millimeter ("mm2") of silicon has increased at an annual rate of 3.04% (from 0.49W per mm2 to 0.63W/mm2). Assuming the trend of increasing energy consumption per chip size continues over the forecast period and holding each AI-related semiconductor chip's silicon size the same,[39] the same MW of data center-driven demand growth over time will require fewer AI-related semiconductor chips.

LEI assumed that 41% of demand for energy from data centers is consumed for non-computing activities, and 59% is used for computing. This is based on the low end of Lawrence Berkeley National Laboratory's ("LBNL") estimation of 59% to 76%,[40] while the remaining energy would be consumed for other equipment such as data storage and networking. LEI's assumption is conservative because the smaller share of energy used for AI-related computing means a smaller implied number of chips would be required per MW of data-center related load.

After applying these calculations, LEI arrived at an energy consumption per AI-related semiconductor chip that starts at a little over 1,500 W in 2025 and rises to over 1,750 W by 2030 (see Figure 6). The increase is driven by an increasing density of transistors per surface area of a chip, which is discussed in more detail in Appendix A.

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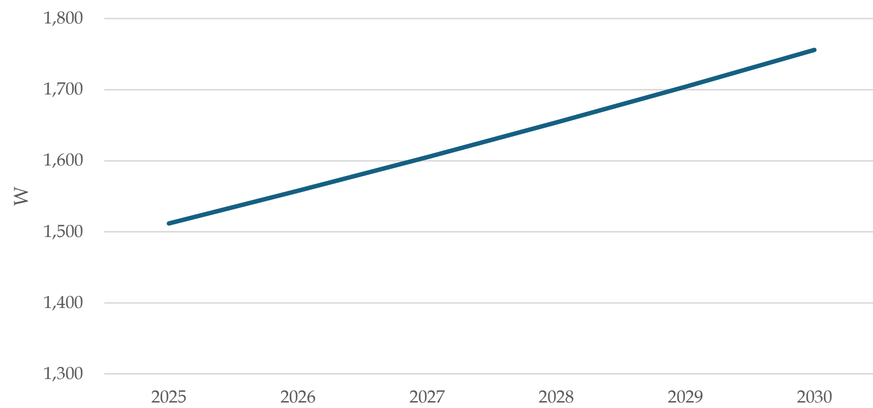
37 Nvidia's P100 DGXS, launched in April 2016. <https://www.techpowerup.com/gpu-specs/tesla-p100-dgxs.c3285>

38 Nvidia's B200 SXM, launched in 2024. <https://www.techpowerup.com/gpu-specs/b200-sxm-192-gb.c4210>

39 The global supply of semiconductors is measured by the number of specific sized wafers that can be produced over a period of time. This means the size of the AI-related chip does not impact the overall limitation on chip output – if the AI-related chip is bigger, it means a smaller number of chips can be produced over a period of time, and vice versa.

40 Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M.A.B., Holecek, B., Koomey, J., Masanet, E., Sartor, D. 2024. *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, Berkeley, California. LBNL-2001637. P. 53. Figure 5.6.

Figure 6. Estimated energy demand (W) per AI-related chip

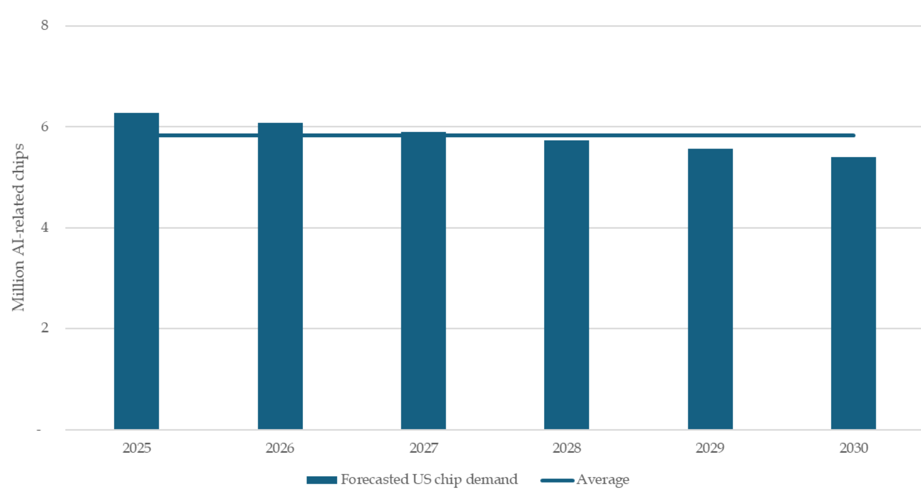


Note: Adjusted for power usage efficiency ("PUE") of 1.2. PUE refers to the share of energy consumption for computing versus other data center usage, and is discussed in detail in Appendix A.
Source: LEI analysis.

3.1.3 Step 3: Estimate the number of chips implied by aggregated data center demand forecasts

Dividing the annual data center-driven load growth from Step 1 by the energy demand impact of an AI-related semiconductor chip results in the number of new semiconductor chips which need to be installed in the US each year if the data center electricity demand outlook for RTO/ISO/BAs is to be achieved. This demand averages 5.8 million chips per year for the outlook period (see Figure 7).

Figure 7. Total annual incremental AI chip demand implied by RTO/ISO/BA US data center outlook



3.1.4 Step 4: Estimate global production capacity based on industry growth

Will the world have enough production capacity for AI-related chips to meet this demand?

LEI began by estimating the baseline annual production capacity by examining the largest individual chip producers. Nvidia held a 98% market share in 2023, shipping 3.76 million units of data-center graphic processing units ("GPUs"). Including the other two main suppliers, Advanced Micro Devices ("AMD") and Intel Corporation ("Intel"), data center-related chip shipments in 2023 totaled 3.85 million.[41] Note that this 3.85 million is already much less than the projected need of the RTO/ISO/BA forecast for over 6 million in 2025—which indicates that even near-term outlooks for data center demand are too high.

However, the chip manufacturing industry has been growing and LEI examined its prospects for further growth. Industry observers have noted that the fast growth seen in 2024 and 2025 (over 15% annually) is likely to slow somewhat for the longer term:

- Deloitte reported that the semiconductor industry had a robust 2024 with growth of about 19% in sales revenue, and it expects this to slow somewhat for 2025, with sales revenue growth projected at about 11%.[42] PricewaterhouseCoopers forecasts that the logic product sector (used for AI and cloud computing) of the global semiconductor industry would grow by a CAGR of 6.28% from 2024 to 2030.[43]
- The Semiconductor Industry Association projects that global semiconductor capacity (as measured by wafer starts per month) will increase by 108% from 2022 to 2032, implying a 7.6% CAGR.[44]
- IDC, a consultancy, forecasts growth in worldwide foundry (semiconductor chip manufacturing facility) market revenue of about 10% from 2026 through 2028.[45]

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[41] Shah, Agam. *Nvidia Shipped 3.76 Million Data-center GPUs in 2023, According to Study*. June 10, 2024. <https://www.hpcwire.com/2024/06/10/nvidia-shipped-3-76-million-data-center-gpus-in-2023-according-to-study/>

[42] Kusters, J., Bhattacharjee, D., Bish, J., Nicholas, J.T., Stewart, D., Ramachandran, K. "2025 global semiconductor industry outlook." Deloitte. February 4, 2025. <https://www2.deloitte.com/us/en/insights/industry/technology/technology-media-telecom-outlooks/semiconductor-industry-outlook.html>

[43] PricewaterhouseCoopers, *State of the Semiconductor Industry*. November 28, 2024. P. 3. Exhibit 1. <https://www.pwc.com/gx/en/industries/technology/state-of-the-semiconductor-industry-report.pdf>

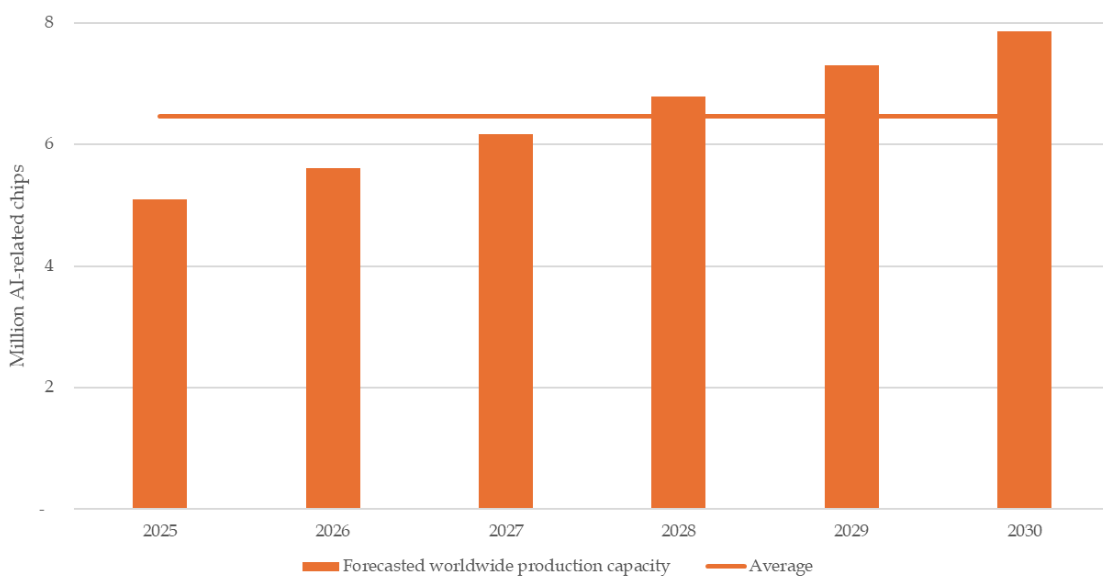
[44] Semiconductor Industry Association. "Strengthening the U.S. Semiconductor Supply Chain." May 2024. <https://www.semiconductors.org/wp-content/uploads/2024/05/Emerging-Resilience-in-the-Semiconductor-Supply-Chain_SIA-Summary.pdf>

[45] IDC. "Global Semiconductor Market to Grow by 15% in 2025, Driven by AI." December 12, 2024. <https://my.idc.com/getdoc.jsp?containerId=prAP52837624>

- Another industry source projects that the global market for silicon wafers was valued at $17.27 billion in 2023, and would reach $22.1 billion by 2030, a CAGR of 5.77%.[46]

To be conservative (i.e., to project higher AI chip production capacity), LEI applied high end industry estimates and forecasts: a 19% growth rate applied to the 3.85 million AI-related semiconductor chips shipped in 2023, to estimate worldwide manufacturing capacity in 2024,[47] followed by 11% growth for 2025, 10% growth for 2026 to 2028, and 7.6% thereafter. This results in world-wide production capacity of reach 7.9 million AI-related semiconductor chips by 2030 (an average growth rate of 10.7% from 2023 to 2030) (see Figure 8).

Figure 8. LEI outlook for AI-related global chip capacity



3.2 Findings

The bottom line is derived by comparing total global capacity for AI-related semiconductor chips (which reaches 7.9 million chips by 2030 as noted above) with the implied growth in chip demand from the RTO/ISO/BA outlook. Over the six-year period from 2025 to 2030, US demand for

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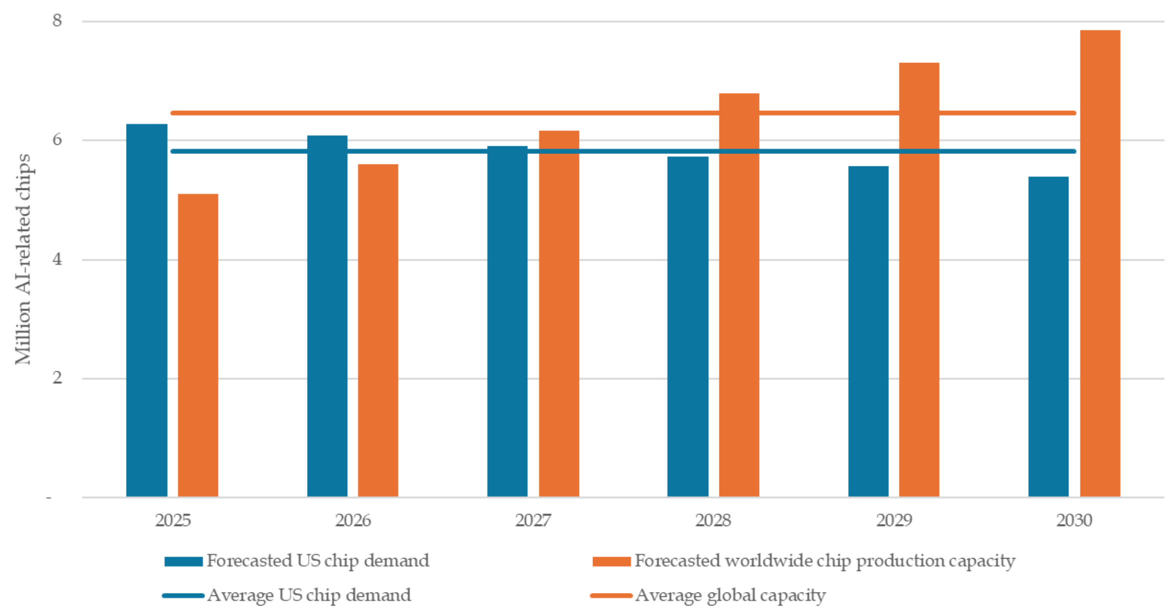
[46] Semiconductor Insight. "Silicon Wafer Market Size, Share, Trends, Market Growth and Business Strategies 2025-2032." (Undated). <https://semiconductorinsight.com/report/silicon-wafer-market/>

[47] Morgan Stanley research also estimated that AI-related chip sales from Nvidia to be 4 million in 2024 (Source: Morgan Stanley Research. "AI Supply Chain – The Latest about NVDA GB200 Superchip." P. 10. Exhibit 11. May 13, 2024). Note that, for conservativeness, LEI assumed that most of the capacity expansion would happen by 2025, reflecting the commissioning of the Taiwan Semiconductor Manufacturing Company's Arizona Fab (Source: Taiwan Semiconductor Manufacturing Company Limited. *TSMC Arizona.* <https://www.tsmc.com/static/abouttsmcaz/index.htm>).

semiconductor chips would account for 90% of global capacity if all the projected US data center demand were to materialize (see Figure 9, blue line and orange line).

Figure 9. Implied incremental AI-related chips needed in the US vs forecasted worldwide AI-related chip production capacity



It is not realistic to expect that the United States will be able to acquire 90% of the global supply of chips over the next six years. Global chip manufacturing capacity will be sought after by customers around the globe, not just in the United States, and not just in the US jurisdictions surveyed. As noted previously in Section 2.1.1, the IEA estimated that in 2024, US data center installed capacity amounted to 42 GW, while other countries' data center installed capacity amounted to approximately 55 GW, so that US data center electricity installed capacity accounted for less than half of the world's total.[48] Nvidia (the world's largest chip maker) reported that revenue from sales to customers outside of the United States accounted for 69%, 56%, and 53% of total revenue for fiscal years 2023, 2024, and 2025 respectively."[49]

[48] IEA. *Energy and AI*. April 2025. P. 259. Table A.2. <https://www.iea.org/reports/energy-and-ai>

[49] Nvidia. *Form 10-K. Annual Report for fiscal year ended January 26, 2025*. P. 79.

There is demand for AI-related semiconductor chips in other parts of the world, such as Europe (e.g. France[50] and Ireland[51]), Canada,[52] China, East/Southeast Asian countries such as Japan, South Korea, Singapore and Malaysia; and the Middle East.[53] These regions and countries are competing for the global supply of chips. Recent talks between the US administration and Saudi Arabia indicate the kingdom's expectations for strong growth in data centers and associated demand for chips and plans for other Middle Eastern nations for purchase of chips from the United States (as well as plans to invest in data centers in the United States).[54]

This implies that the total data center electricity demand growth projections over the next five years for the US are unrealistic compared to the chip manufacturing industry's own projections for how fast it can ramp up production. It is unlikely that sufficient equipment could be manufactured globally to create such electricity demand unless a large (perhaps government-funded) project was put into place to do so. Hence, it is not reasonable to believe that data centers in the United States will procure almost all the output of production capacities for AI-related semiconductor chips from the entire world over a six-year period.

As noted in Section 2.1 of this report, data center requests for service across the United States have some level of duplication. Even ignoring the possibility of duplication, LEI's analysis of chip supply suggests that some data center developers will be unable to build as intended in the next few years and therefore not all the electricity demand growth from data centers reflected in the aggregate US outlook will be realized.

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50 *France 24*. "Microsoft, Amazon to invest billions in French tech." December 5, 2024. <https://www.france24.com/en/live-news/20240512-amazon-plans-to-invest-1-2-bn-euros-in-france-macron-s-office?ref=frenchtechjournal.com>

51 Burns, John. "Up to 82 data centres operating in Republic." *Irish Times.* June 17, 2023. <https://www.irishtimes.com/business/2023/06/17/up-to-82-data-centres-operating-in-republic/>

52 Microsoft. "Microsoft Expands Digital Footprint In Quebec With Usd$500 Million Investment In Infrastructure And Skilling Initiatives." November 2023. <https://news.microsoft.com/en-ca/2023/11/22/microsoft-expands-digital-footprint-in-quebec-with-usd500-million-investment-in-infrastructure-and-skilling-initiatives/>

53 Countries in the Middle East, including Saudi Arabia and UAE, have plans to attract investments from technology companies such as Google and Microsoft to develop data centers in their region. For example, see: MIT Sloan Management Review. "How the Middle East is Emerging as a Data Center Powerhouse Amid Booming AI Demand." October 2024. <https://www.mitsloanme.com/article/how-the-middle-east-is-emerging-as-a-data-center-powerhouse-amid-booming-ai-demand/>

54 Gunia, Amy. "Will 'massive' Gulf deals cement the US lead in the race for global AI dominance?" CNN. May 22, 2025. <https://www.cnn.com/2025/05/22/business/trump-gulf-deals-global-ai-race-hnk-spc-intl>

4 Uncertainty around electricity demand from data center growth poses risks for ratepayers

As LEI examined in detail in this report, future growth of data centers is highly uncertain, especially projections for growth in any given location. As detailed, there are a number of drivers of this uncertainty, and they all point to upside bias in demand outlooks.

The uncertainty surrounding data centers impacts utility load projections because data centers are such large customers. As such, they can create substantial risk for other customers if risks are left unmitigated. As discussed earlier in this report, vertically integrated utilities build infrastructure and pass along costs to customers; such costs could include new system assets (generation and transmission capacity), as well as related costs such as the cost of firm transportation on natural gas pipelines intended to serve new gas-fired generating plants. If the expected electric demand does not appear, the costs incurred to prepare the system to serve the demand would typically be re-allocated to the remaining customers, and if the expected electric demand is large compared to the demand of the existing customer base, the impact could be substantial. Even if the new large load materializes, it may not be on the system long enough to cover the costs of system assets which it caused to be built to serve it. The loss of any kind of customer (including residential customers if population migrates, or an industrial customer who shuts down a factory) would impact the remining customers, but with data centers, the risk posed by a single customer is amplified because data centers can be such large loads.

If a utility takes a more conservative approach and underestimates future load, then it will have to tell some new customers that they must wait for service. This may not be ideal for the utility, because it may mean that these customers may move their plans to other regions, causing the utility to forgo expansion of its customer base and associated rate base growth. But when the scale of investment needed is in the hundreds of millions of dollars, utilities and their regulators will want to avoid basing expansion plans on speculative requests.

In short, over-forecasting based on speculative data center load can expose existing customers to potentially unnecessary and avoidable costs. An approach that prioritizes verified commitments, investment planning to meet demonstrated need, and ensuring ESA or other contract and tariff terms require financial commitments from potential data center customers would help reduce risk to existing customers.

5 Conclusion

The purpose of this analysis was to examine the projections of data center electric load growth in the United States. These projections are beset by uncertainty, and the factors driving the uncertainty (the incentive for duplication of requests by data centers, evidence of attrition in data center announcements, utility investment incentives, long lead times for new generation equipment, the typical timeline for adding infrastructure to the grid, and limits to global semiconductor chip supplies) all point to the greater risk of over-forecasting demand than under-forecasting demand. This bias may prove costly to existing customers, especially in regions where vertically integrated utilities own generation and pass costs to customers under cost-of-service regulation and the incremental cost of investment far exceeds the average embedded cost.

LEI's approach to this analysis was two-pronged:

First, LEI examined factors on the demand side (on the part of data center developers) and the supply side (on the part of utilities) which impact forecasts of data center electricity load. LEI found that these factors not only contribute to uncertainty in load forecasts, but they also add upside bias to such forecasts.

Second, LEI performed a reality check on data center forecasts for the United States by comparing them to the global capacity to manufacture the number of chips required to support such forecasts. LEI found that global chip manufacturing is likely to be far short of what would be needed to supply the US demand forecast based on LEI's tally of RTO/ISO/BA data center projections.

LEI concludes that the risk to the overall outlook for data center demand at the US level contains upside bias: it is more likely that the outlook implied by the RTO/ISO/BA tally is higher than what will materialize in the time frame of the forecast. If this happens and utilities have invested in incremental generation and transmission based on unrealistic projections of data center demand, costs will increase for other utility customers.

Appendix A: The basics of data center electricity demand

A data center is a physical facility composed of computing and data storage equipment, and software. At a high level, its three main components are:

- **Network infrastructure** to connect servers (physical and virtual) that perform data center services and provide storage internally in the data center, and to provide external connectivity to end-user locations;
- **Storage infrastructure** to hold data; and
- **Computing resources** to provide the processing, memory, local storage, and network connectivity that drive applications.[55]

The data center industry is shifting towards larger companies and larger data centers known as hyperscale centers, which provide cloud computing and data management services to organizations that require extensive infrastructure for large-scale data processing and storage. In 2024 about 80% of North American demand for data centers was driven by hyperscalers.[56] Not only is the size of an individual hyperscale data center much larger, but the companies that use them are huge—large cloud service providers include Amazon Web Services, Google Cloud, Microsoft Azure, International Business Machines Corporation ("IBM") Cloud, Baidu, and Alibaba Cloud.[57] By 2023, hyperscale centers accounted for about 95% of computing instances (a measure of computing resources), meaning that these hyperscale centers now dominate the data center industry (see dotted line in Figure 10). The computing power needed for AI workloads is the driver of the growth in the size of data centers and a factor in the ongoing expansion of the data center industry.[58]

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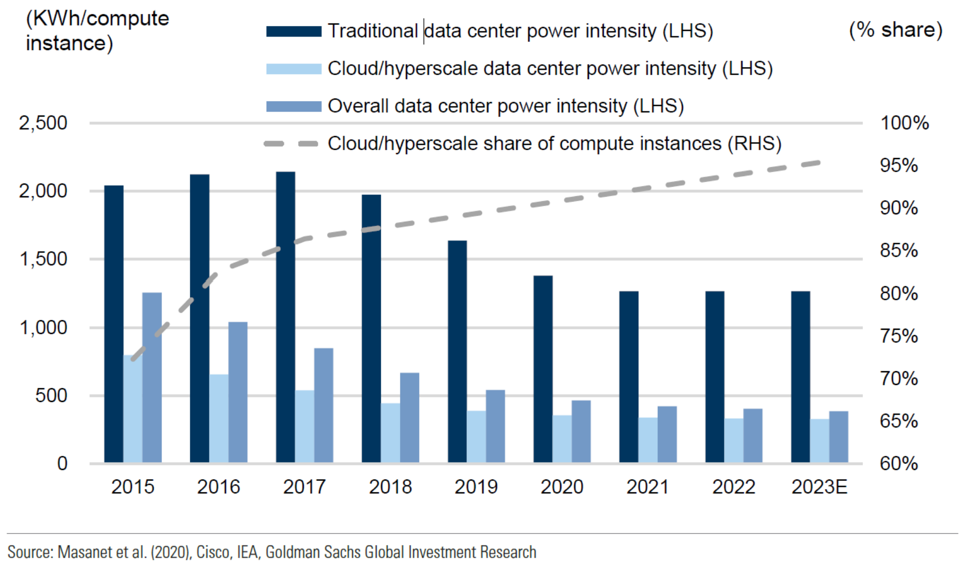
[55] Cisco. "What Is a Data Center?" <https://www.cisco.com/c/en/us/solutions/data-center-virtualization/what-is-a-data-center.html>

[56] Collier's. "Accessing Power and Capital: The Future of Data Center Development 2024." P. 3. <https://www.colliers.com/en/research/dmc-nrep-uscm-usdc-colliers-data-center-white-paper-2024 >

[57] New England States Committee on Electricity. "Data Centers and the Power System: A Primer." Soring 2024. P. 6. <https://nescoe.com/resource-center/data-centers-primer/>

[58] McKinsey & Company. "AI power: Expanding data center capacity to meet growing demand." October 29, 2024. <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand>

Figure 10. Hyperscale share of data center computing growth over time, and energy efficiency (global estimates)



Source: Goldman Sachs. "AI, data centers and the coming US power demand surge." April 28. 2024. P. 13. Exhibit 10. <https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf>

Note: A compute instance is a measure of computing resources (independent of the hardware used). In cloud computing, a compute instance is a server resource provided by a third-party cloud service and supported by the hardware in the data center (Source: Amazon Web Services. *What is an Instance In Cloud Computing?* <https://aws.amazon.com/what-is/cloud-instances/>).

Hyperscale data centers are more energy efficient than smaller data centers in the sense that a larger share of the energy used by hyperscale data centers goes into actual computing work. This is referred to as power usage effectiveness ("PUE"). As of 2023, the estimated average PUE for all data centers in the United States was 1.4[59] (meaning 40% of energy is not used for computing), while AI-specialized data centers had an average PUE of 1.14.[60] Future improvements in this kind of efficiency might be limited because the most efficient data centers are already close to a PUE of 1, which is the lowest possible. However, computation efficiency can still improve.

5.1 Data centers support a variety of businesses, some with volatile markets

There are three overarching categories of business which data centers support:

[59] Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M.A.B., Holecek, B., Koomey, J., Masanet, E., Sartor, D. 2024. *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, Berkeley, California. LBNL-2001637. P. 47.

[60] Ibid. P. 47. Figure 4.5.

- **Cloud computing data centers** support a broad spectrum of information technology ("IT") services that involve storing and processing data;
- **AI data centers** (including for training and inference) are specifically designed to handle the computationally intensive tasks of artificial intelligence and machine learning, requiring specialized and advanced equipment; and
- **Crypto mining centers** have the narrowest function: they validate blockchain transactions and generate new cryptocurrency units.

Cloud computing and AI data centers are expected to have more stable operations and earnings compared to crypto mining centers activities and earnings which are more volatile, rising and falling with the value of cryptocurrencies and energy prices.[61] The EIA estimated that electricity demand associated with cryptocurrency mining operations in the United States likely accounted for 0.6% to 2.3% of US electricity consumption by January 2024, but EIA has not updated these numbers.[62] These ranges compare to the 4.4% energy use estimated for data centers overall for 2023.[63]

5.2 Computational and design efficiency can impact electricity demand from data centers

The energy efficiency of data centers is an important factor in driving future data center load growth.

There are two measures of efficiency for data centers. First, as noted previously, PUE measures the ratio of power (MW of electricity) used by the data center as a whole and the power used by the computing equipment. Power used by non-computing equipment can include cooling, power transformation (from high voltage to low voltage), and conversion (from alternating current to direct current). The PUE of a data center depends on factors including the data center's design, the capacity factor of the data center, location of the data center (which reflects the ambient temperature and impacts the efficiency of cooling equipment), among others.

Over the past decade, the PUE of hyperscale data centers owned by large companies has improved. For example, Google reports its improvement in PUE since 2008, when the PUE for all large-scale Google data centers was above 1.2. Google's PUE improved over time, falling to 1.10

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61 Cloudnium LLC. *Understanding the Differences Between Data Centers and Crypto Mining Facilities*. July 12, 2024. <https://cloudnium.net/understanding-the-differences-between-data-centers-and-crypto-mining-facilites/>

62 EIA. "Tracking electricity consumption from U.S. cryptocurrency mining operations." February 1, 2024. <https://www.eia.gov/todayinenergy/detail.php?id=61364#>

63 Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M.A.B., Holecek, B., Koomey, J., Masanet, E., Sartor, D. 2024. *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, Berkeley, California. LBNL-2001637. P. 52.

by Q3 2024.[64] Similarly, Amazon Web Services announced that it reached a global PUE of 1.15 across its data centers in 2023, while Meta (the parent company of Facebook) has an average PUE of around 1.08.[65] In theory, a perfect data center (where all the energy used goes to computing) would have a PUE of 1. This means there is only very limited room for the PUE of hyperscale data centers to improve further. At best, the PUE of hyperscale data centers owned by large companies can only improve by approximately 10%, as the PUE value cannot drop below 1.

The other measure of data center efficiency is the energy required per unit of computing work. Generally, computing work is measured in floating point operations ("FLOPS"). A FLOP is a calculation done by a computer. The energy efficiency of a data center's computing power can be measured by FLOPS/W, with a higher number indicating greater efficiency. Floating point operations differ by the level of precision: the more precise the calculation, the more inefficient the calculation process.

The FLOPS/W of a data center depends largely on the semiconductor chips used by the data center. In recent years, AI-related work has been typically carried out by GPUs, which are semiconductor chips originally designed for graphics-related purposes, but which were found to be very efficient in doing the type of calculations needed for AI.[66]

5.3 Semiconductor chips have become more efficient, but also more energy dense

The IEA reported that a modern AI-related computer chip uses 99% less electricity to perform the same calculations than a similar chip in 2008, and the efficiency of these chips doubled roughly every two-and-a-half to three years (see Figure 11). This implies an average improvement of 30% in FLOPS/W per year.[67]

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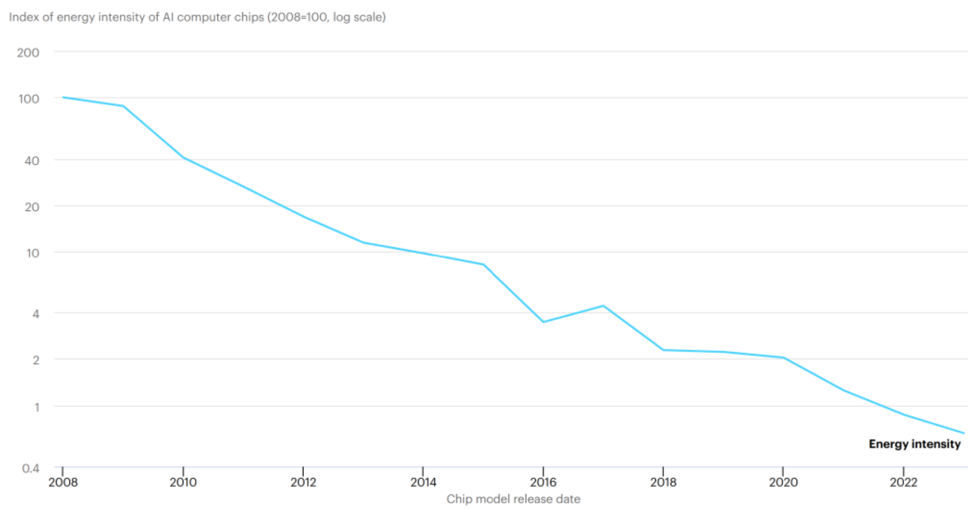
64 Google Data Centers. "Growing the internet while reducing energy consumption." <https://www.google.com/about/datacenters/efficiency/>

65 Butler, Georgia. Data Center Dynamics. *AWS global data centers achieved PUE of 1.15 in 2023.* December 4, 2024. <https://www.datacenterdynamics.com/en/news/aws-global-data-centers-achieved-pue-of-115-in-2023/>

66 Khan, Saif M. and Alexander Mann. *AI Chips: What They Are and Why They Matter*. Center for Security and Emerging Technology. April 2020. P. 20. <cset.georgetown.edu/research/ai-chips-what-they-are-and-why-they-matter>

67 A 100% improvement over 2.75 years translate to an average of 28.7% improvement every year.

Figure 11. Energy intensity of AI computer chips



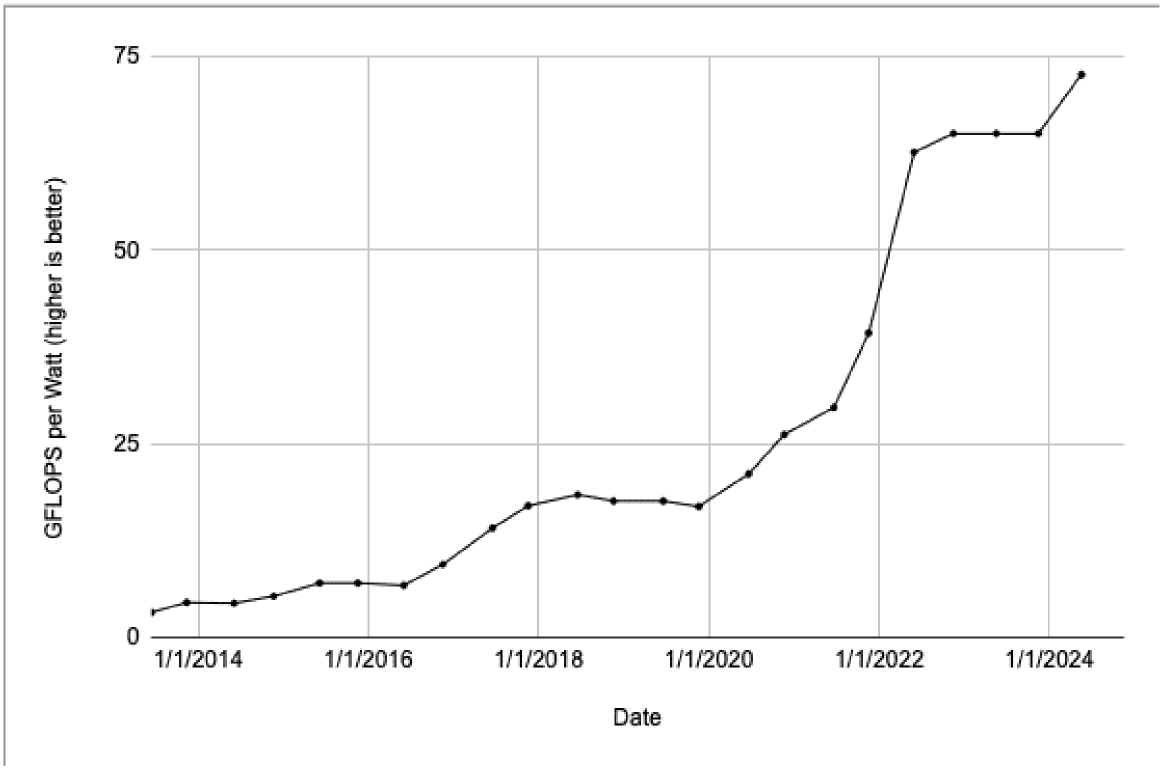
Source: IEA. "What the data centre and AI boom could mean for the energy sector." October 18, 2024. https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector.

Similarly, Nvidia, using data from the top 500 supercomputers in the world, showed that the FLOPS/W of supercomputers improved from approximately 4 giga-FLOPS ("GFLOPS")/W[68] in late 2013, to 74 GFLOPS/W in 2024 (see Figure 12). This also implies an approximately 30% average improvement in energy efficiency per year, similar to the IEA calculation.[69]

[68] A GFLOP means a giga-FLOP, which is approximately a billion FLOPS.

[69] A 18.5-fold improvement over approximately 11 years translate to 30.4% improvement every year.

Figure 12. Energy efficiency gains over time for the most efficient supercomputer



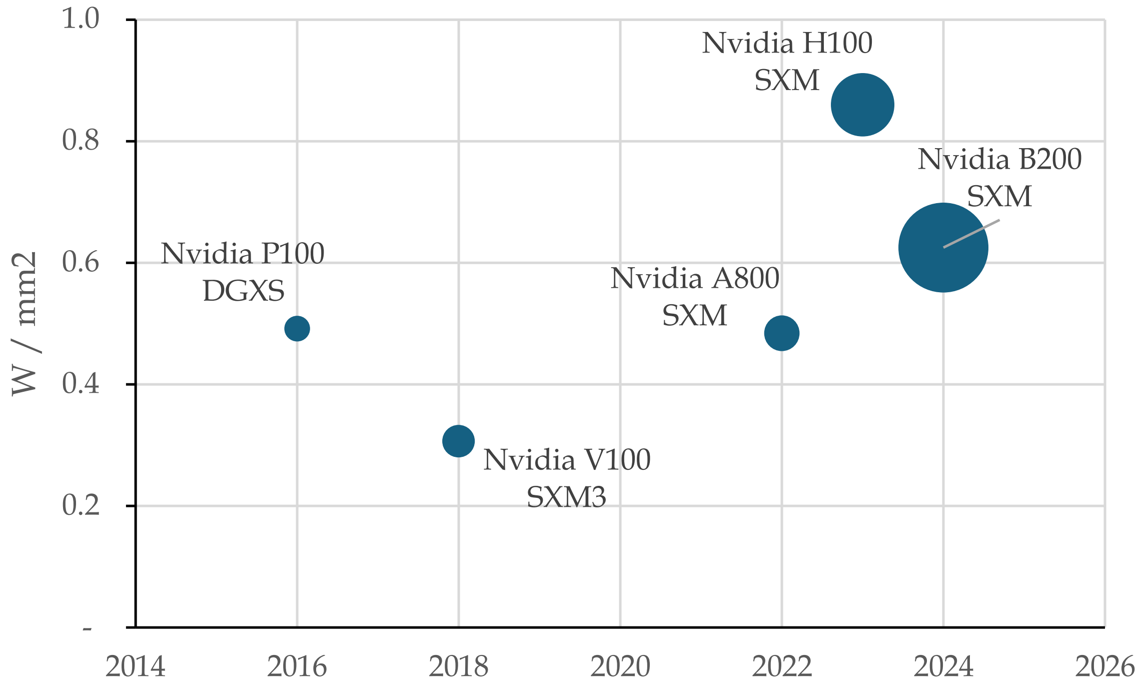
Source: Harris, Dion. *Sustainable Strides: How AI and Accelerated Computing Are Driving Energy Efficiency*. July 22, 2024. Nvidia. <https://blogs.nvidia.com/blog/accelerated-ai-energy-efficiency/>

Note: Data is from Top500.org, an organization which compiles statistics about supercomputers.

Another key driver of data center-related electricity demand growth is that AI-related chips use more energy than conventional chips because AI-related chips have more computing power packed into a single chip. While the FLOPS/W of AI-related chips has improved over time, each new generation of AI-related chips has more FLOPS packed into an individual chip than the previous generation. This has resulted in an overall increase in energy demand per chip. Based on LEI's comparison of five generations of AI-related chips specifically designed for data centers, the energy density of AI-related chips (measured in watt per square millimeter, or W/mm2 of chip size) has increased at an average rate of 3.04% per year (see Figure 13).

While PUE efficiencies are near the maximum, the computational efficiency of AI chips is still improving at a fast clip, even as the electricity demand associated with AI chips has risen. Each chip has more GFLOPS packed in the same chip surface area, so each chip may still demand an increasing level of electricity.

Figure 13. Energy density (W/mm2 of die size) of generations of AI-related chips



Note 1: Size of the dot represents the relative FLOPS (for FP32 calculations) of the generation of AI-chips presented.

Note 2: The "die size" of a chip is the size of the silicon wafer where the circuit is printed, while the energy demand is based on the total design power of the chip.

Note 3: The Nvidia P100 DGXS is chosen as the first AI-related chip in the analysis as it is the first Nvidia GPU that uses a Server PCI Express Module ("SXM"), indicating that this chip is designed specifically for data centers (as opposed to workstation or gaming computer purpose).

Sources: TechPowerUp GPU database, IEEE Spectrum, LEI analysis.

5.4 Innovations in AI show dramatically lower energy demand

The overall energy demand associated with operating generative AI may prove significantly lower than previously expected. The launch of DeepSeek-R1 in early 2025 took the AI and energy industry by surprise—it is an event which might be considered a "wild card." It is a more energy-efficient AI model, and it raises important questions about how much electricity would be needed to support an AI-driven economy.[70] For example, the efficiency of DeepSeek-R1 implies less energy is needed. However, the model is also cheaper to train and use, and this could translate to lower prices charged to AI users; the lower prices could incentivize new and broader applications for AI, which would increase demand for data centers and the electricity to power them. The net effect on electricity demand is still unknown.

[70] DeepSeek uses the same kinds of chips as other AI programs.

Wherever data centers do locate, questions remain around how much energy they will require, and what kinds of resources will be necessary to meet that demand. Specifically, trends in AI suggest that gains in energy efficiency and demand response may significantly lessen the electricity requirements of future data centers. On the other hand, lower electricity costs could expand AI into new market applications.

Appendix B: Works cited

Amazon Web Services. *What is an Instance In Cloud Computing?* <https://aws.amazon.com/what-is/cloud-instances/>

Bain and Company. "Utilities Must Reinvent Themselves to Harness the AI-Driven Data Center Boom." October 2024. <https://www.bain.com/insights/utilities-must-reinvent-themselves-to-harness-the-ai-driven-data-center-boom/>

Bloomberg News. "How 'Load Shifting' May Help Improve Data Center Sustainability." *Data Center Knowledge.* February 26, 2024. <https://www.datacenterknowledge.com/sustainability/how-load-shifting-may-help-improve-data-center-sustainability>.

Burns, John. "Up to 82 data centres operating in Republic." *Irish Times.* June 17, 2023. <https://www.irishtimes.com/business/2023/06/17/up-to-82-data-centres-operating-in-republic/

Butler, Georgia. Data Center Dynamics. *AWS global data centers achieved PUE of 1.15 in 2023.* December 4, 2024. <https://www.datacenterdynamics.com/en/news/aws-global-data-centers-achieved-pue-of-115-in-2023/>

Casey, Simon. "GE Vernova CEO Sees Order Backlog Stretching Into 2028." *Bloomberg.* March 11, 2025. <https://www.bloomberg.com/news/articles/2025-03-11/ge-vernova-ceo-sees-order-backlog-stretching-into-2028?utm_source=chatgpt.com&embedded-checkout=true>

Chow, Andrew R. "Why Tax Breaks for Data Centers Could Backfire on States." April 25, 2025. <https://time.com/7280058/data-centers-tax-breaks-ai/>

Cisco. "What Is a Data Center?" <https://www.cisco.com/c/en/us/solutions/data-center-virtualization/what-is-a-data-center.html>

Cloudnium LLC. *Understanding the Differences Between Data Centers and Crypto Mining Facilities*. July 12, 2024. <https://cloudnium.net/understanding-the-differences-between-data-centers-and-crypto-mining-facilites/>

Collier's. "Accessing Power and Capital: The Future of Data Center Development 2024." <https://www.colliers.com/en/research/dmc-nrep-uscm-usdc-colliers-data-center-white-paper-2024?utm_source=google&utm_medium=paidsearch&utm_campaign=2023-colliers-leadgen-reports-nat-datacenter&utm_content=PID0001-v3&utm_term=hyperscale%20data%20centers&gad_source=1&gclid=CjwKCAiA1eO7BhATEiwAm0Ee-BlRD5_fY_rrI31camaws8bQjoF2ql4P796L6-FBwNxOHfyQSiCcgBoCiEAQAvD_BwE>

Cunningham, Nicholas. "Costs to build gas plants triple, says CEO of NextEra Energy." Gas Outlook. March 25, 2025. <https://gasoutlook.com/analysis/costs-to-build-gas-plants-triple-says-ceo-of-nextera-energy/>

Data Center Knowledge. "Hyperscalers in 2024: Where Next for the World's Biggest Data Center Operators?" February 28, 2024. <https://www.datacenterknowledge.com/hyperscalers/hyperscalers-in-2024-where-next-for-the-world-s-biggest-data-center-operators->

Dominion Energy. "Schedule 10 Large General Service." Virginia State Corporation Commission. December 20, 2022.

Dominion Energy. "Terms and Conditions Section IX. Deposits." Virginia State Corporation Commission. December 12, 2013.

Dominion Energy. "Terms and Conditions Section XXII. Electric Line Extensions and Installations." Virginia State Corporation Commission. March 8, 2024.

Eddy, Nathan. "Local Opposition Hinders More Data Center Construction Projects." Data Center Knowledge. May 15, 2025. <https://www.datacenterknowledge.com/regulations/local-opposition-hinders-more-data-center-construction-projects>

EIA. "Tracking electricity consumption from U.S. cryptocurrency mining operations." February 1, 2024. <https://www.eia.gov/todayinenergy/detail.php?id=61364#>

Energy Intelligence. "US Gas Companies Temper Data Center Demand Expectations." *Natural Gas Week* 41.11 (March 14, 2025).

IDC. "Global Semiconductor Market to Grow by 15% in 2025, Driven by AI." December 12, 2024. https://my.idc.com/getdoc.jsp?containerId=prAP52837624

FERC Docket No. EL25-49-000. Public Interest Organizations. *Comments of Public Interest Organizations in response to PJM Interconnection, L.L.C.'s 03/24/2025 Answer to FERC's 02/20/2025 Order under EL25-49*. April 23, 2025. P. 16-17. <https://elibrary.ferc.gov/eLibrary/filedownload?fileid=9D23F7BC-5484-CAA9-9030-96645FF00000>.

France 24. "Microsoft, Amazon to invest billions in French tech." December 5, 2024. <https://www.france24.com/en/live-news/20240512-amazon-plans-to-invest-1-2-bn-euros-in-france-macron-s-office?ref=frenchtechjournal.com>GPC. "Rules Regulations and Rate Schedules for Electric Service." Georgia Public Service Commission. January 2023.

Giacobone, Bianca. "Phantom data centers are flooding the load queue." March 26, 2025. <https://www.latitudemedia.com/news/phantom-data-centers-are-flooding-the-load-queue/>

Goldman Sachs. "AI, data centers and the coming US power demand surge." April 28. 2024. <https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf>

Google Data Centers. "Growing the internet while reducing energy consumption." <https://www.google.com/about/datacenters/efficiency/>

GPC. "Customer Renewable Supply Procurement Schedule CRSP-1." Georgia Public Service Commission. October 2023.

GPC. "Power and Light Large Schedule PLL-18." Georgia Public Service Commission. April 2025.

GPC. "Rules Regulations and Rate Schedules for Electric Service," Georgia Public Service Commission. January 2023.

GPSC Docket No. 44280. GPC. *Rules & Regulations Update 12-11-2024*. December 2024. <https://psc.ga.gov/search/facts-document/?documentId=220667>

GPSC Docket No. 55378. *Comments on Georgia Power's 2023 Integrated Resource Plan Update*. Microsoft. Docket No. 55378 at 1, 4. Apr. 1, 2024. <https://psc.ga.gov/search/facts-document/?documentId=218199>

GPSC Docket No. 56002. GPC. "2025 IRP Technical Appendix Volume 1: B2025 Load and Energy Forecast." January 2025. <https://psc.ga.gov/search/facts-document/?documentId=221233>

Gunia, Amy. "Will 'massive' Gulf deals cement the US lead in the race for global AI dominance?" CNN. May 22, 2025. <https://www.cnn.com/2025/05/22/business/trump-gulf-deals-global-ai-race-hnk-spc-intl>

Harris, Dion. *Sustainable Strides: How AI and Accelerated Computing Are Driving Energy Efficiency*. July 22, 2024. Nvidia. <https://blogs.nvidia.com/blog/accelerated-ai-energy-efficiency/>

IEA. "Average lead times to build new electricity grid assets in Europe and the United States, 2010-2021." January 12, 2023. <https://www.iea.org/data-and-statistics/charts/average-lead-times-to-build-new-electricity-grid-assets-in-europe-and-the-united-states-2010-2021>

IEA. "Average power generation construction time (capacity weighted), 2010-2018." April 14, 2019. <https://www.iea.org/data-and-statistics/charts/average-power-generation-construction-time-capacity-weighted-2010-2018>

IEA. *Energy and AI*. April 2025. <https://www.iea.org/reports/energy-and-ai>.

IEA. "What the data centre and AI boom could mean for the energy sector." October 18, 2024. <https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector>

Indiana Michigan Power Company. "Submission of Unopposed Settlement Agreement and Unopposed Motion for Acceptance of Out of Time Filing." Indiana Utility Regulatory Commission Cause No. 46097. November 22, 2024.

Judge, Peter. "Google tests system to cut data center power use during grid problems." *Data Center Dynamics*. October 5, 2023. <https://www.datacenterdynamics.com/en/news/google-tests-system-to-cut-data-center-power-use-during-grid-problems/>

Khan, Saif M. and Alexander Mann. *AI Chips: What They Are and Why They Matter*. Center for Security and Emerging Technology. April 2020. <cset.georgetown.edu/research/ai-chips-what-they-are-and-why-they-matter>

Koomey, Jonathan (Koomey Analytics), Schmidt, Zachary (Koomey Analytics), and Das, Tania (Bipartisan Policy Center). *Electricity Demand Growth and Data Centers: A Guide for the Perplexed*. February 2025. <https://bipartisanpolicy.org/download/?file=/wp-content/uploads/2025/02/BPC-Report-Electricity-Demand-Growth-and-Data-Centers-A-Guide-for-the-Perplexed.pdf>.

Kusters, J., Bhattacharjee, D., Bish, J., Nicholas, J.T., Stewart, D., Ramachandran, K. "2025 global semiconductor industry outlook." Deloitte. February 4, 2025. <https://www2.deloitte.com/us/en/insights/industry/technology/technology-media-telecom-outlooks/semiconductor-industry-outlook.html>

Levitt, Ben. "AI and Energy, the Big Picture." S&P Global. December 2024. <https://www.spglobal.com/en/research-insights/special-reports/look-forward/ai-and-energy>

Malik, Naureen S. "Gas Power Won't Provide an Easy Fix for AI Boom." Bloomberg. January 8, 2025. <https://www.bloomberg.com/news/newsletters/2025-01-08/gas-power-won-t-provide-an-easy-fix-for-ai-boom>

Mannion, Annemarie. "Microsoft Hits Pause Button on $1B in Data Centers in Ohio." *Engineering News-Record*. April 10, 2025. <https://www.enr.com/articles/60572-microsoft-hits-pause-button-on-1b-in-data-centers-in-ohio#:~:text=By%20Annemarie%20Mannion,contractors%20had%20not%20been%20announced>

Martucci, Brian. "A fraction of proposed data centers will get built. Utilities are wising up." *Utility Dive*. May 15, 2025. <https://www.utilitydive.com/news/a-fraction-of-proposed-data-centers-will-get-built-utilities-are-wising-up/748214/>

McKinsey & Company. "AI power: Expanding data center capacity to meet growing demand." October 29, 2024. <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand>

Microsoft. "Microsoft Expands Digital Footprint In Quebec With Usd$500 Million Investment In Infrastructure And Skilling Initiatives". November 2023. <https://news.microsoft.com/en-ca/2023/11/22/microsoft-expands-digital-footprint-in-quebec-with-usd500-million-investment-in-infrastructure-and-skilling-initiatives/>

MIT Sloan Management Review. "How the Middle East is Emerging as a Data Center Powerhouse Amid Booming AI Demand." October 23, 2024. <https://www.mitsloanme.com/article/how-the-middle-east-is-emerging-as-a-data-center-powerhouse-amid-booming-ai-demand/>

Morgan Stanley Research. "AI Supply Chain – The Latest about NVDA GB200 Superchip." May 13, 2024.

New England States Committee on Electricity. "Data Centers and the Power System: A Primer." Spring 2024. <https://nescoe.com/resource-center/data-centers-primer/>

Noelle Walsh, President Microsoft Cloud Operations and Innovation | Nouryon, Non-Executive Director. <https://www.linkedin.com/posts/noelle-walsh-b29356108_microsoftcloud-datacenters-activity-7315439628562423808-W67e/>

Norris, T. H., T. Profeta, D. Patino-Echeverri, and A. Cowie-Haskell. 2025. "Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems." NI R 25-01. Durham, NC: Nicholas Institute for Energy, Environment & Sustainability, Duke University. <https://nicholasinstitute.duke.edu/publications/rethinking-load-growth>

Novet, Jordan. "Amazon has paused some data center lease commitments, Wells Fargo says." CNBC. April 21, 2025. <https://www.cnbc.com/2025/04/21/amazon-has-paused-some-data-center-lease-commitments-wells-fargo.html>

NREL. "Data Center Infrastructure in the United States, 2025." May 2025. <https://docs.nrel.gov/docs/gen/fy25/94502.jpg>

Nvidia. *Form 10-K. Annual Report for fiscal year ended January 26, 2025.*

Patel, Dylan; Jeremie Eliahou Ontiveros; Maya Barkin. "Microsoft's Datacenter Freeze - 1.5GW Self-Build Slowdown & Lease Cancellation Misconceptions." *SemiAnalysis*. April 28, 2025. <https://semianalysis.com/2025/04/28/microsofts-datacenter-freeze/>

PricewaterhouseCoopers, *State of the Semiconductor Industry*. November 28, 2024. <https://www.pwc.com/gx/en/industries/technology/state-of-the-semiconductor-industry-report.pdf>

Seiple, Chris. "Gridlock: The demand dilemma facing the US power industry." *Wood Mackenzie.* October 2024. <https://www.woodmac.com/horizons/gridlock-demand-dilemma-facing-us-power-industry/>

Semiconductor Industry Association. “Strengthening the U.S. Semiconductor Supply Chain.” May 2024. <https://www.semiconductors.org/wp-content/uploads/2024/05/Emerging-Resilience-in-the-Semiconductor-Supply-Chain_SIA-Summary.pdf>

Semiconductor Insight. “Silicon Wafer Market Size, Share, Trends, Market Growth and Business Strategies 2025-2032.” (Undated). <https://semiconductorinsight.com/report/silicon-wafer-market/>

Shah, Agam. *Nvidia Shipped 3.76 Million Data-center GPUs in 2023, According to Study*. June 10, 2024. <https://www.hpcwire.com/2024/06/10/nvidia-shipped-3-76-million-data-center-gpus-in-2023-according-to-study/>

Shehabi, A., Smith, S.J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M.A.B., Holecek, B., Koomey, J., Masanet, E., Sartor, D. 2024. *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, Berkeley, California. LBNL-2001637. Pg 53. Figure 5.6.

Taiwan Semiconductor Manufacturing Company Limited. *TSMC Arizona.* <https://www.tsmc.com/static/abouttsmcaz/index.htm>

TechPowerUp. “NVIDIA B200 SXM 192 GB.” <https://www.techpowerup.com/gpu-specs/b200-sxm-192-gb.c4210>

TechPowerUp. “NVIDIA Tesla P100 DGXS.” <https://www.techpowerup.com/gpu-specs/tesla-p100-dgxs.c3285>

Williams, Kevin. “AI data center boom isn’t going bust but the ‘pause’ is trending at big tech companies.” CNBC. April 27, 2025. <https://www.cnbc.com/2025/04/27/ai-data-center-boom-isnt-going-bust-but-the-pause-is-trending.html>